Kubernetes

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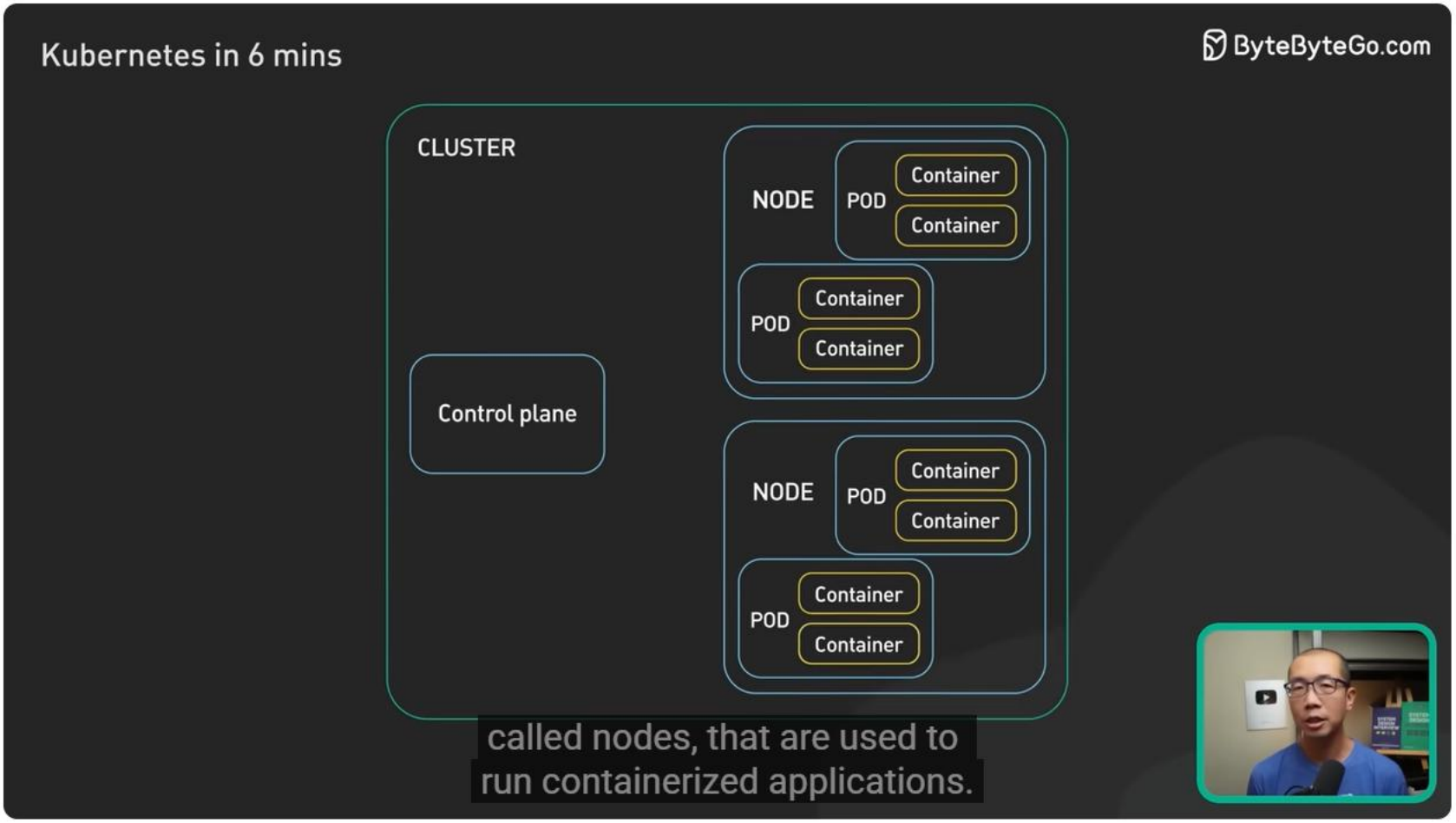
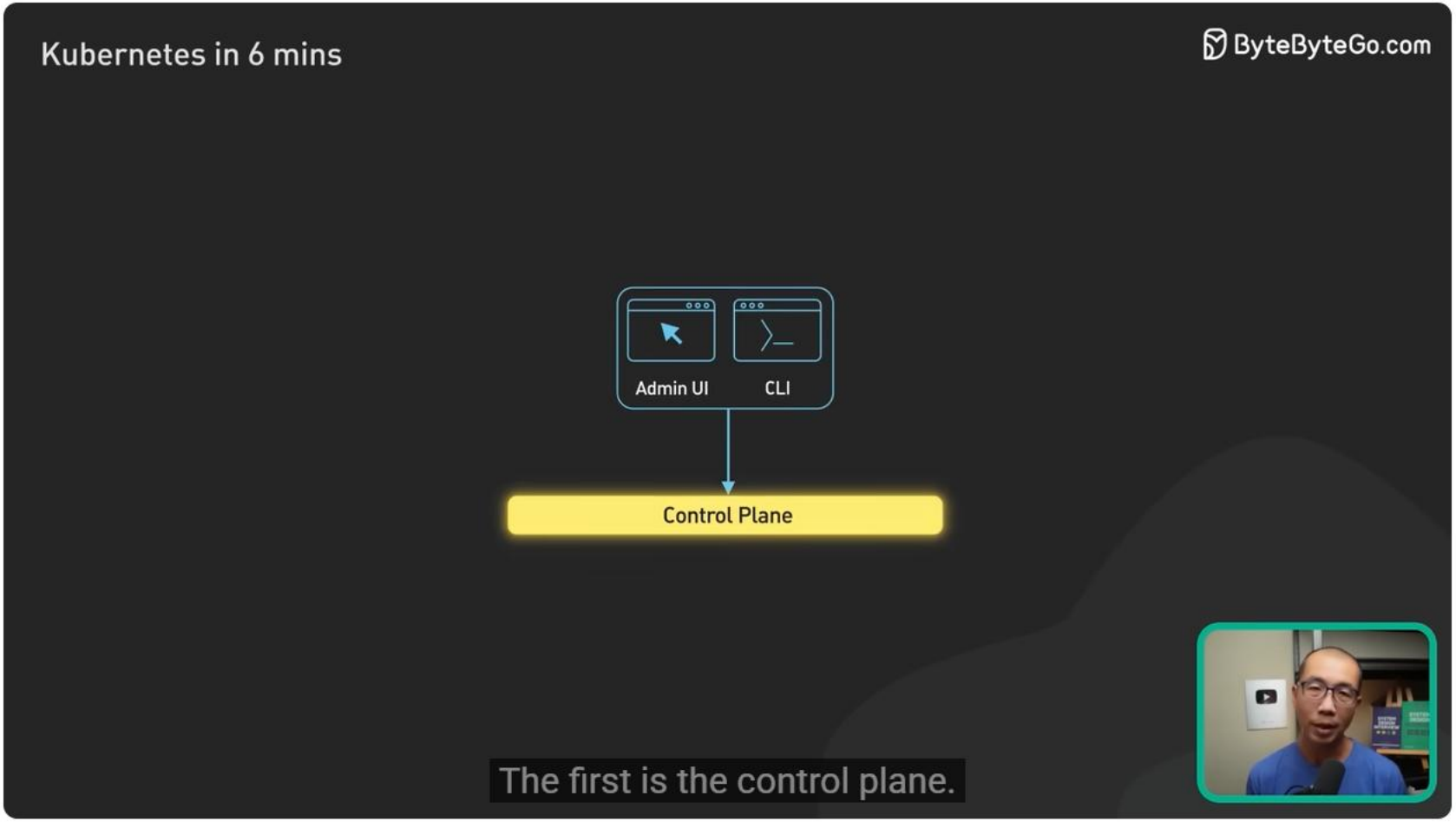
Overview

[Kubernetes Explained in 6 Minutes | k8s Architecture]



Kubernetes Explained in 6 Minutes | k8s Architecture



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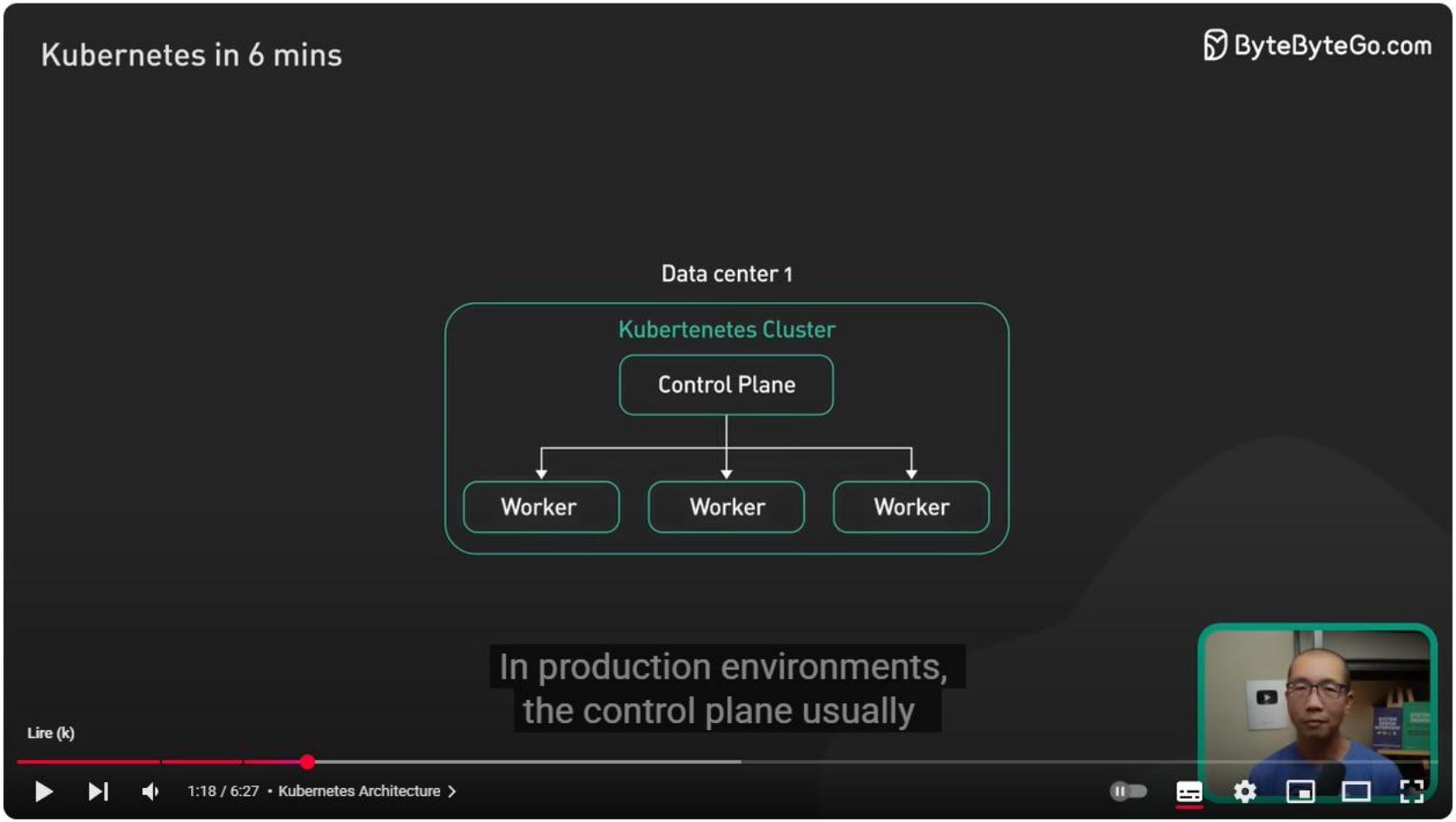

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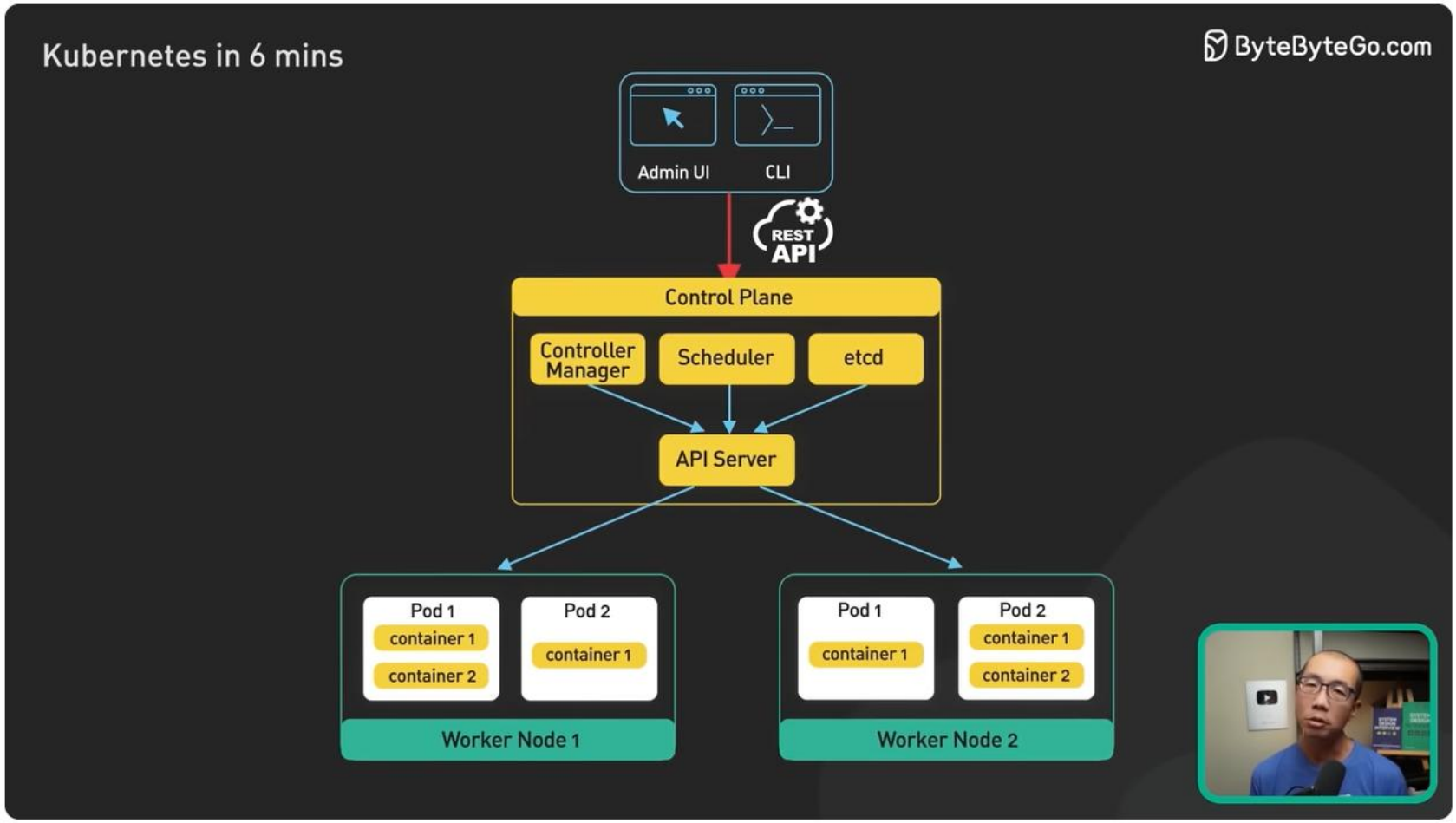

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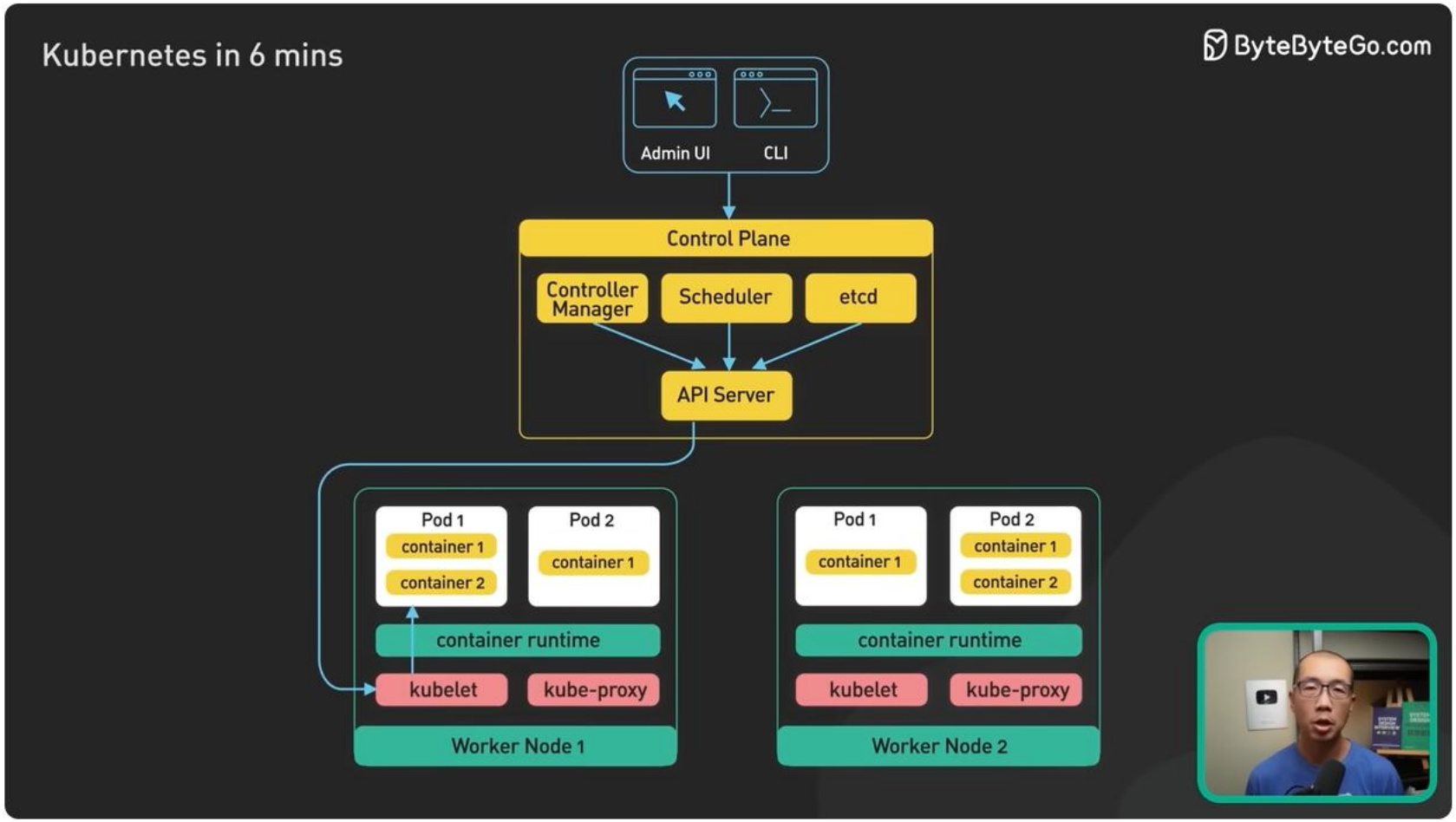

Kubernetes Explained in 6 Minutes | k8s Architecture

- etdc : stores key-value data
- highly scalable
- portable
- operating cost is high

- complex to manage
- costly

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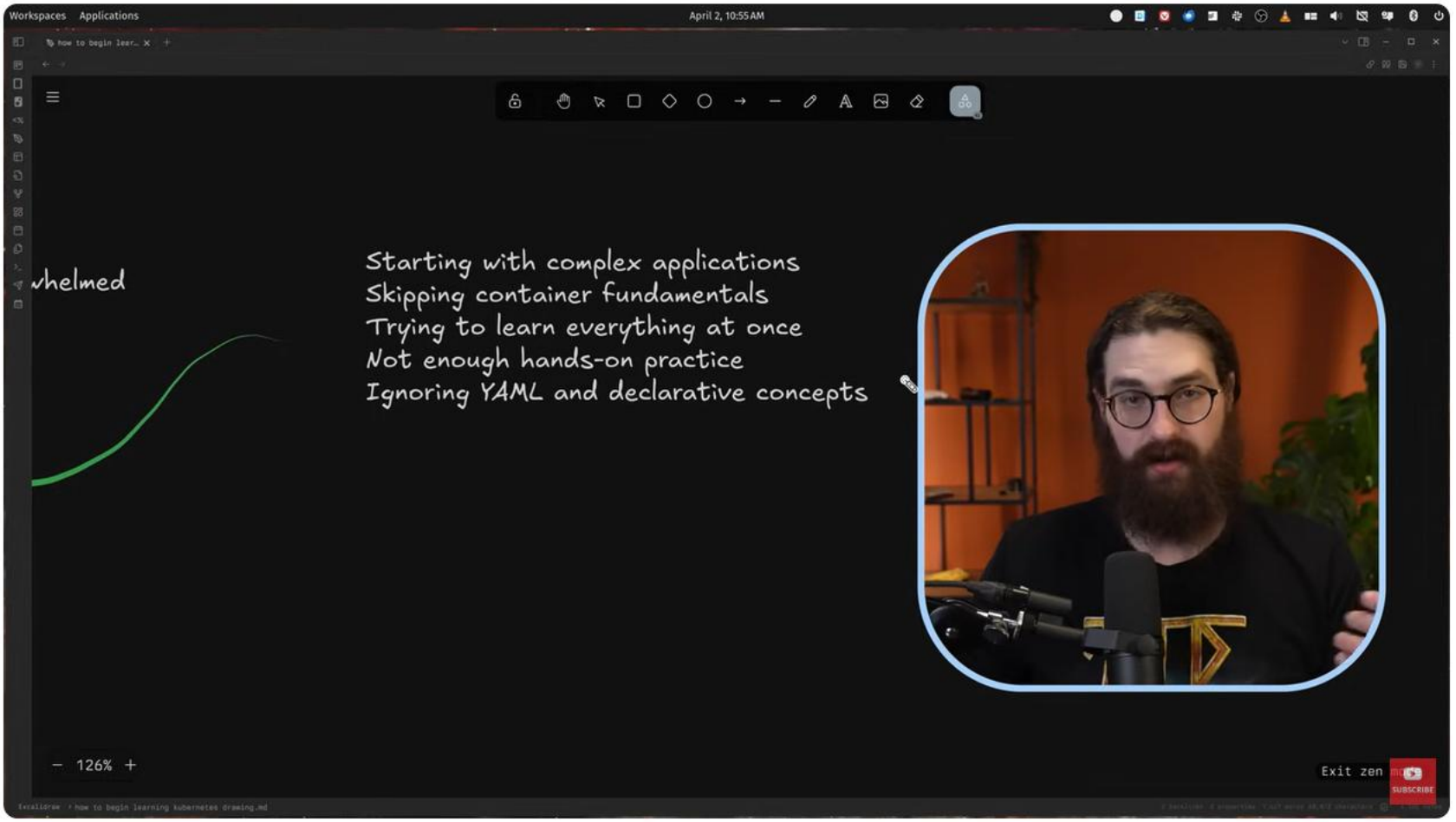
Common mistakes:



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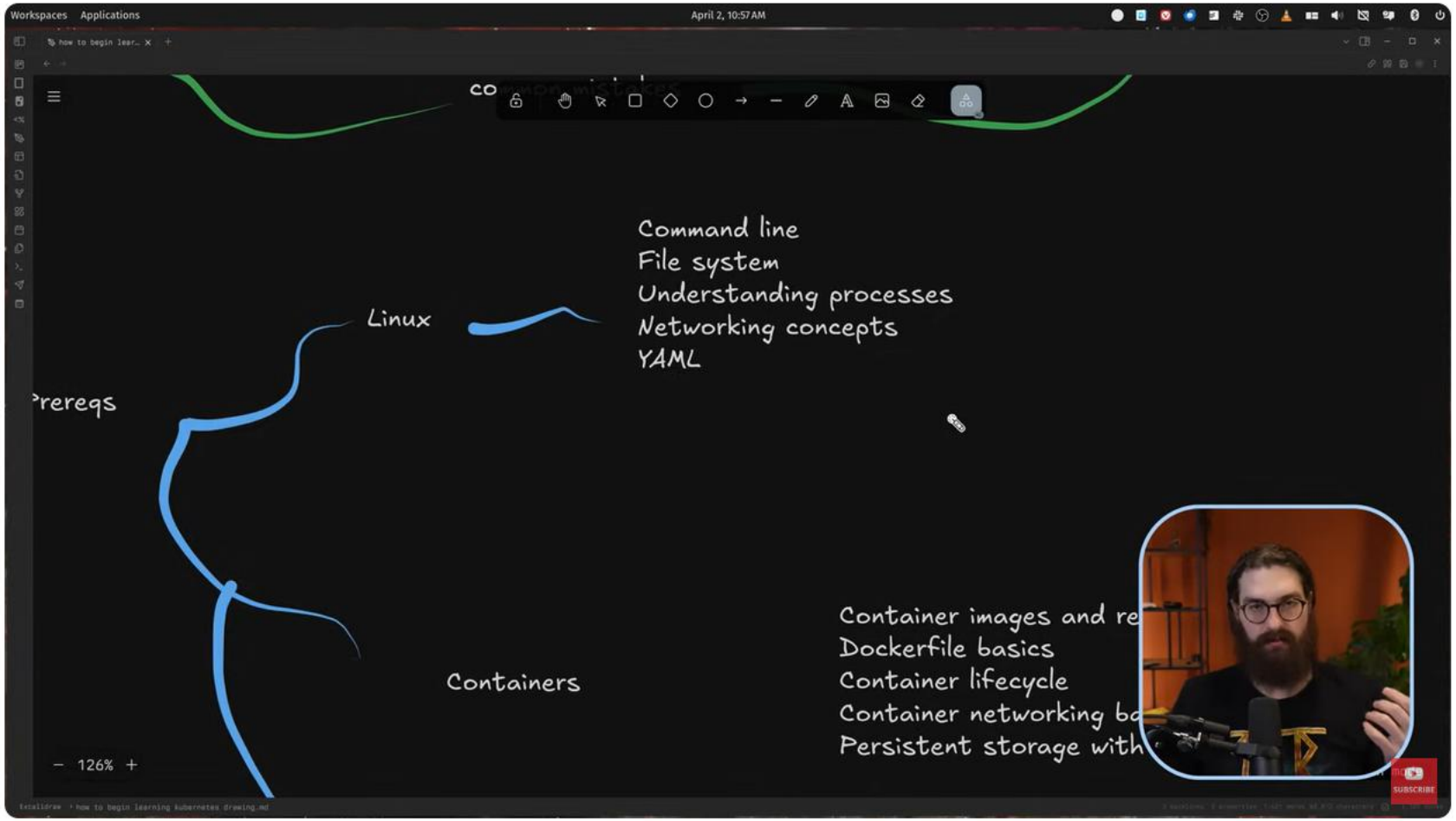
Prerequisites:



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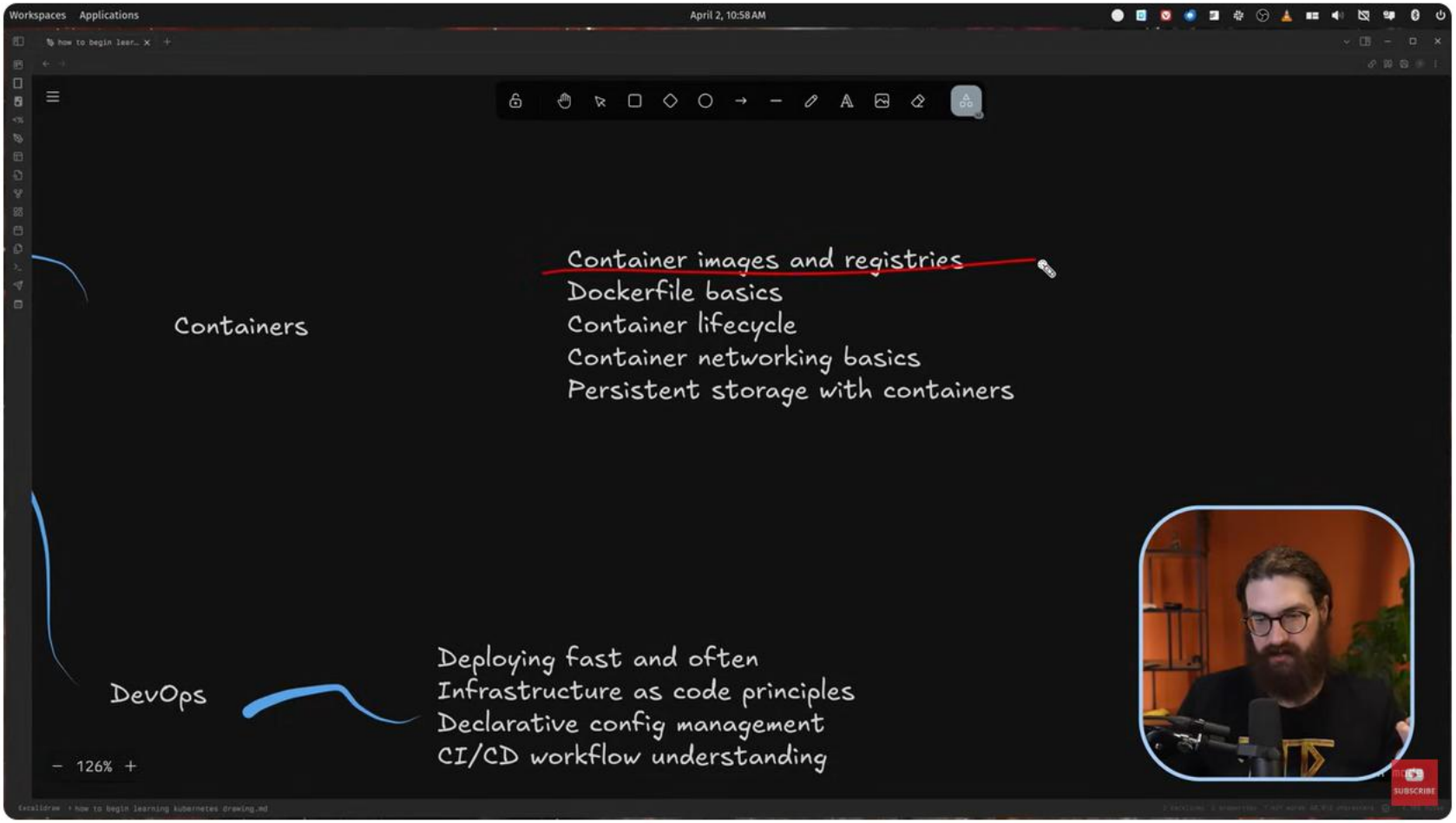

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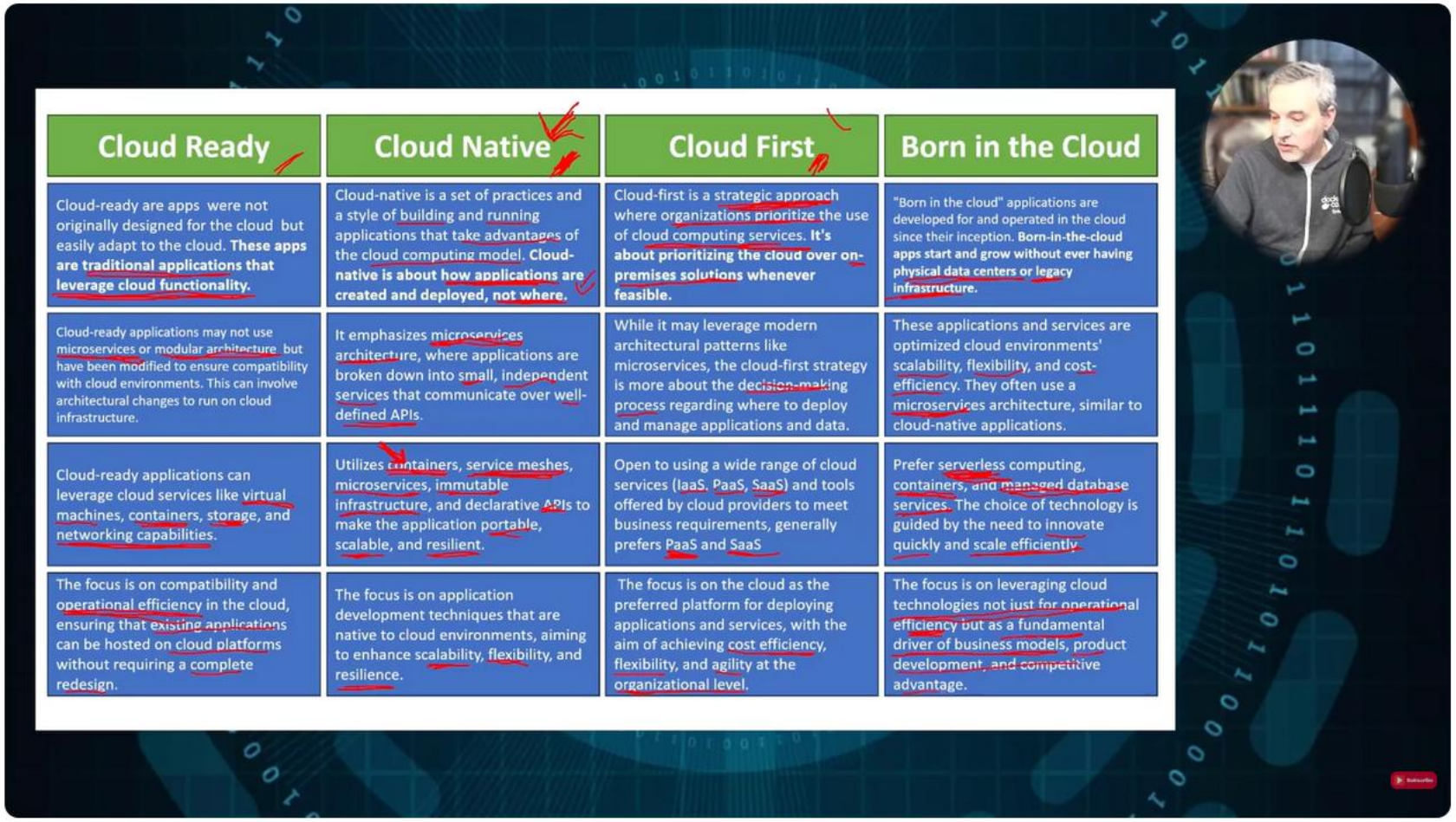
- Github: https://github.com/mischavandenburg
- Learn docker/container
- Learning:
    - https://labs.iximiuz.com/

    - https://www.skool.com/kubecraft/about
- Tools:
    - Ranger desktop (open source)
    - k3s
- isolated environment to train

Cloud Native vs Cloud First vs Cloud Ready vs Born in the Cloud: The Ultimate Showdown



Cloud Native vs Cloud First vs Cloud Ready vs Born in the Cloud: The Ultimate Showdown

Migration

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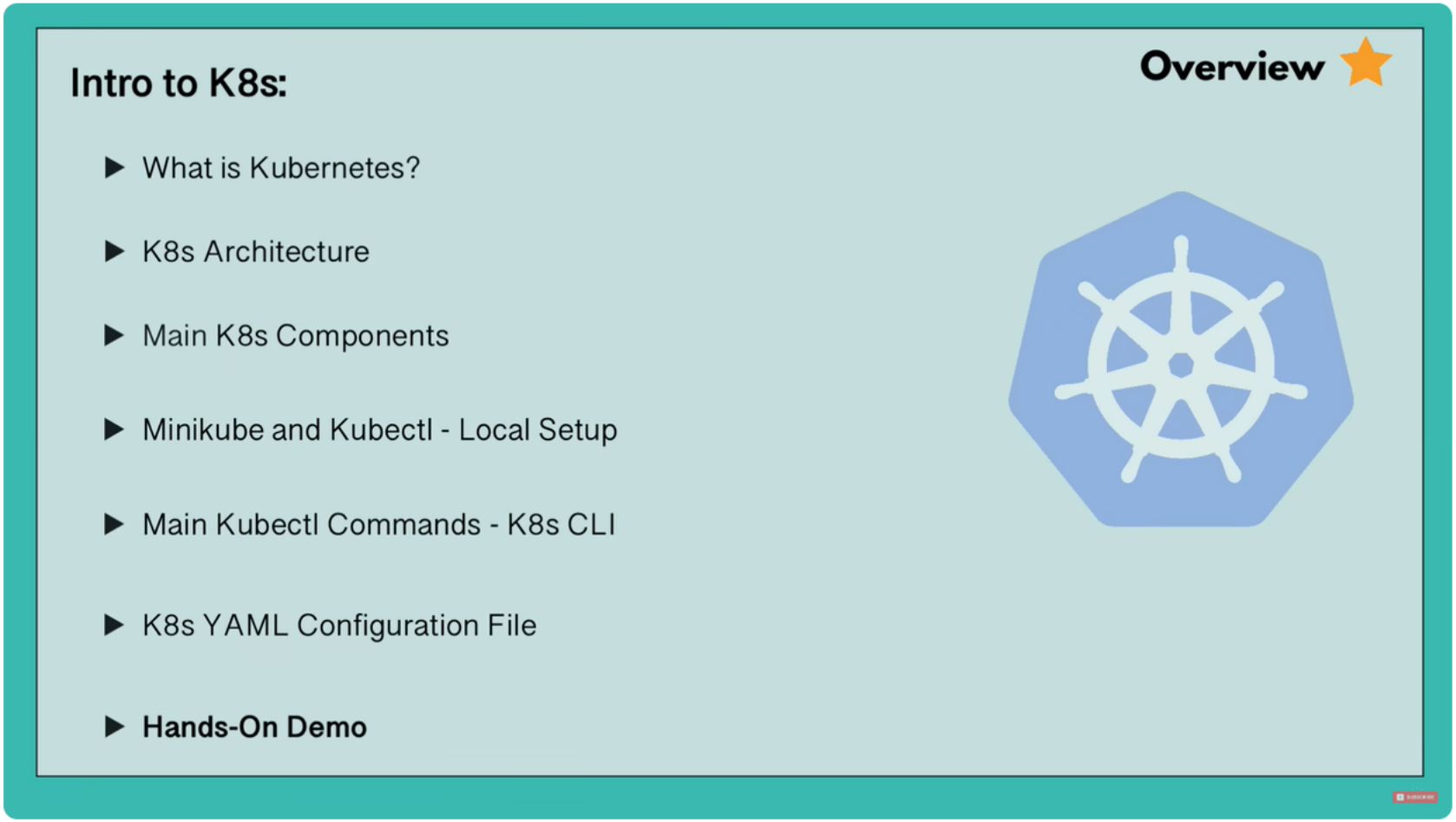
- k8s adds complexity, should be known before moving into k8s
- migration: make the application stateless/cloud native principles
- Ensure that the tests are complete/ready to ensure it is working once deployed
- Not always necessary to split a monolith into micro services if too much tangled/cobbled services, it would become a distributed monolith instead of micro-services
- With containers, sometimes DRY principle should not be too strict (e.g. copy paste into several micro-services for coupling)
- One application One container, Do Not migrate all at once
- Embrace the community and mindset

Migrating to Kubernetes

Learning

Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



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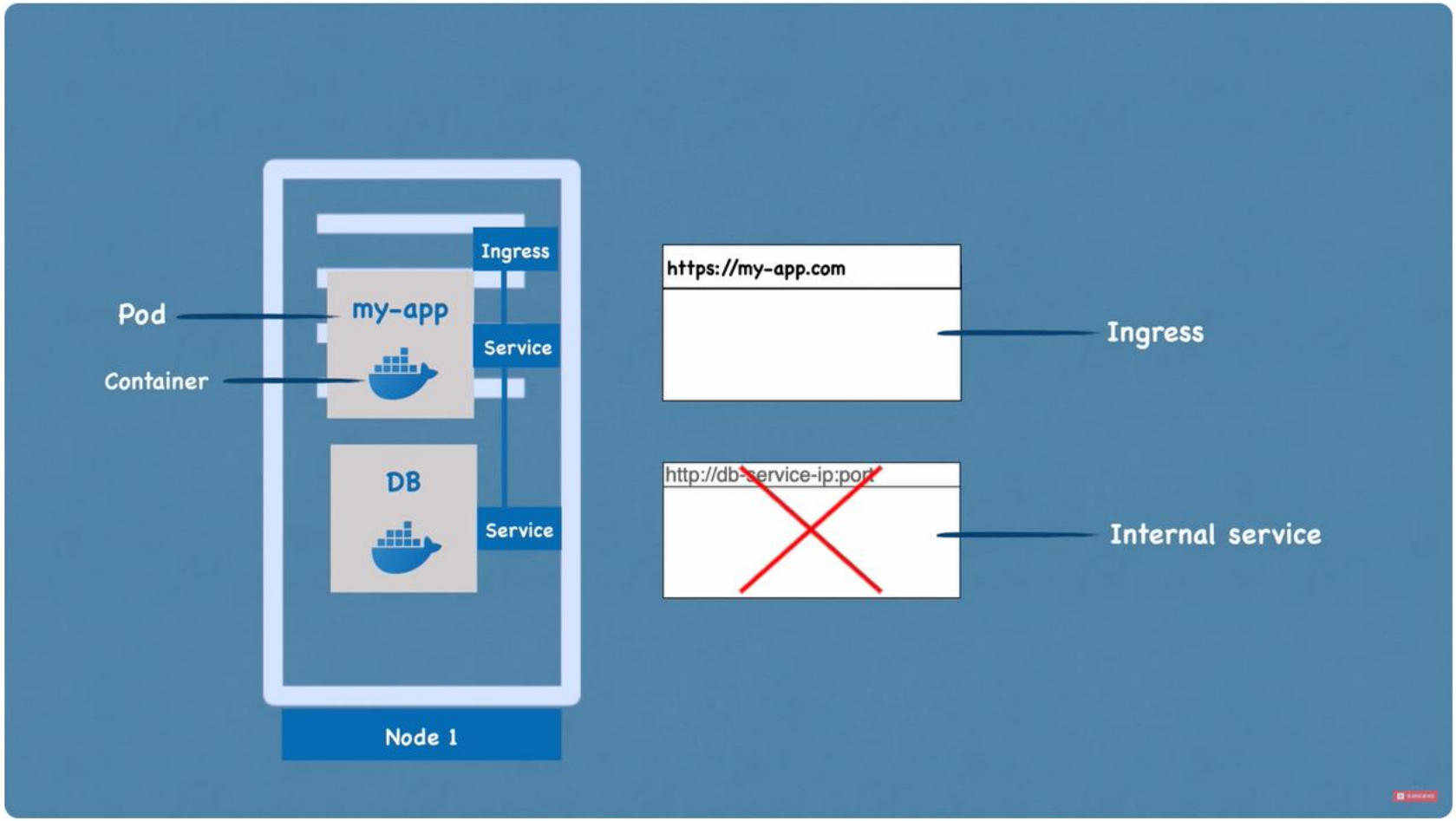

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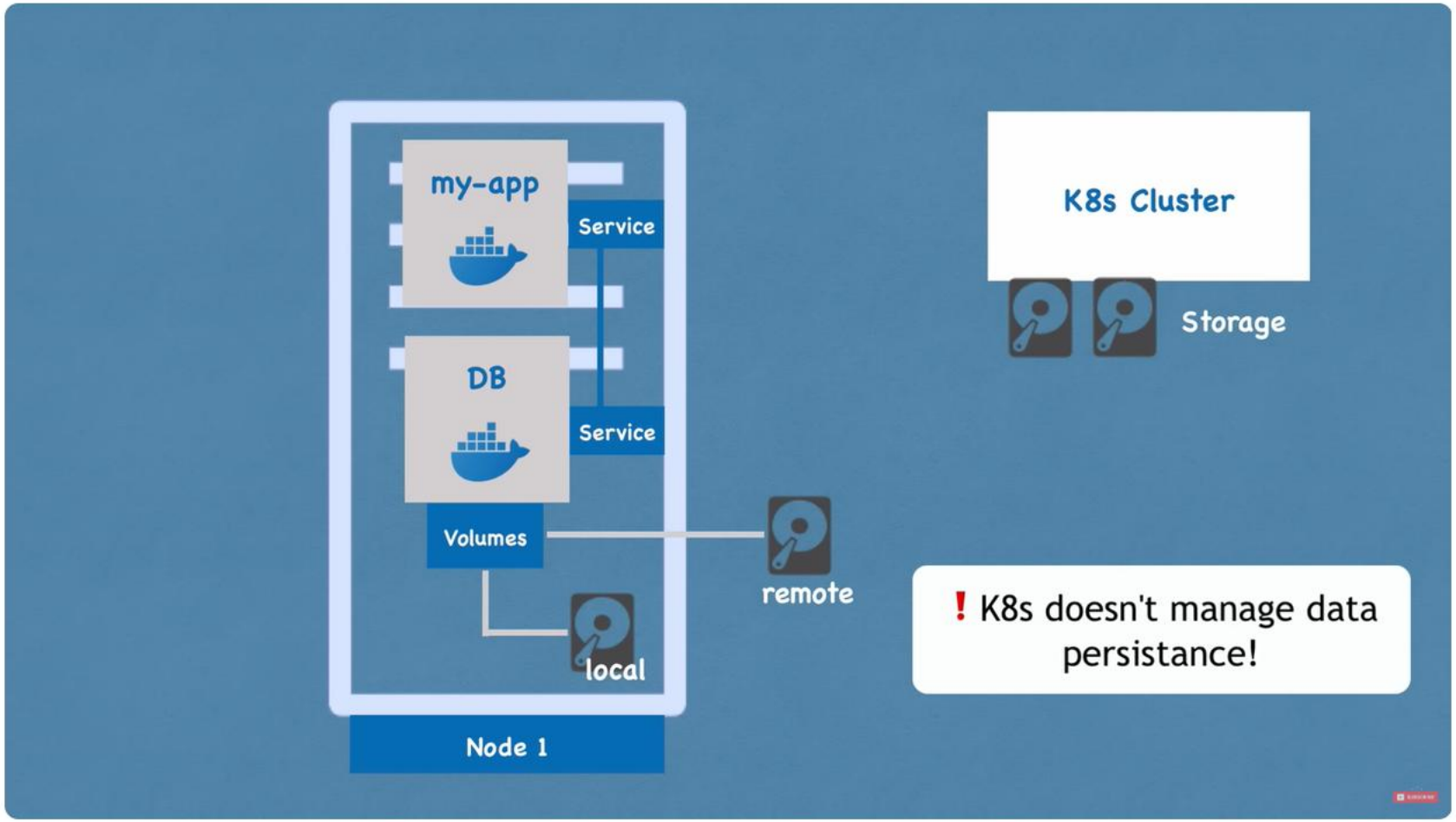

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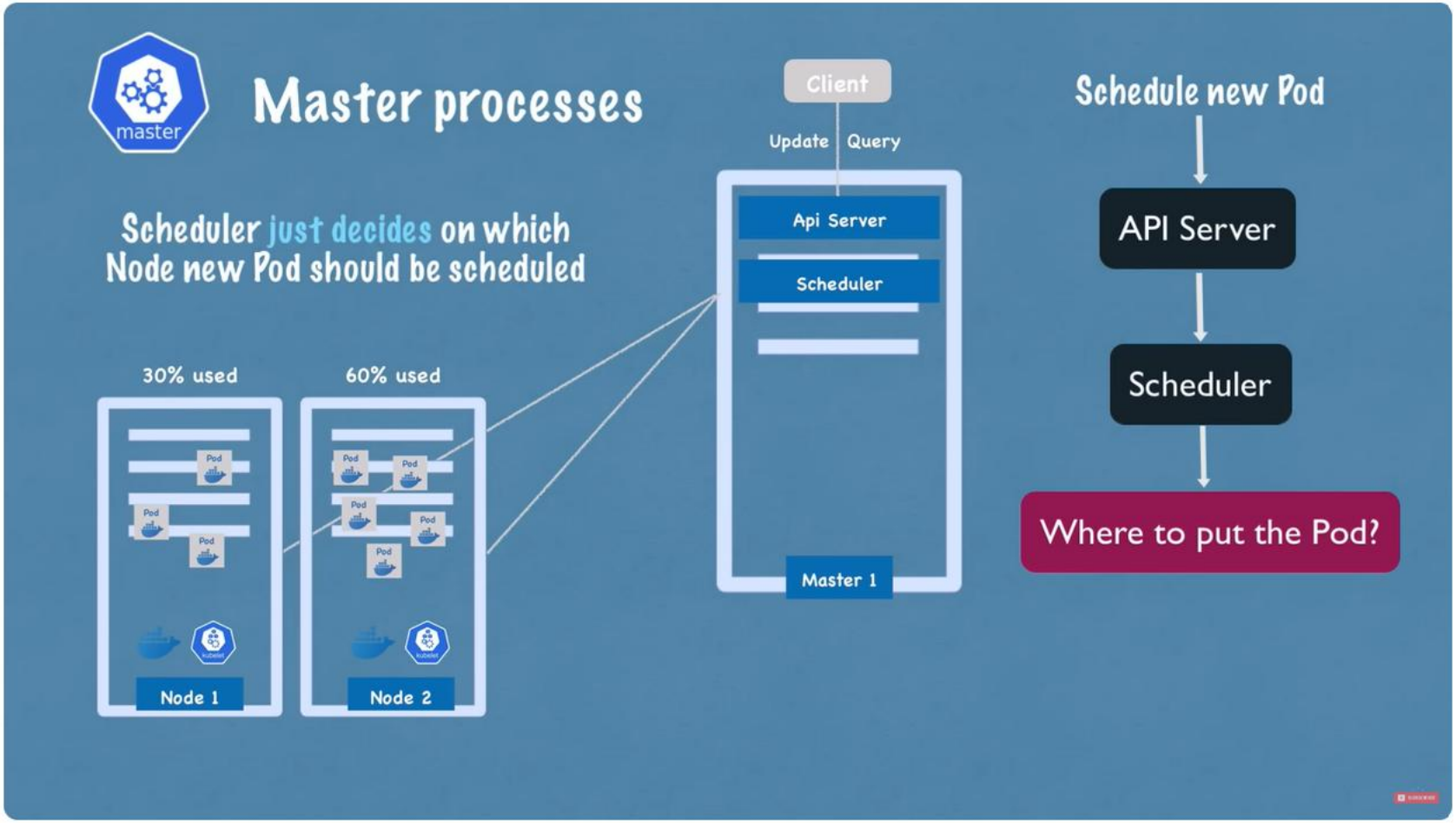

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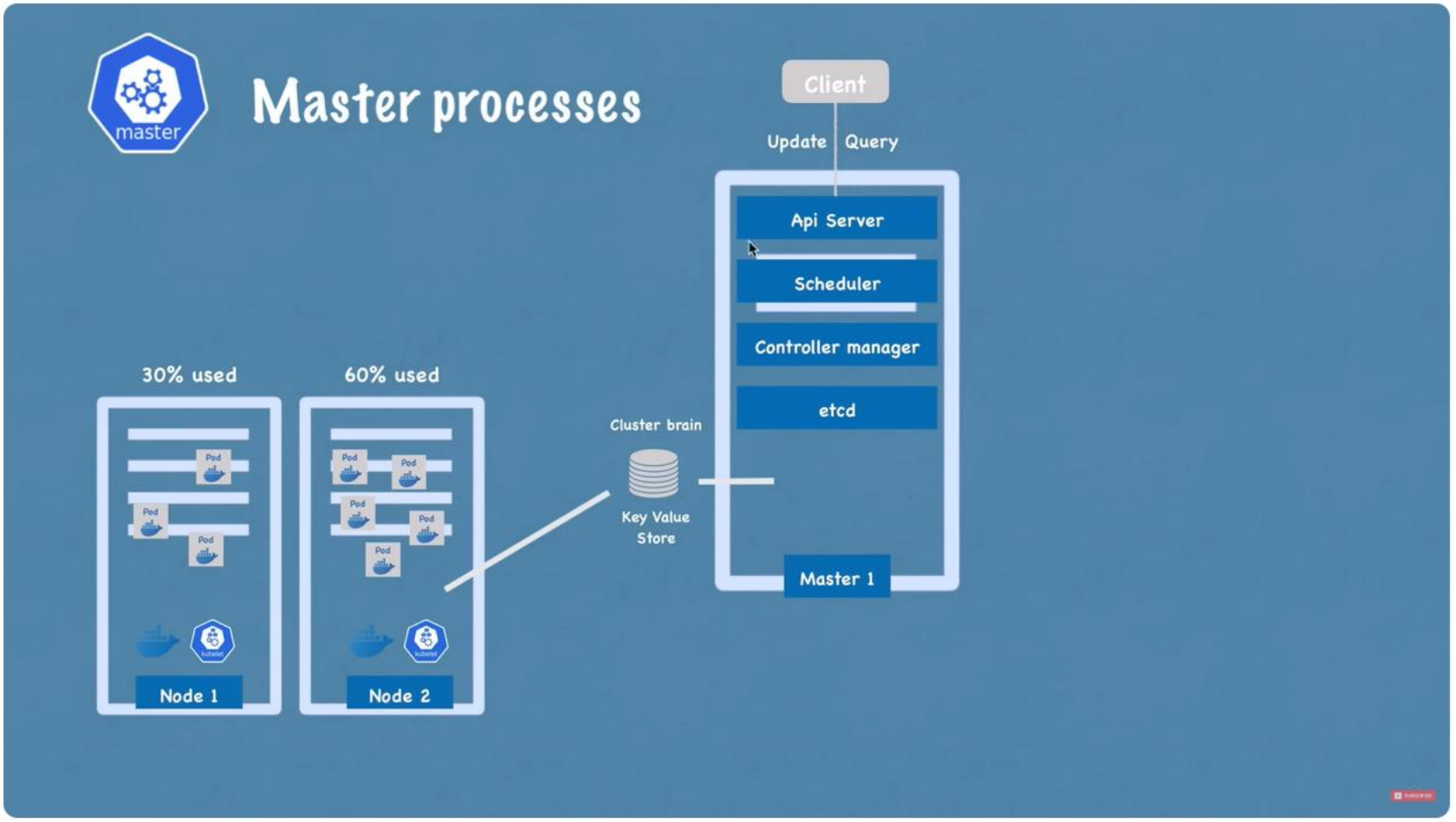

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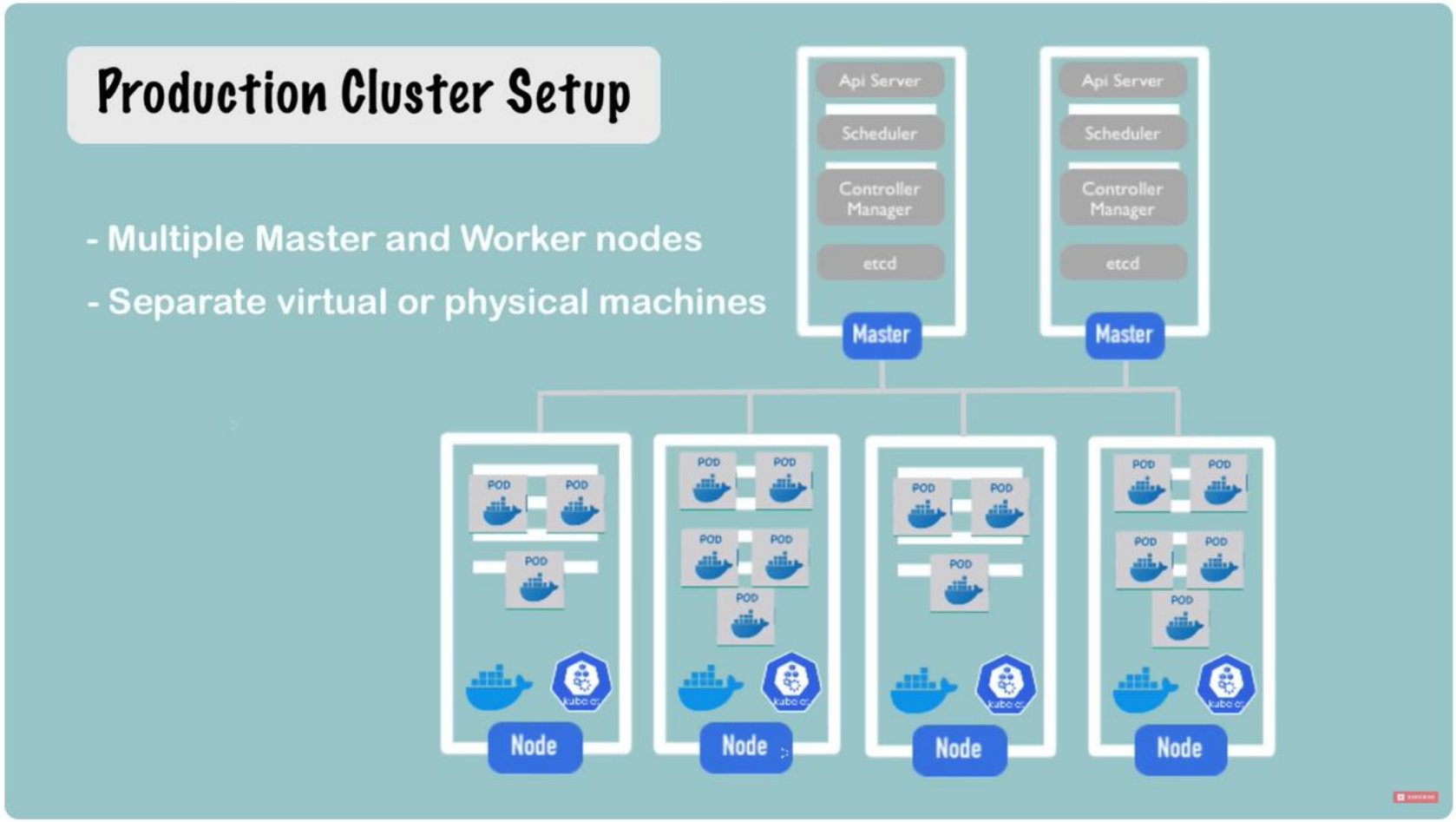

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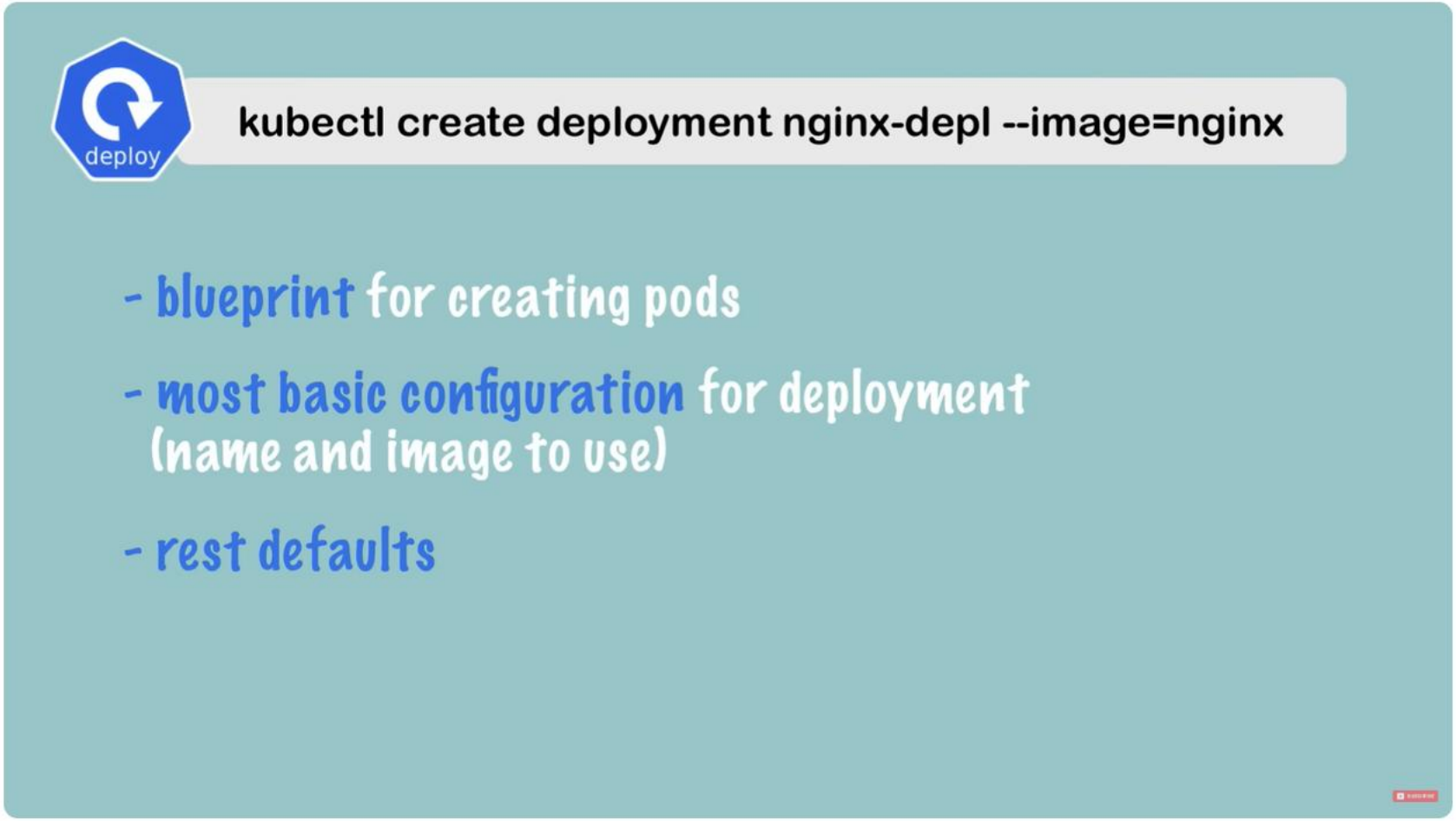

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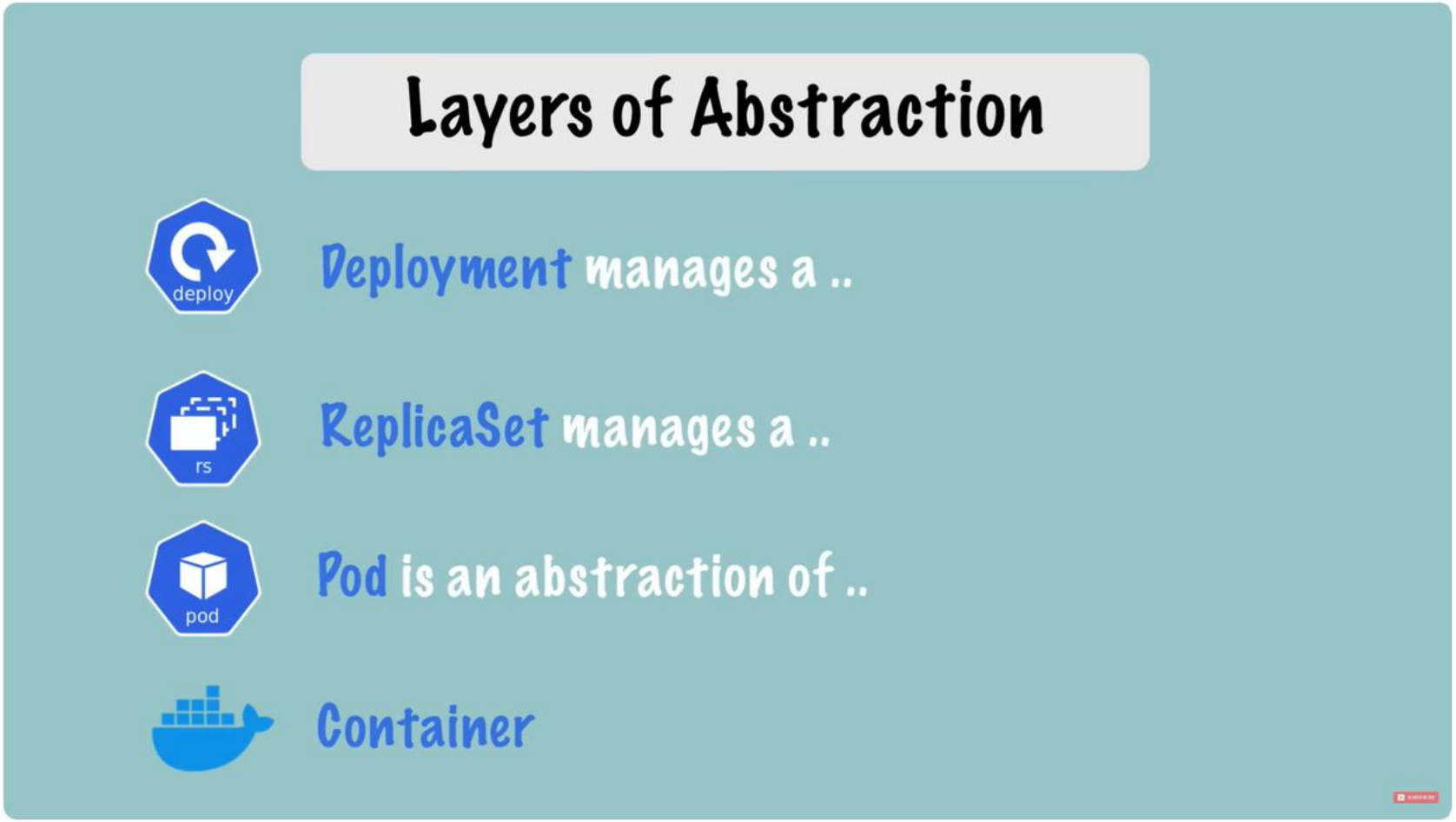

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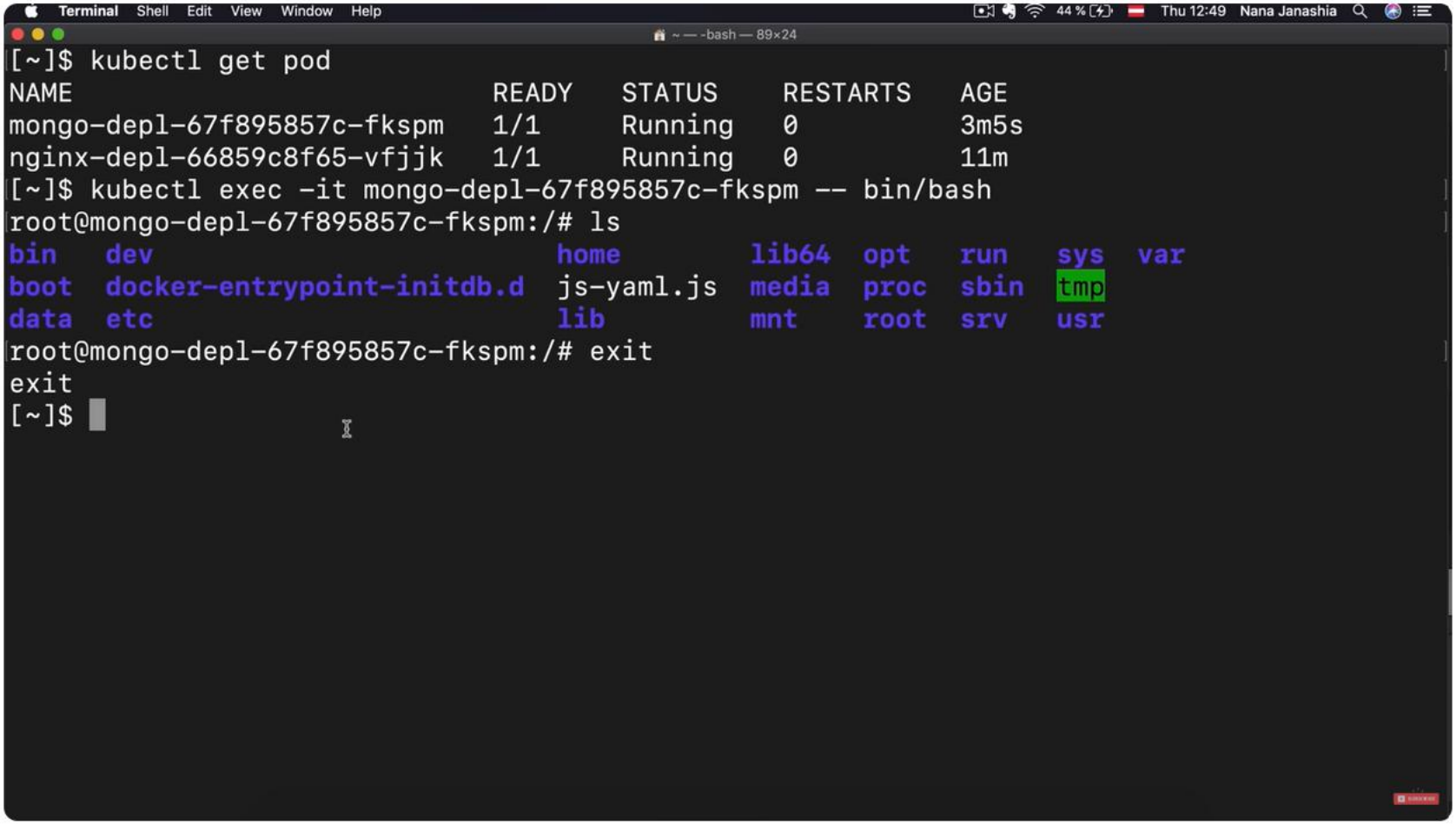

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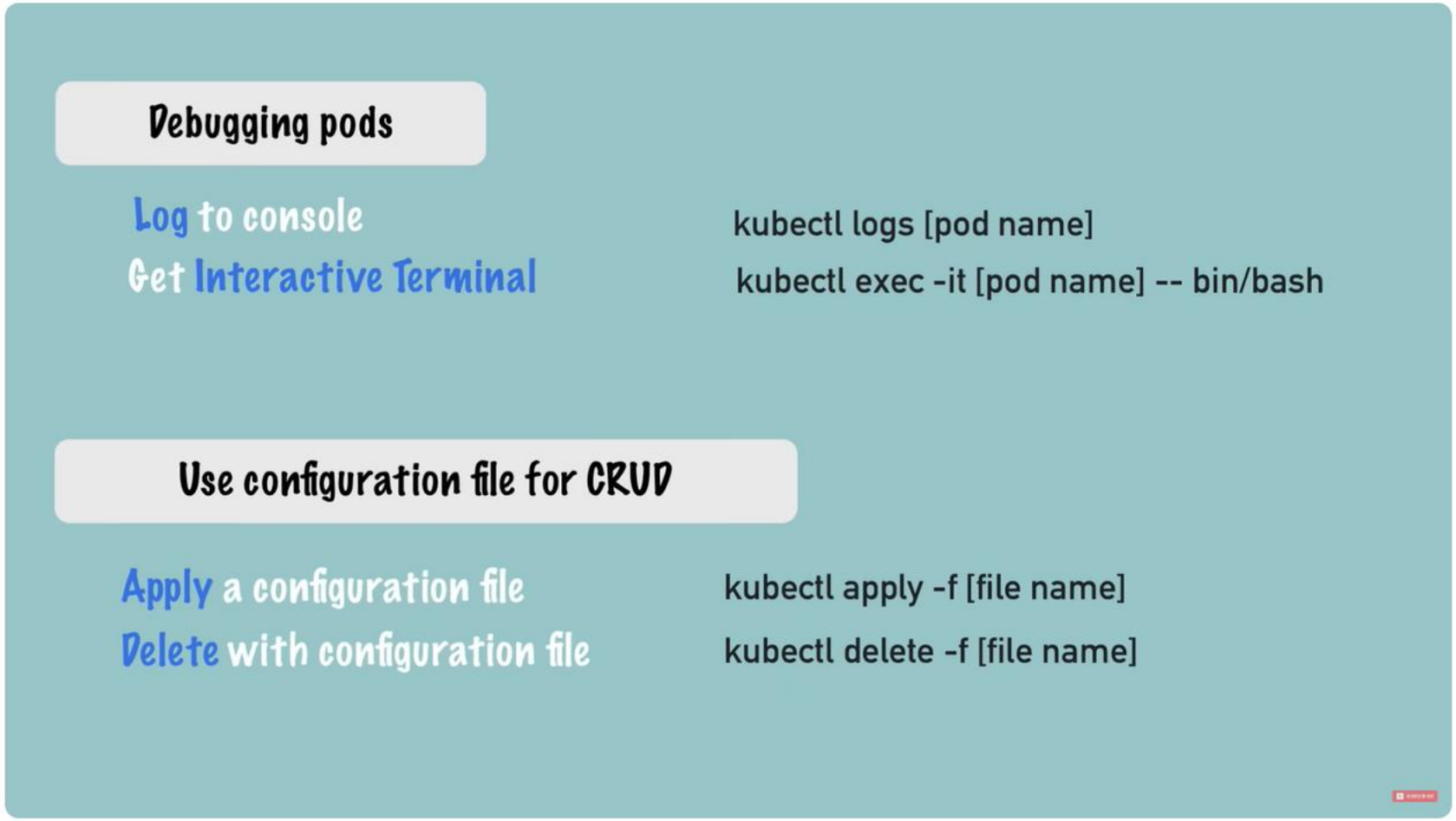

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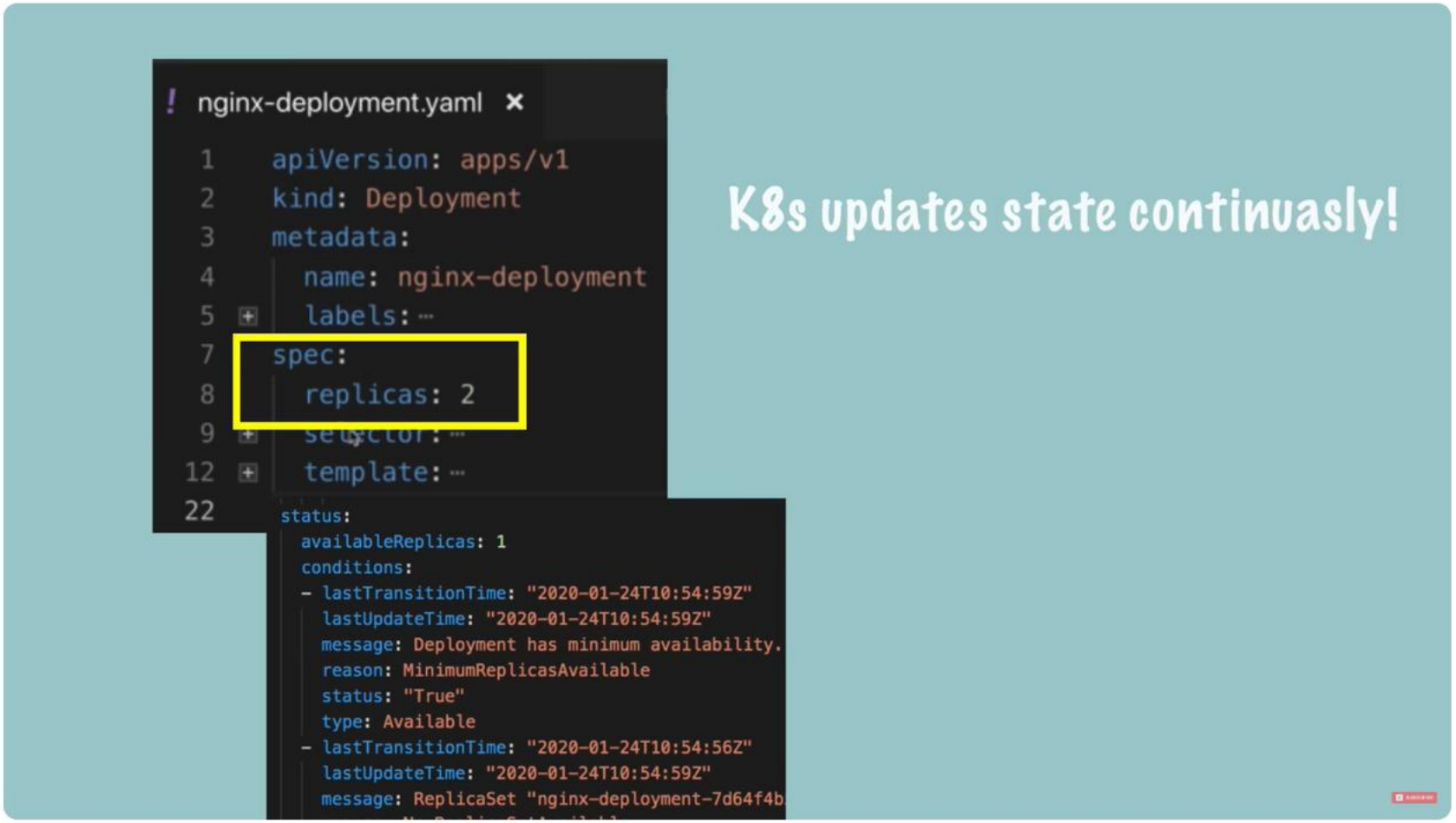

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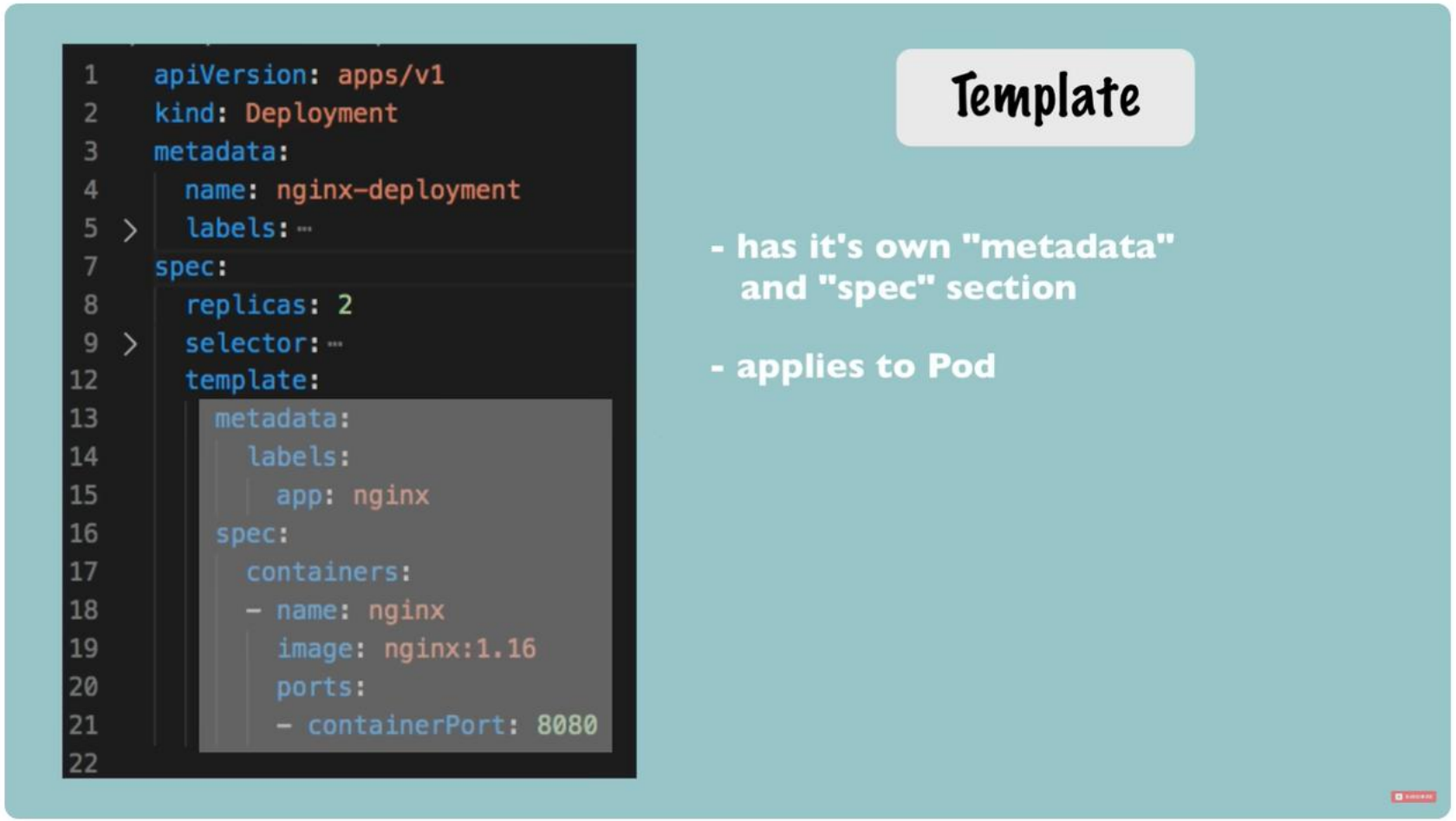

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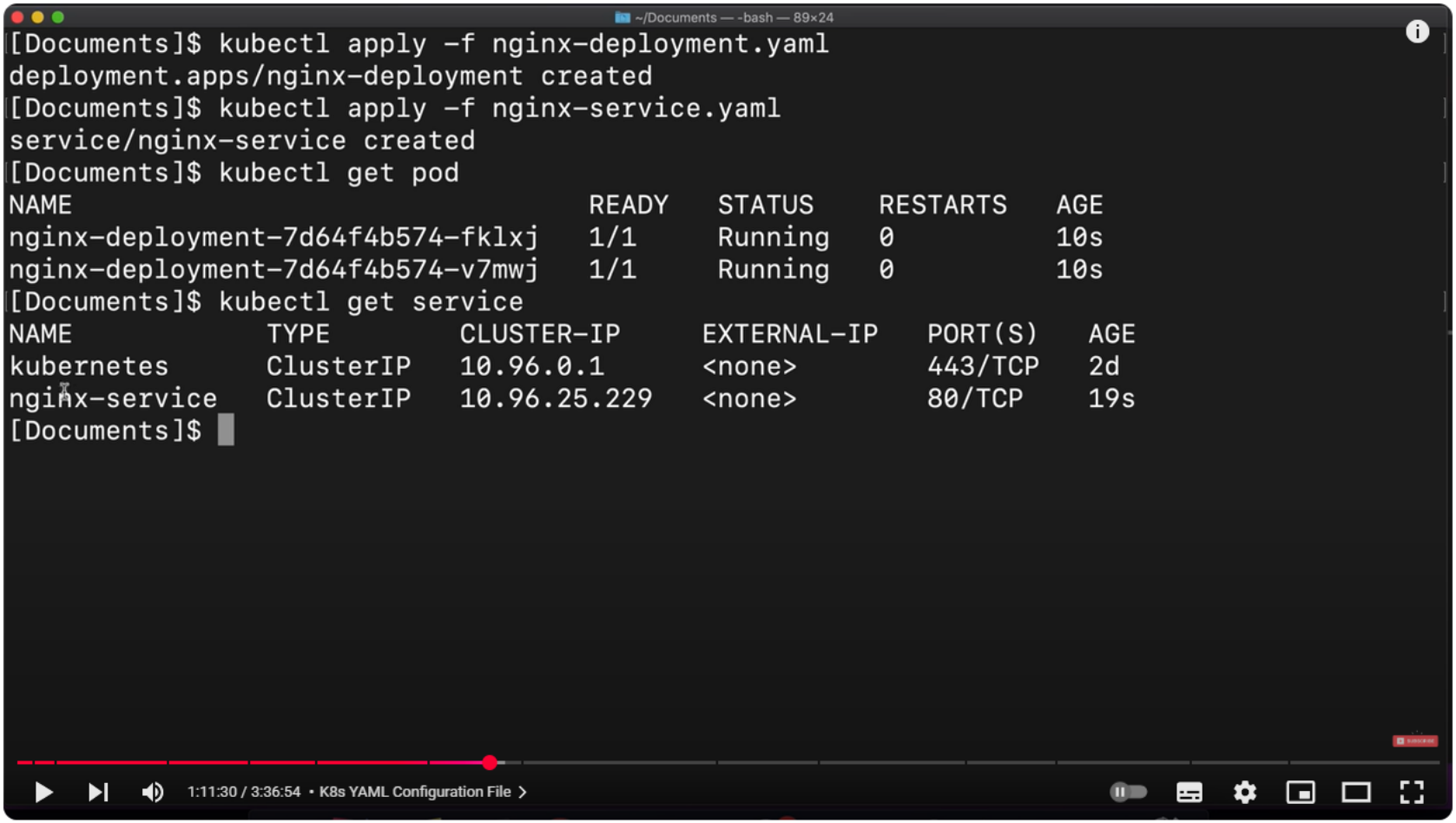

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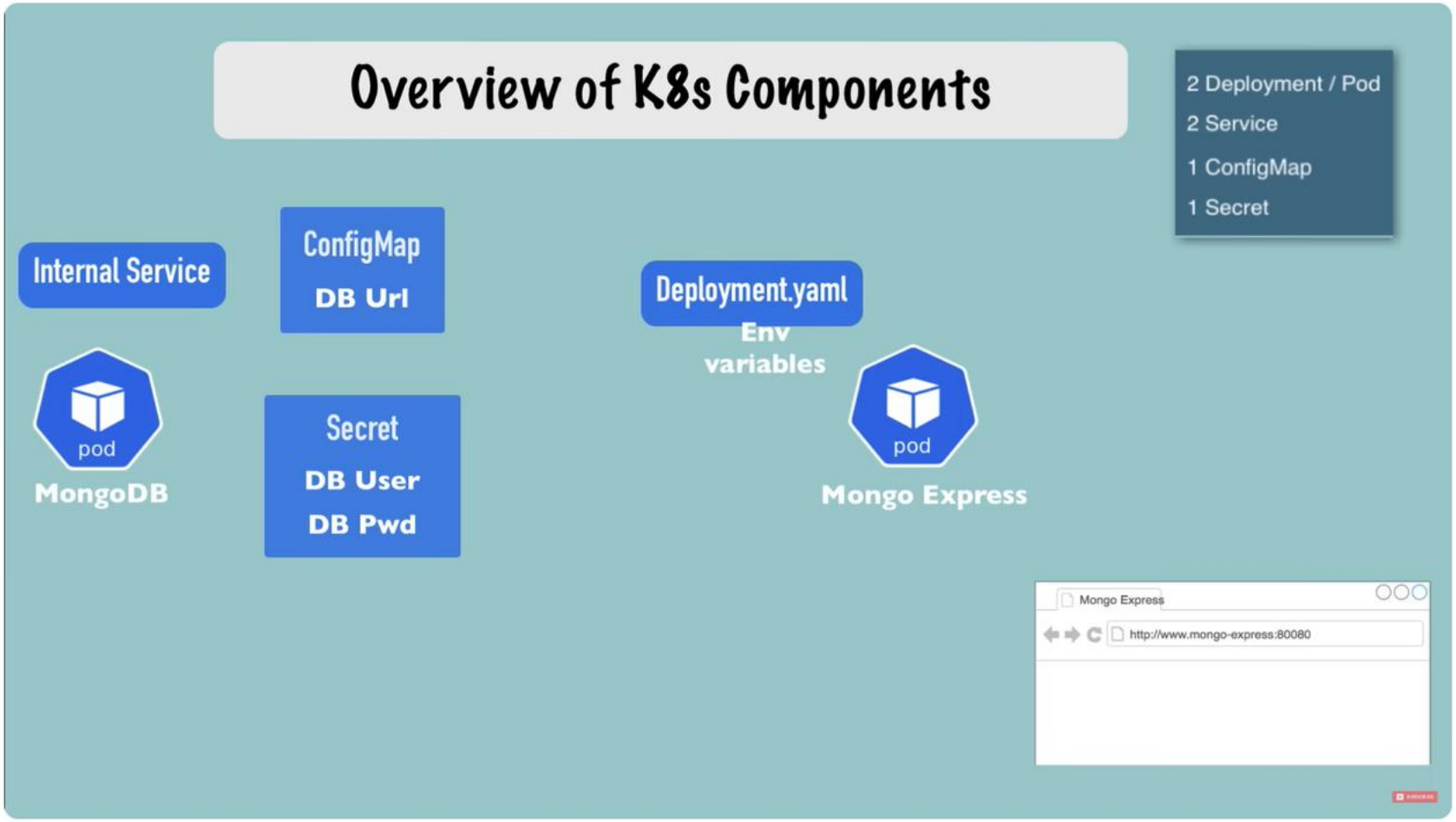

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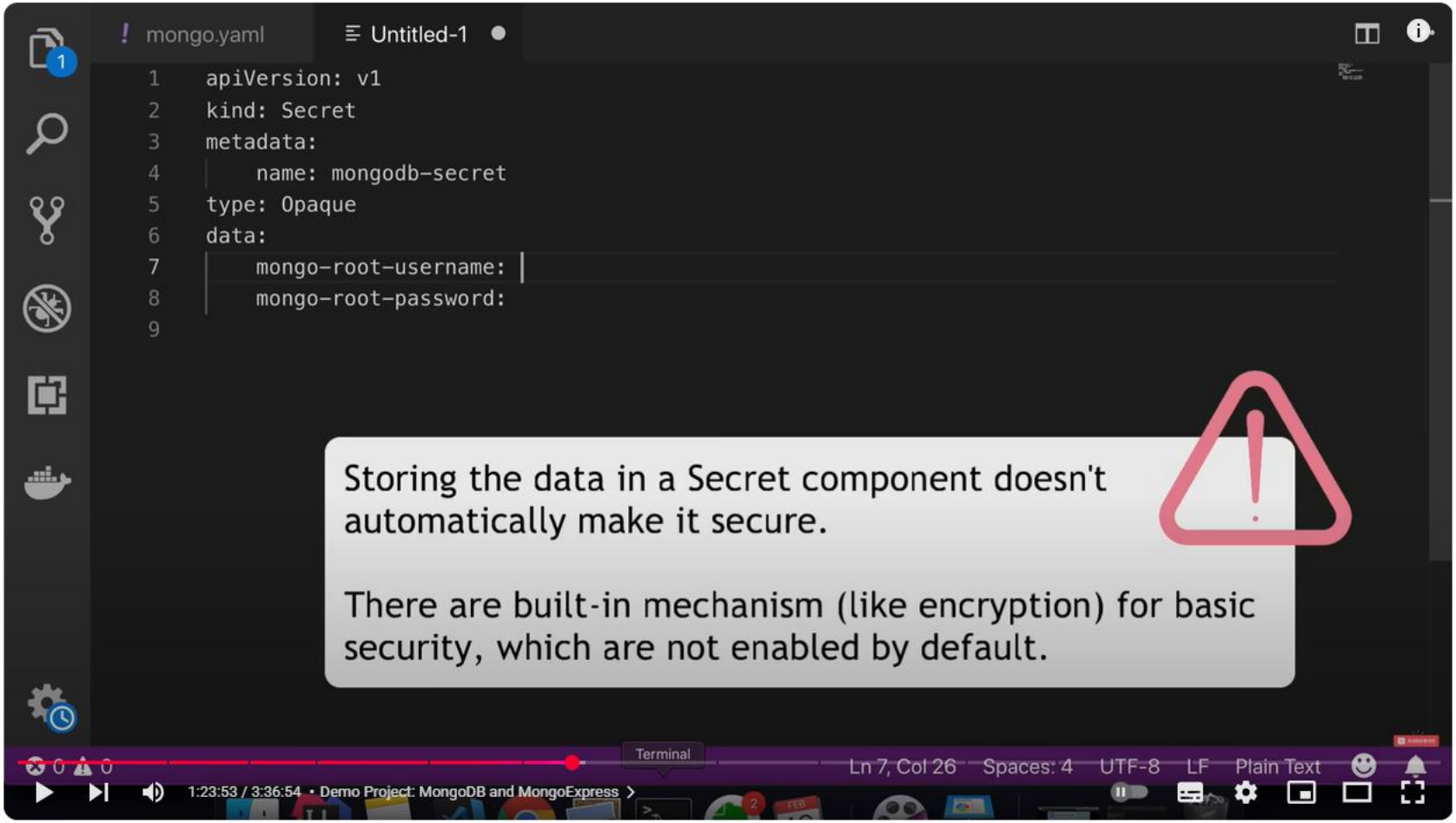

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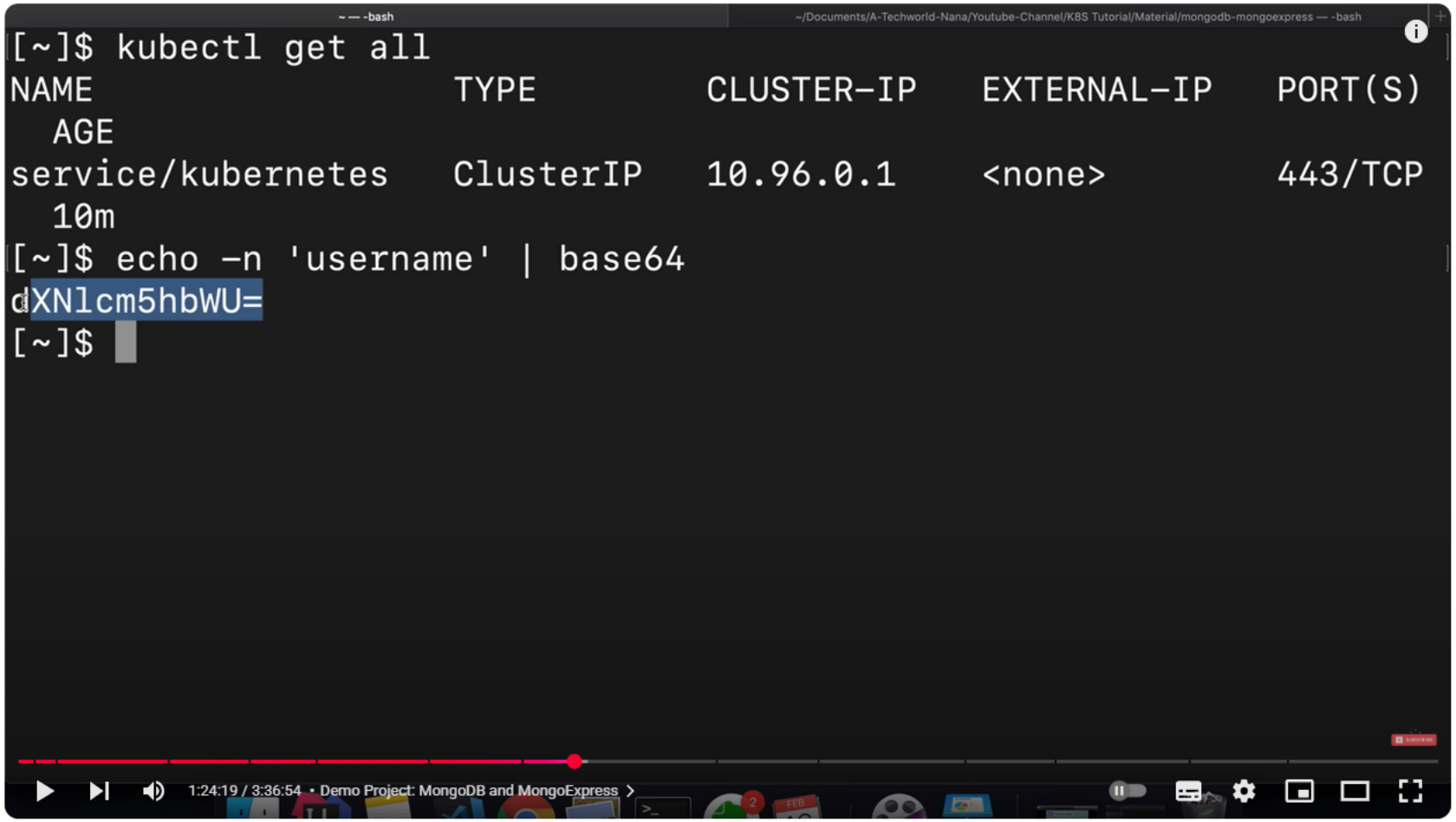

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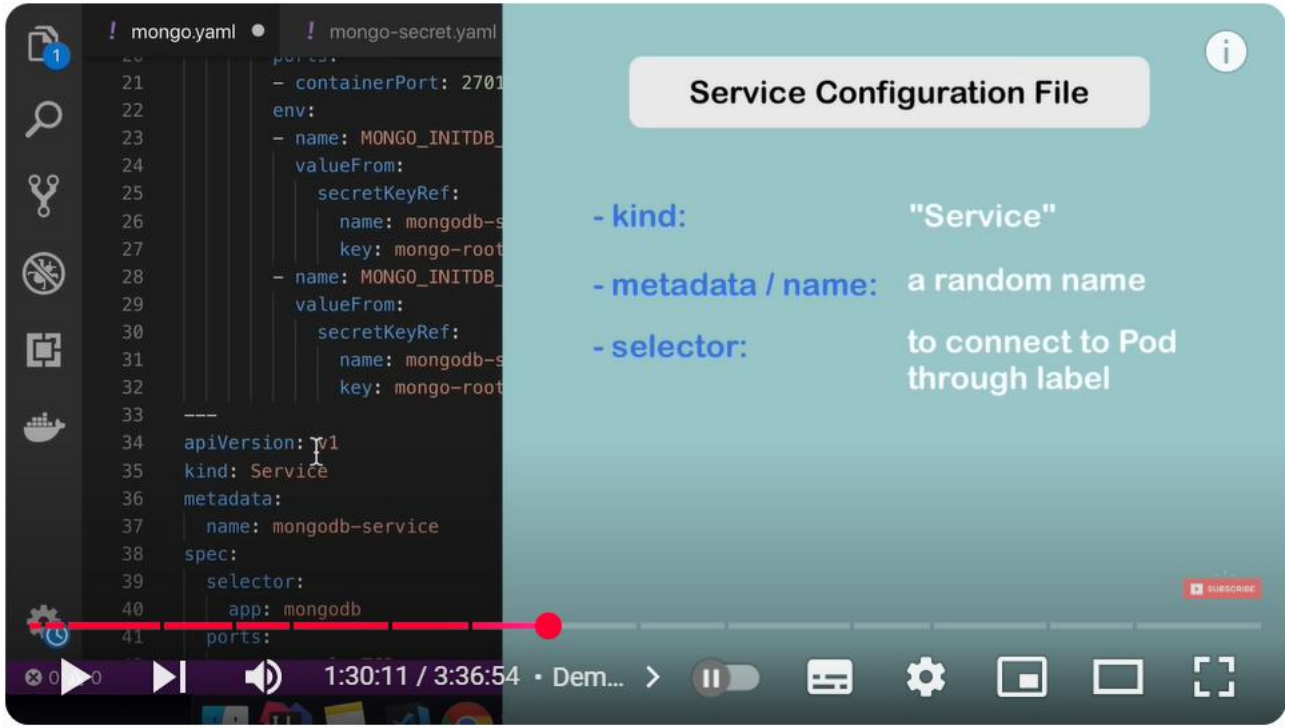

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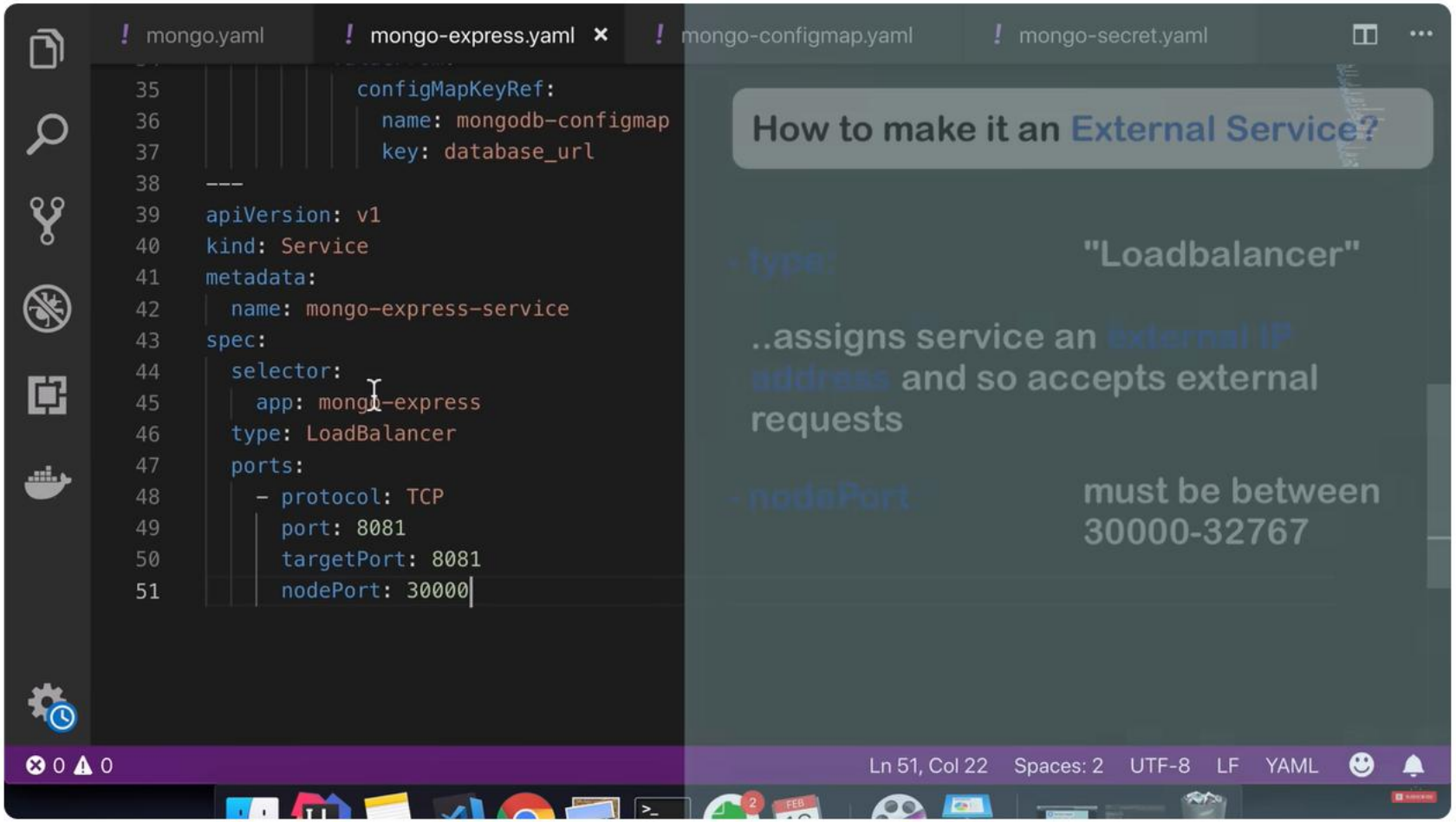

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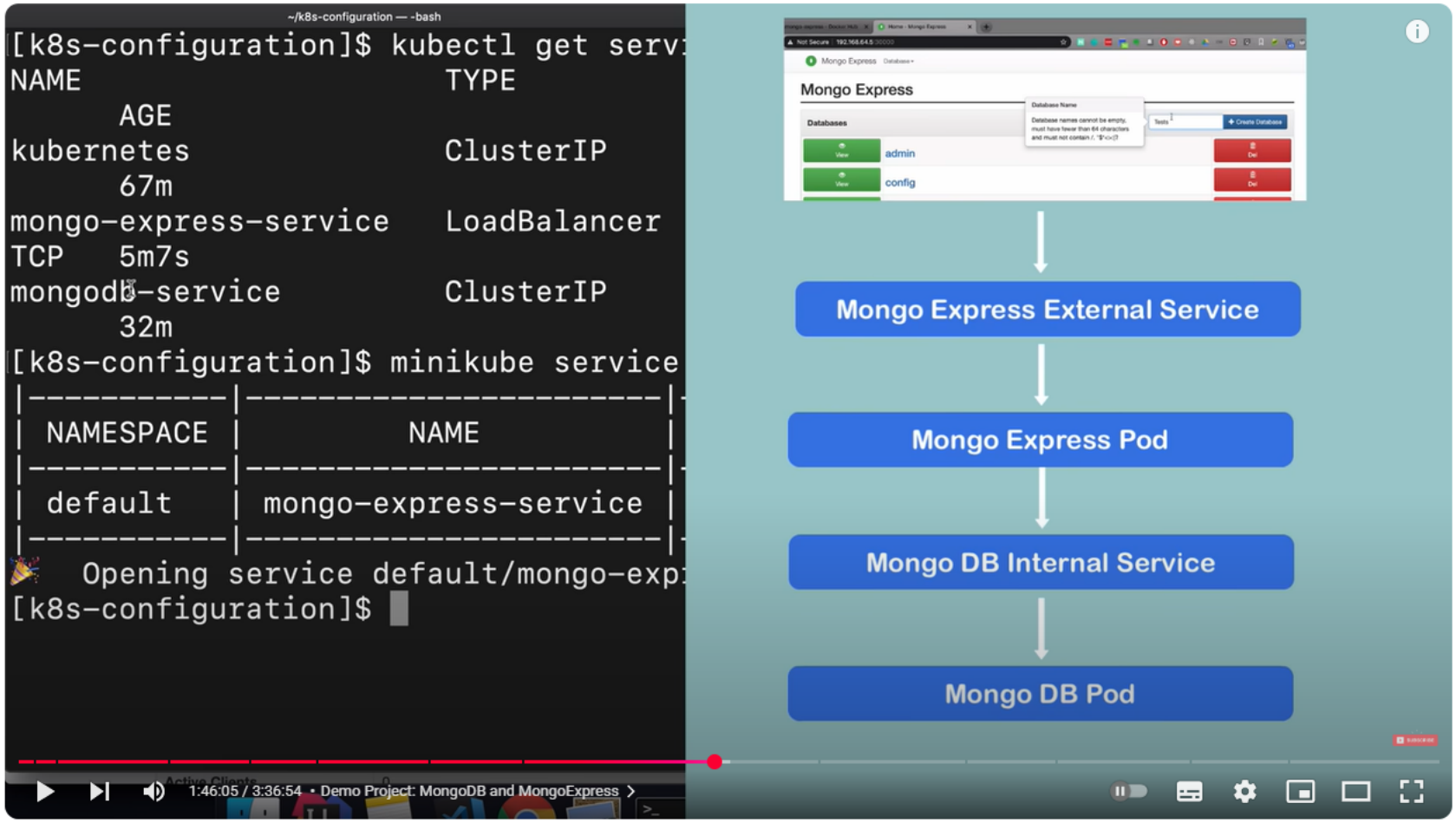

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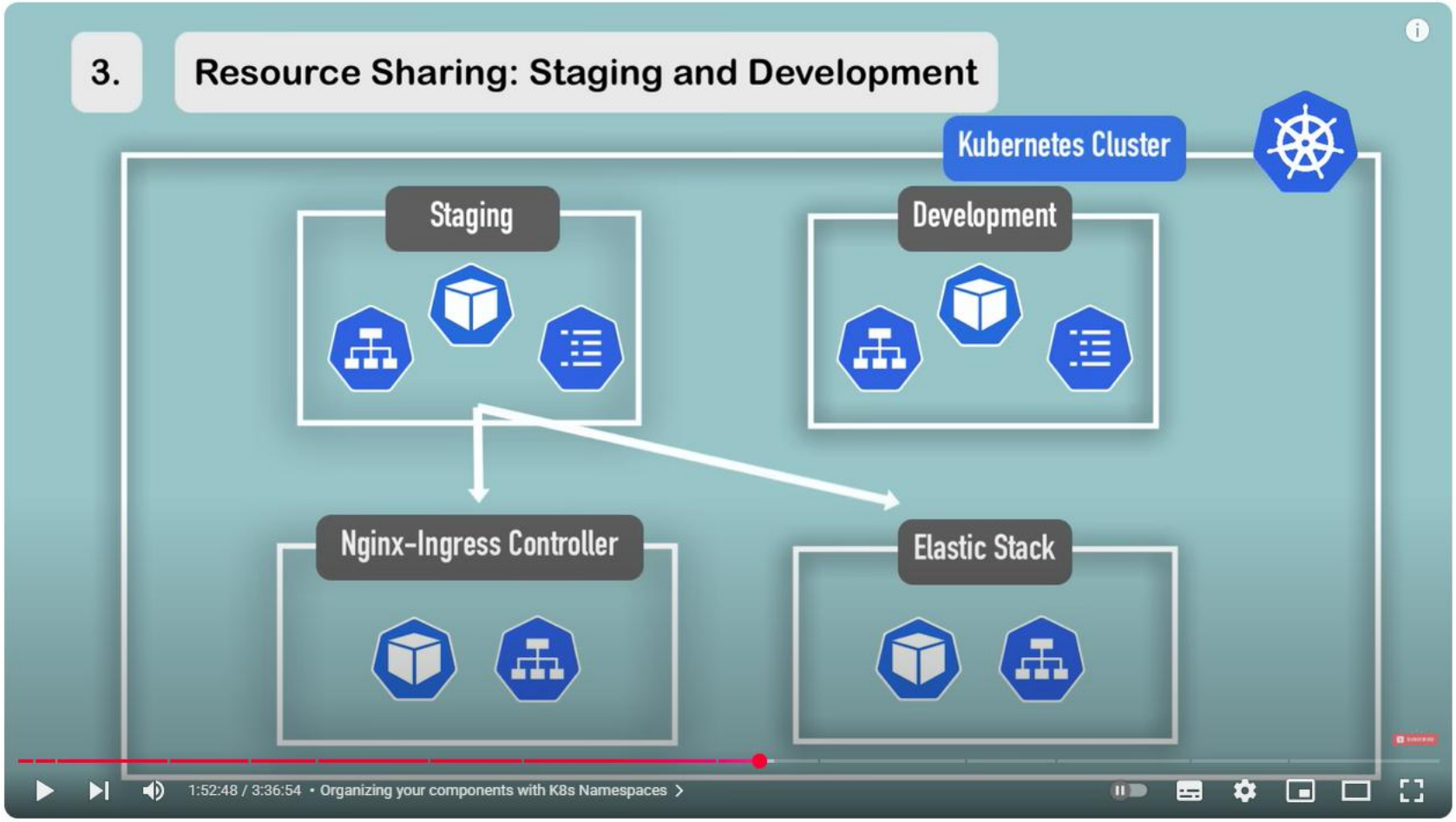

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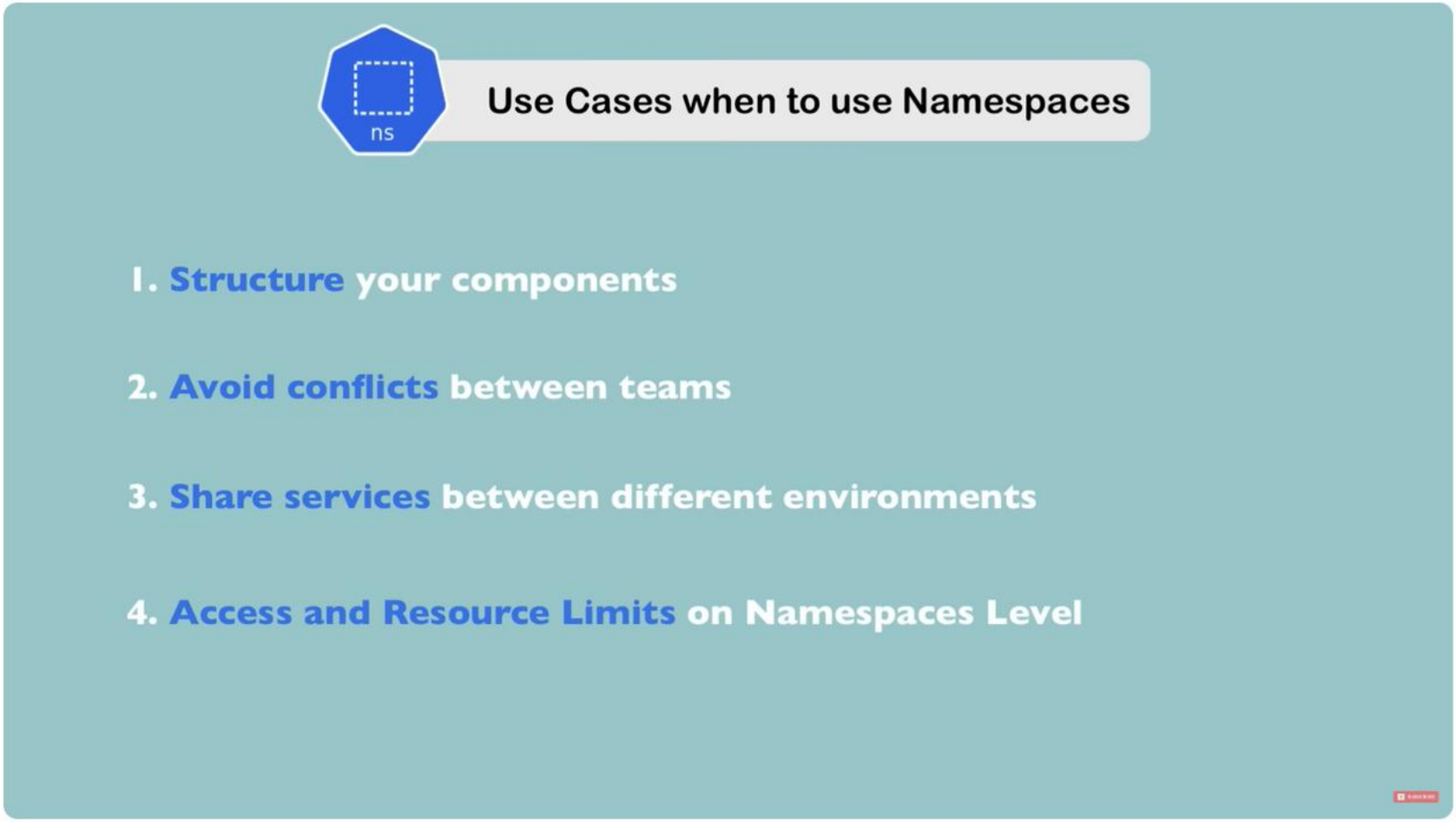

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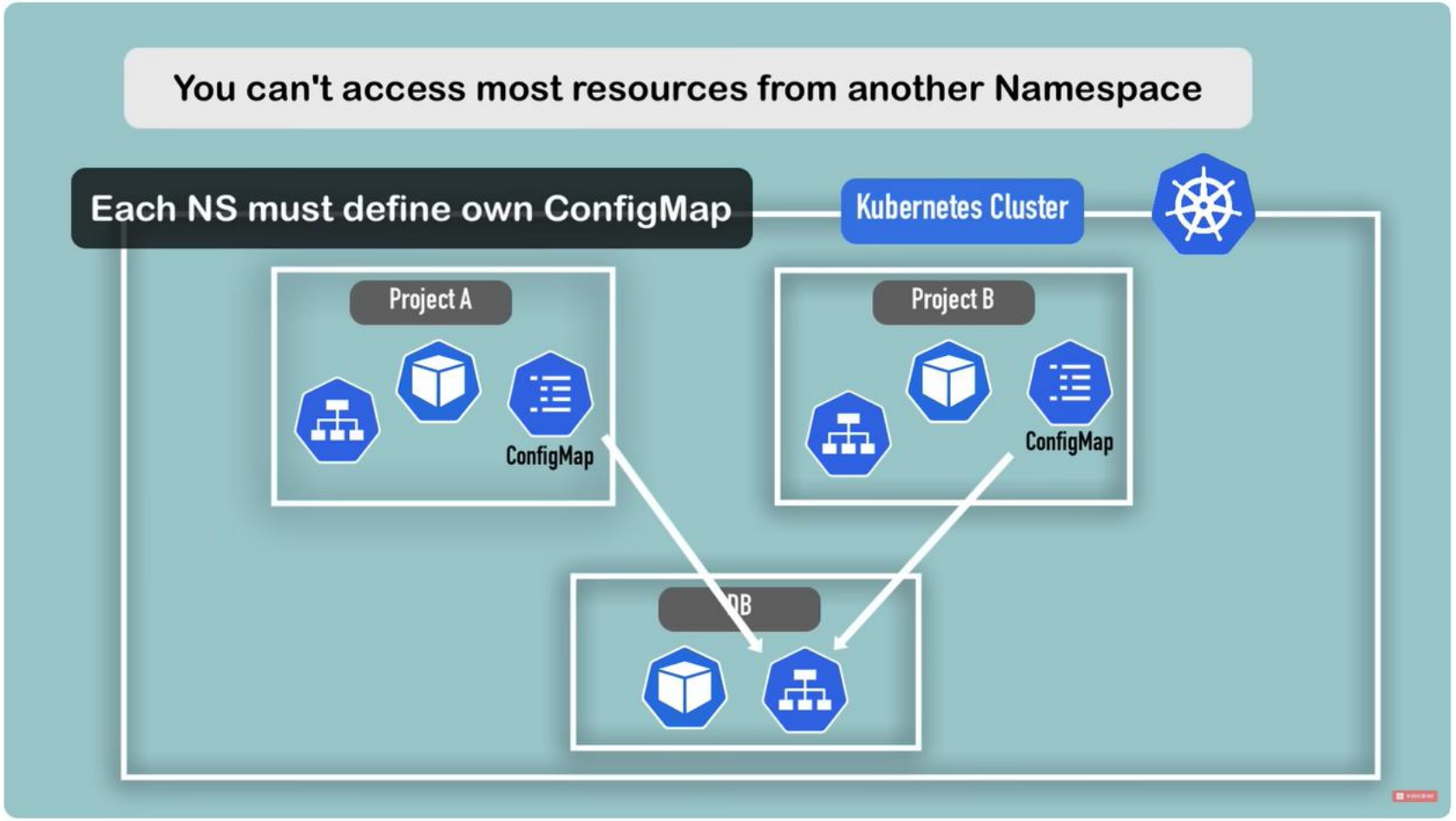

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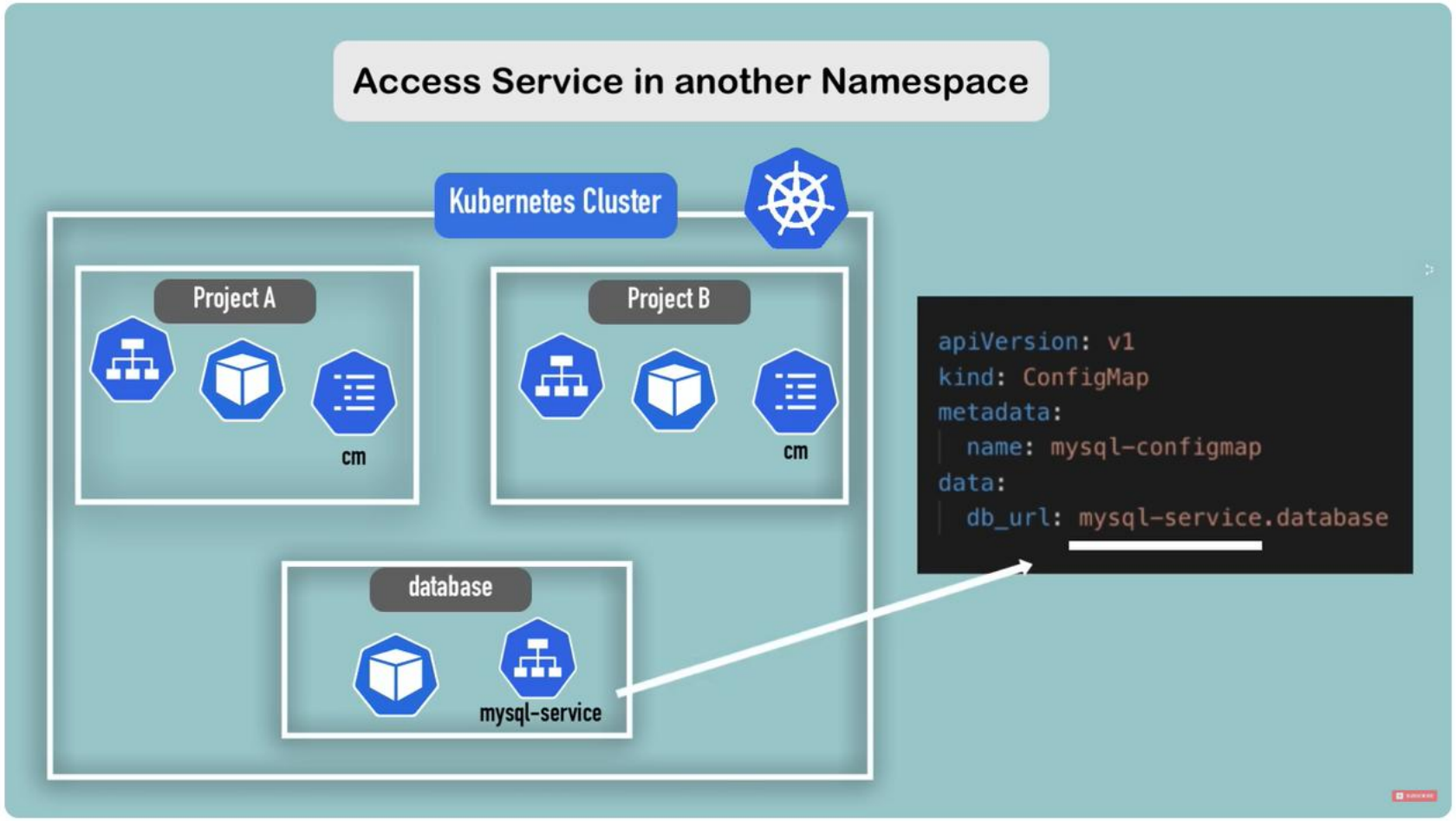

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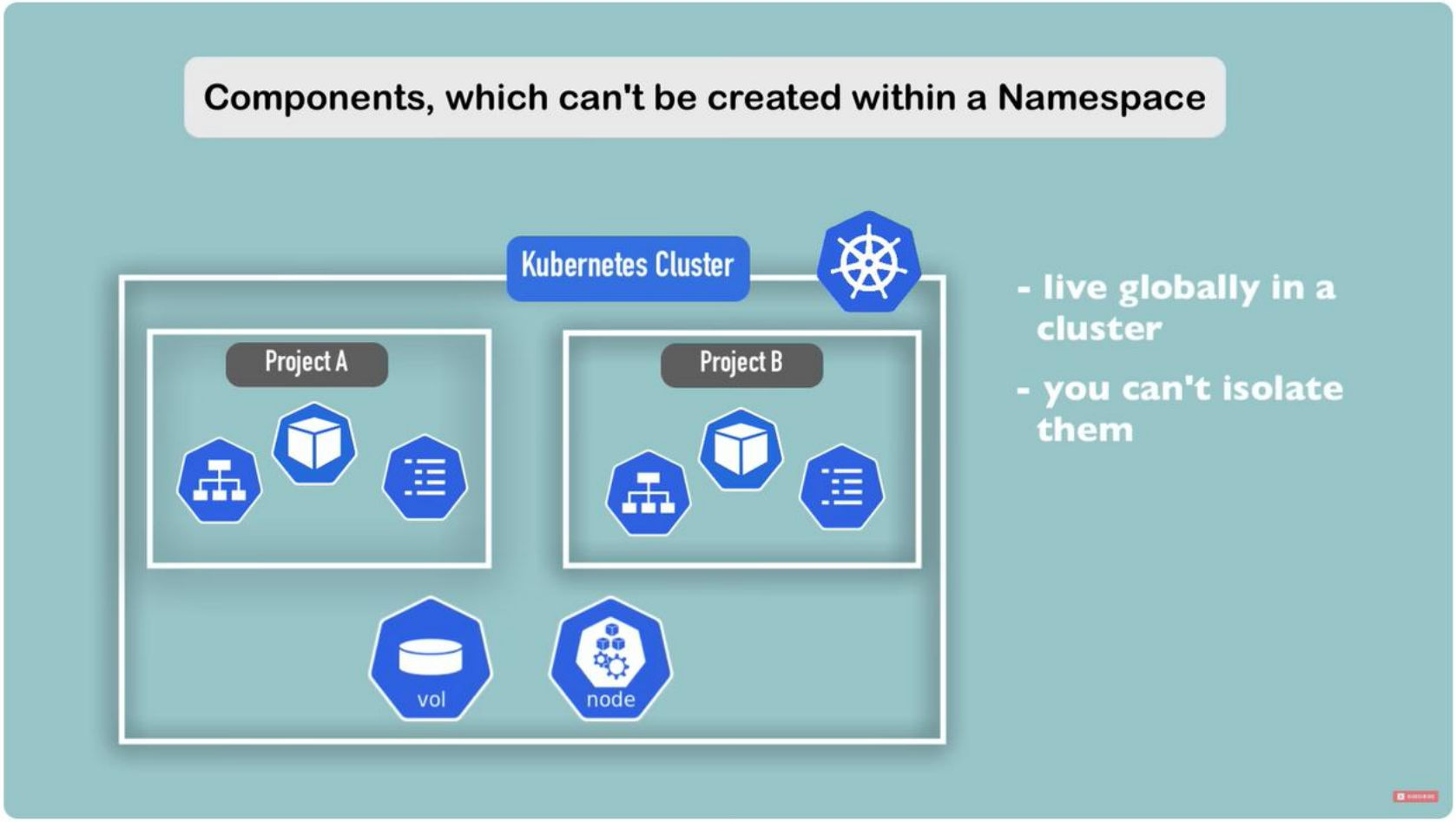

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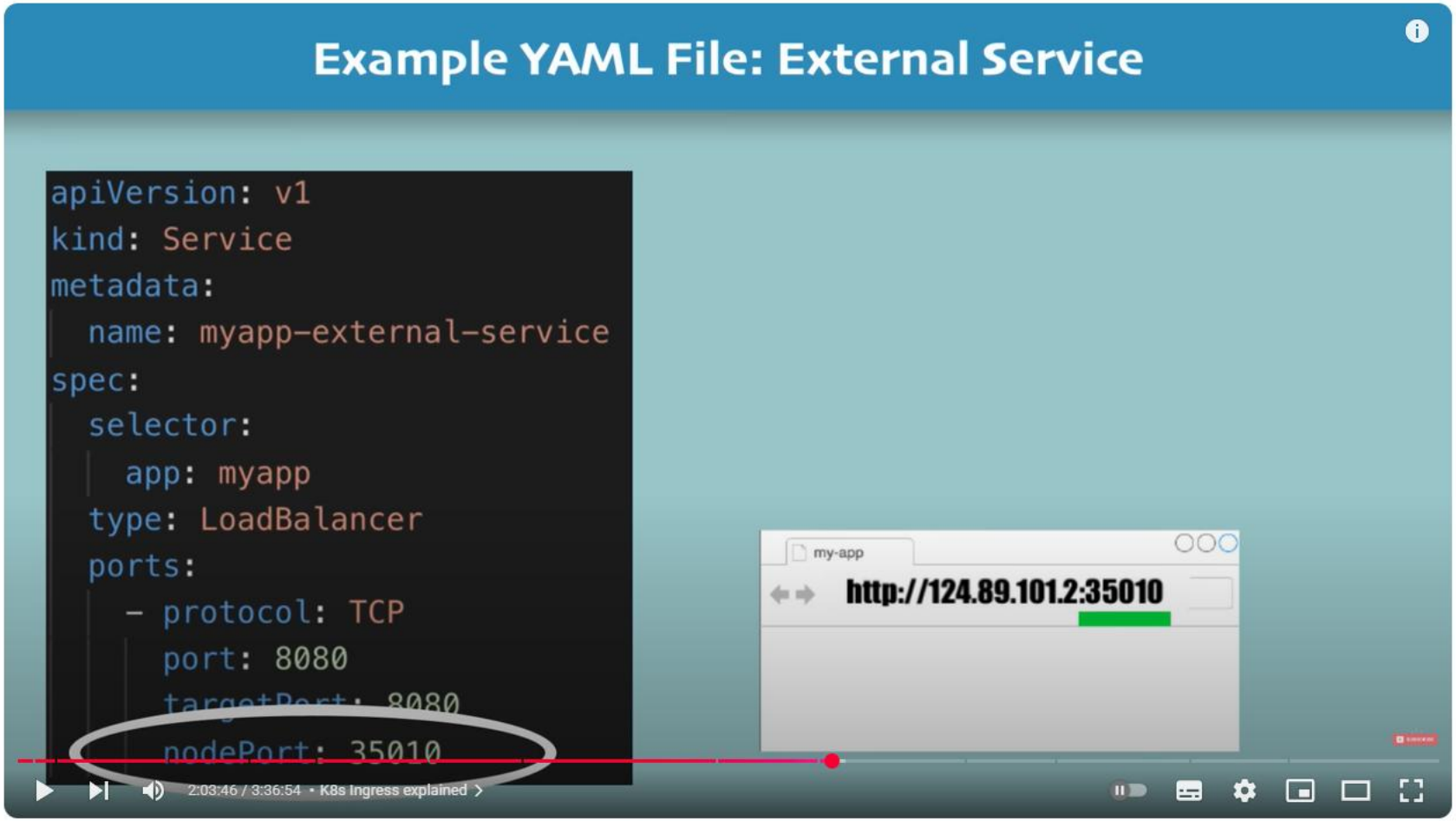

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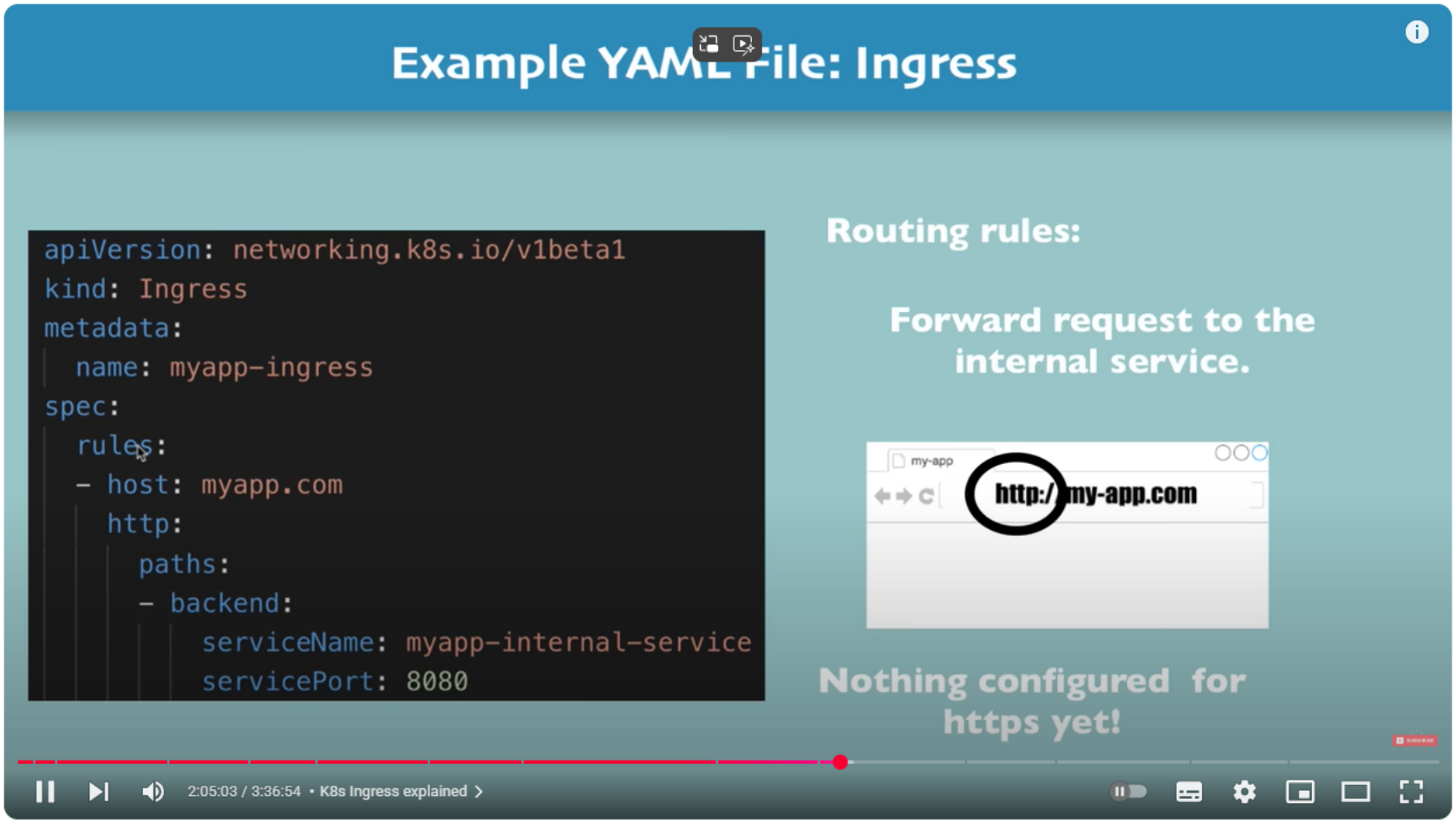

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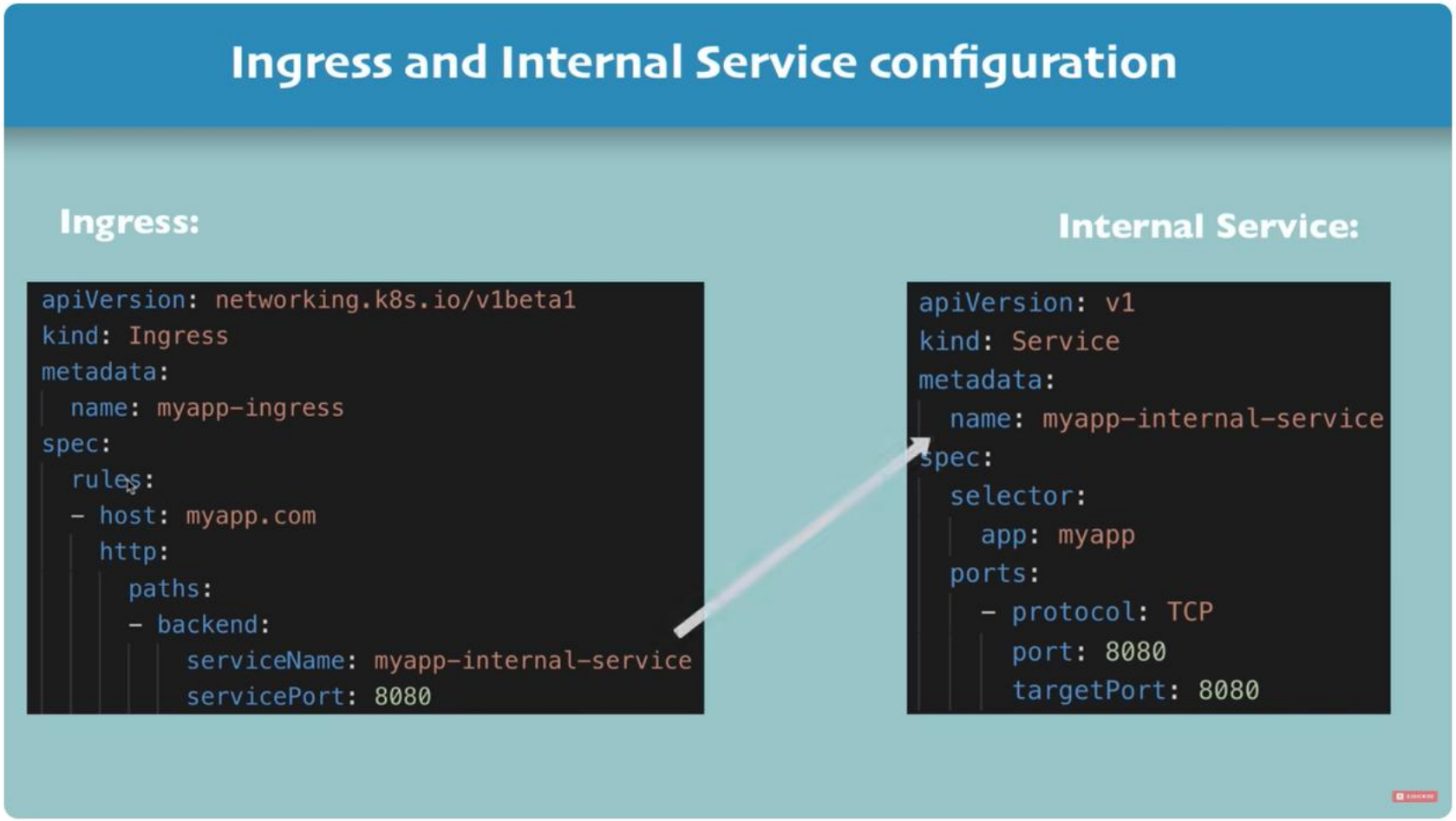

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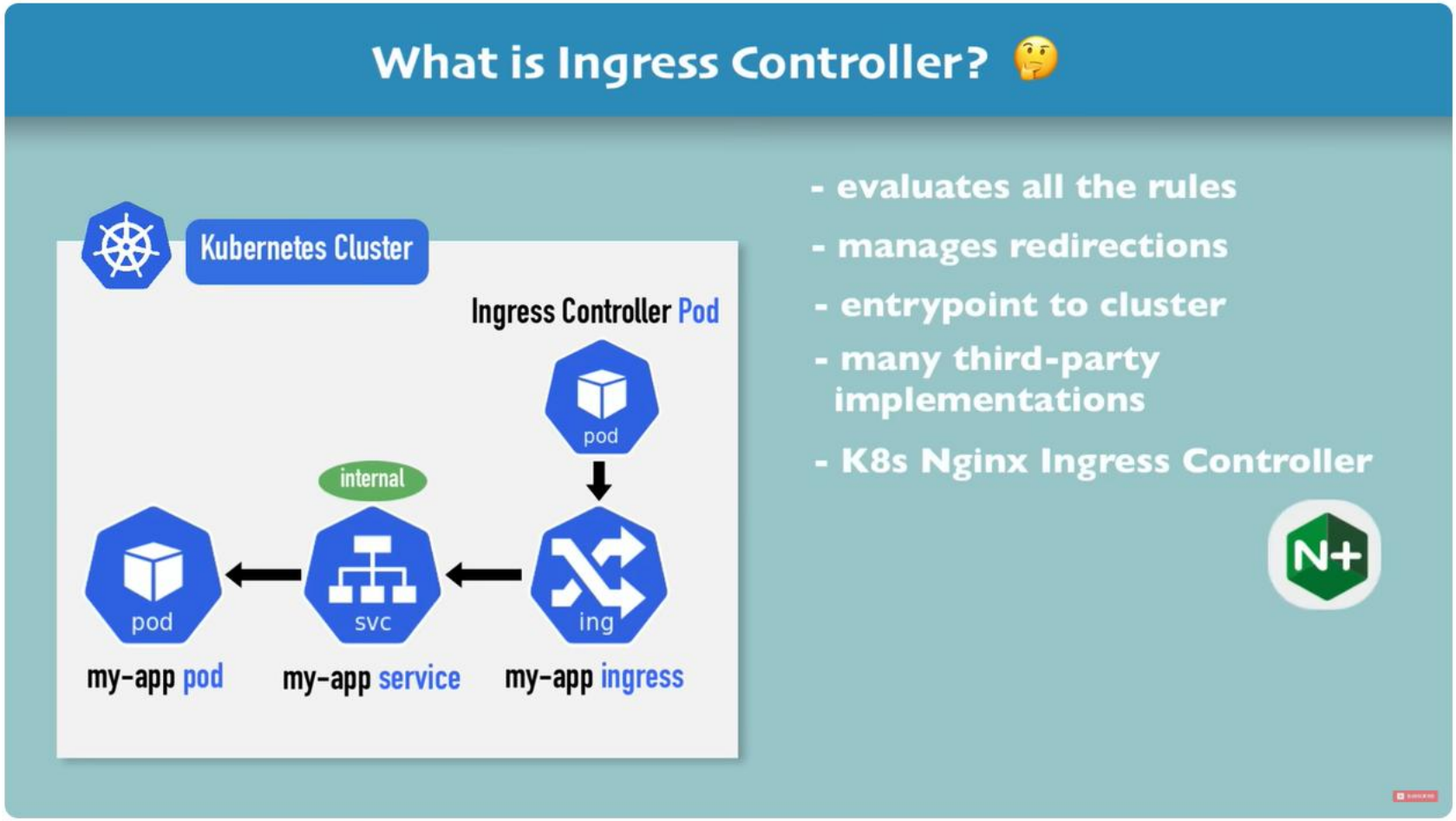

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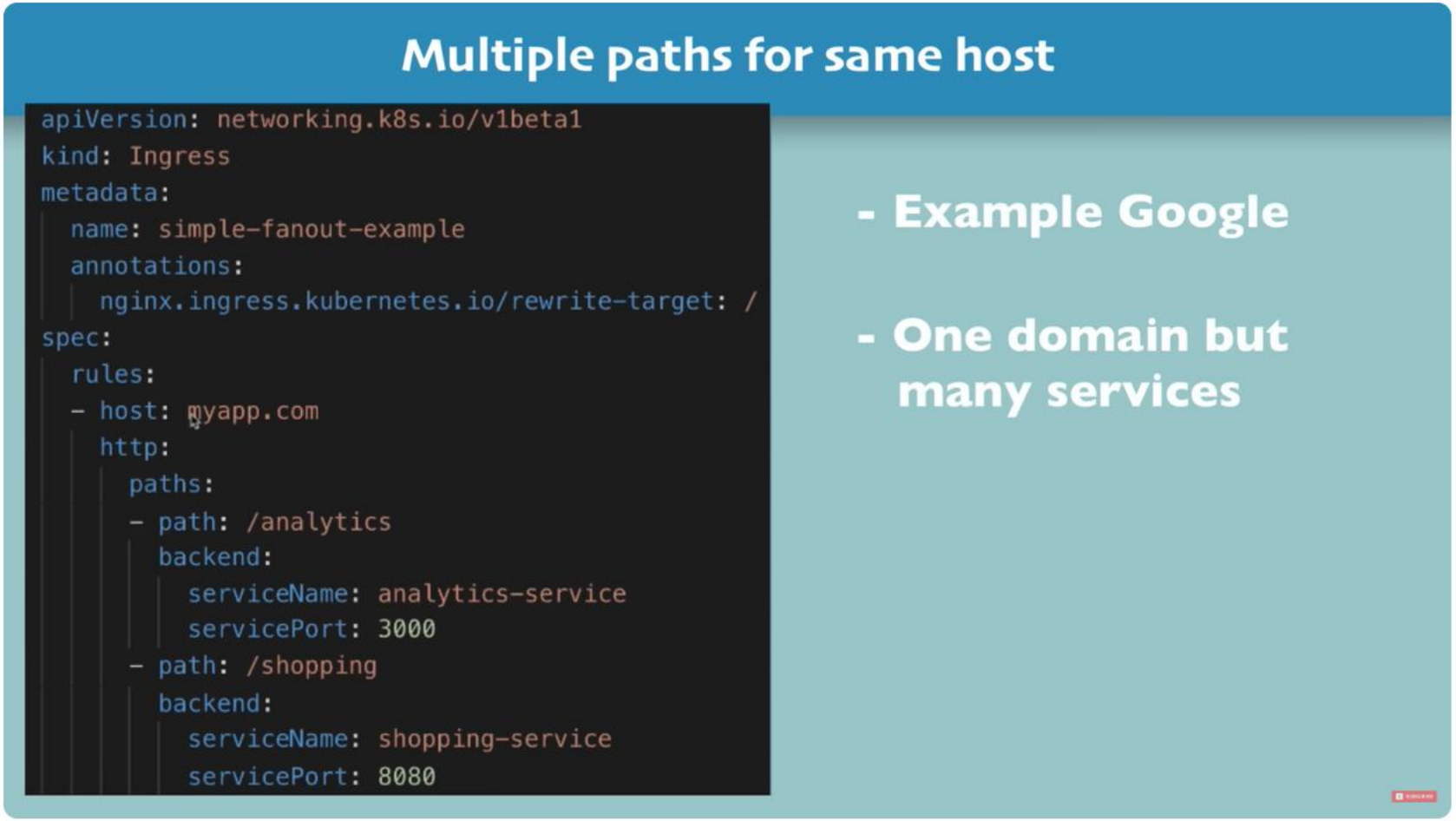


Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

Multiple sub-domains or domains

```yaml
apiVersion: networking.k8s.io/v1beta1
kind: Ingress
metadata:
  name: name-virtual-host-ingress
spec:
  rules:
  - host: analytics.myapp.com
    http:
      paths:
        backend:
          serviceName: analytics-service
          servicePort: 3000
  - host: shopping.myapp.com
    http:
      paths:
        backend:
          serviceName: shopping-service
          servicePort: 8080
```

Instead of:

http://myapp.com/analytics

http://analytics.myapp.com

Multiple sub-domains or domains

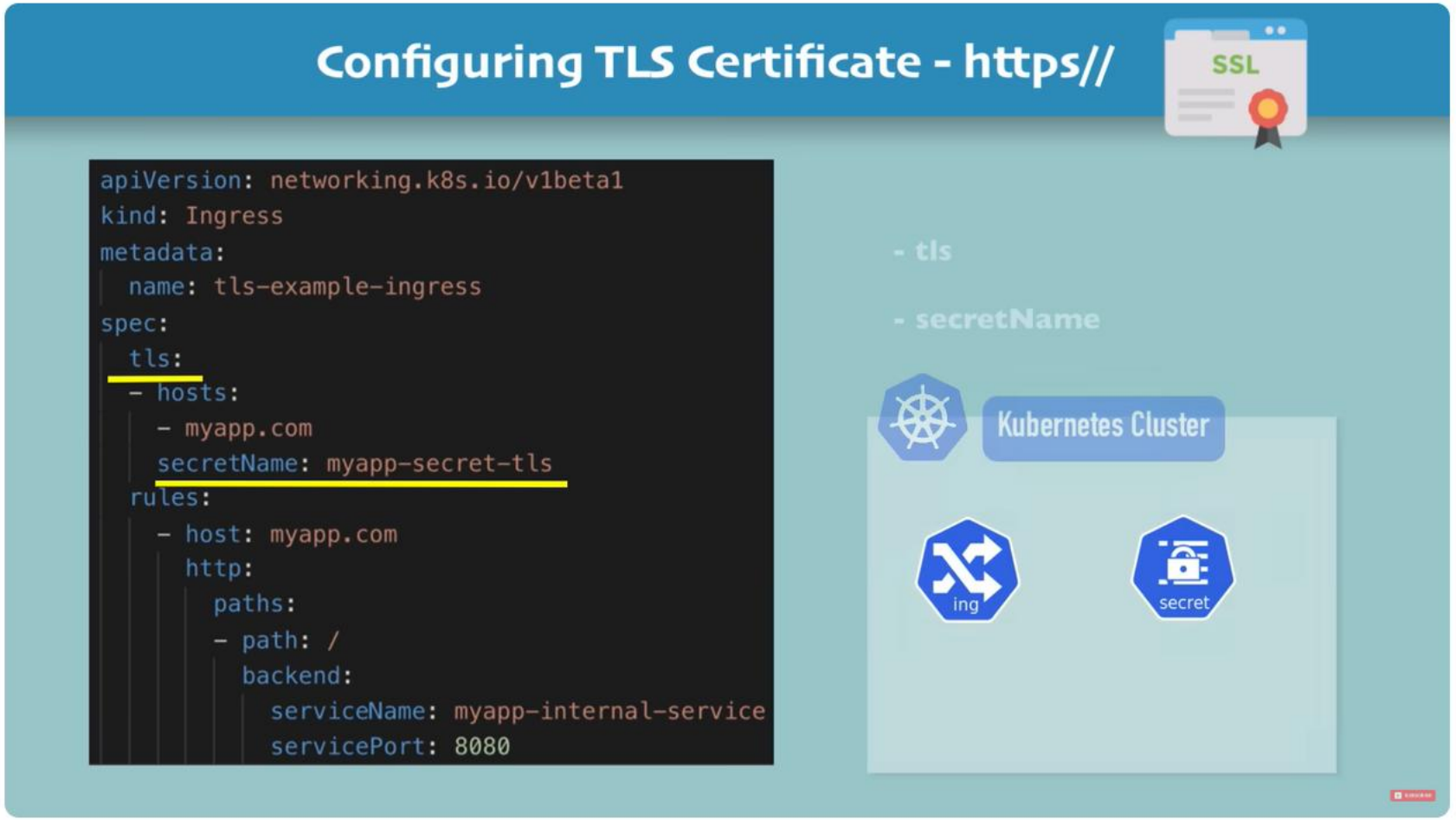
```yaml
apiVersion: networking.k8s.io/v1beta1
kind: Ingress
metadata:
  name: name-virtual-host-ingress
spec:
  rules:
  - host: analytics.myapp.com
    http:
      paths:
        backend:
          serviceName: analytics-service
          servicePort: 3000
  - host: shopping.myapp.com
    http:
      paths:
        backend:
          serviceName: shopping-service
          servicePort: 8080
```

```yaml
  rules:
  - host: myapp.com
    http:
      paths:
      - path: /analytics
        backend:
          serviceName: analytics-service
          servicePort: 3000
      - path: /shopping
        backend:
          serviceName: shopping-service
          servicePort: 8080
```

Instead of 1 host and multiple path

You have now multiple hosts with 1 path. Each host represents a subdomain.

Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



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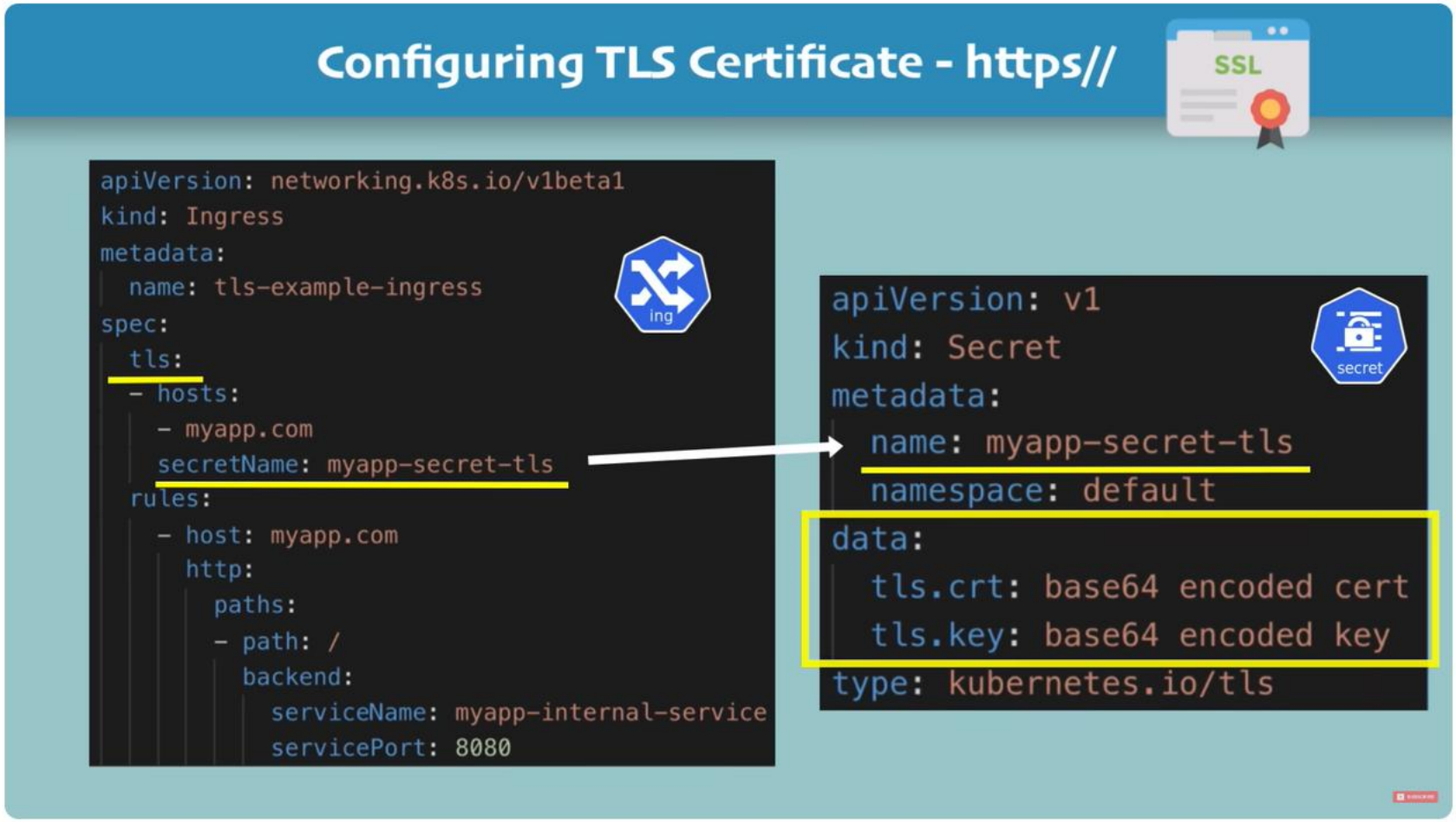
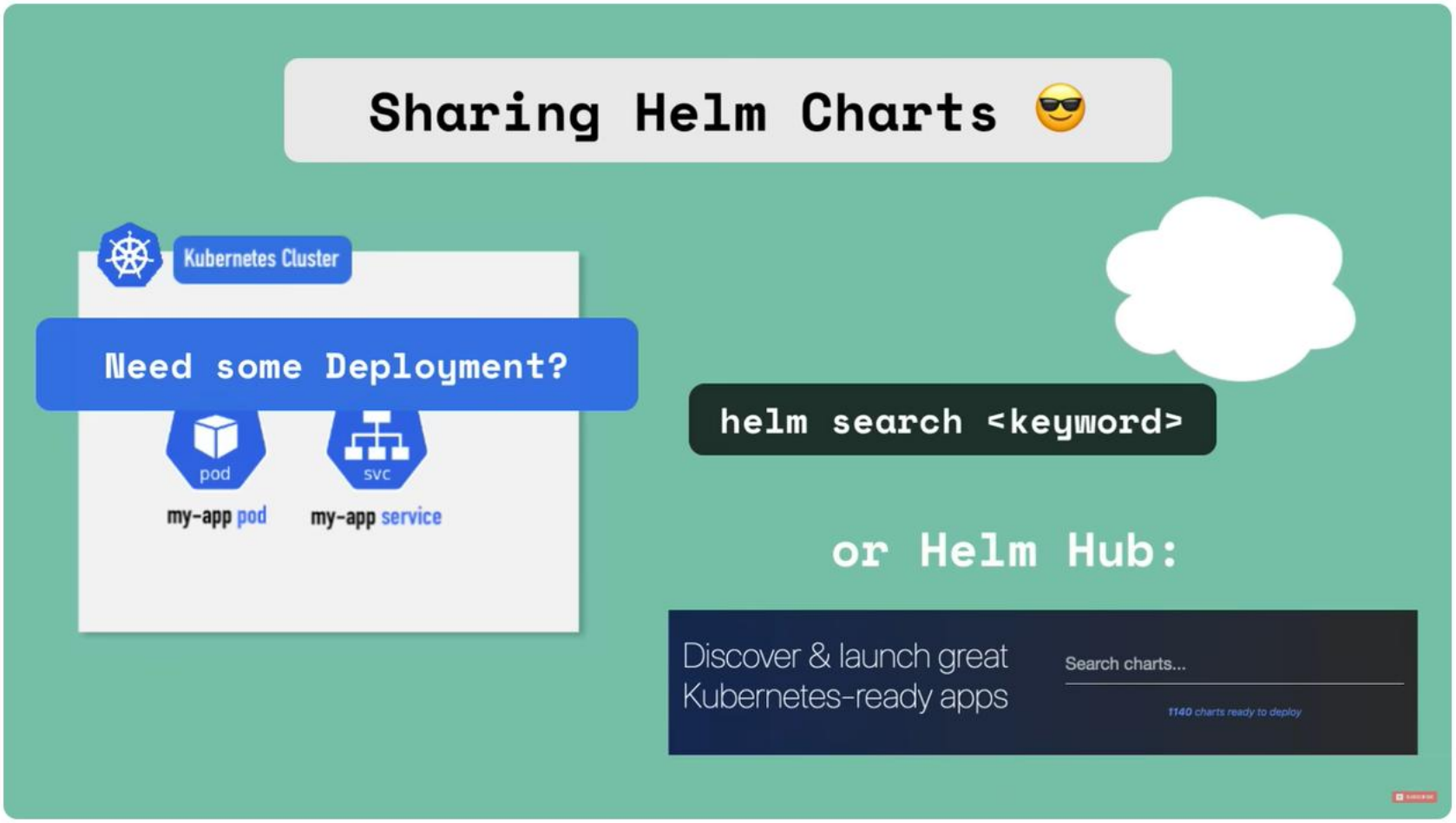


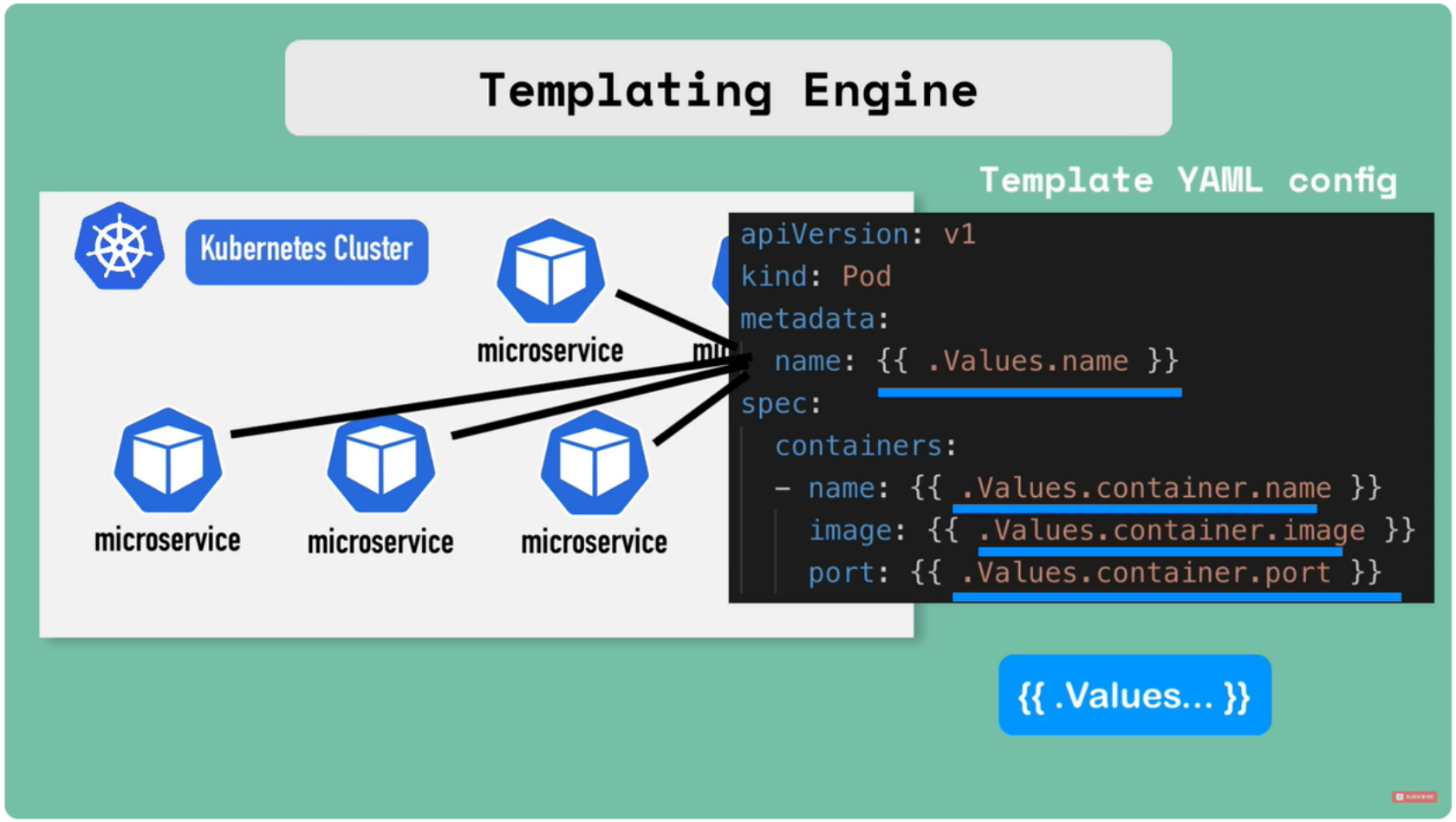
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

Helm: package management for Kubernetes

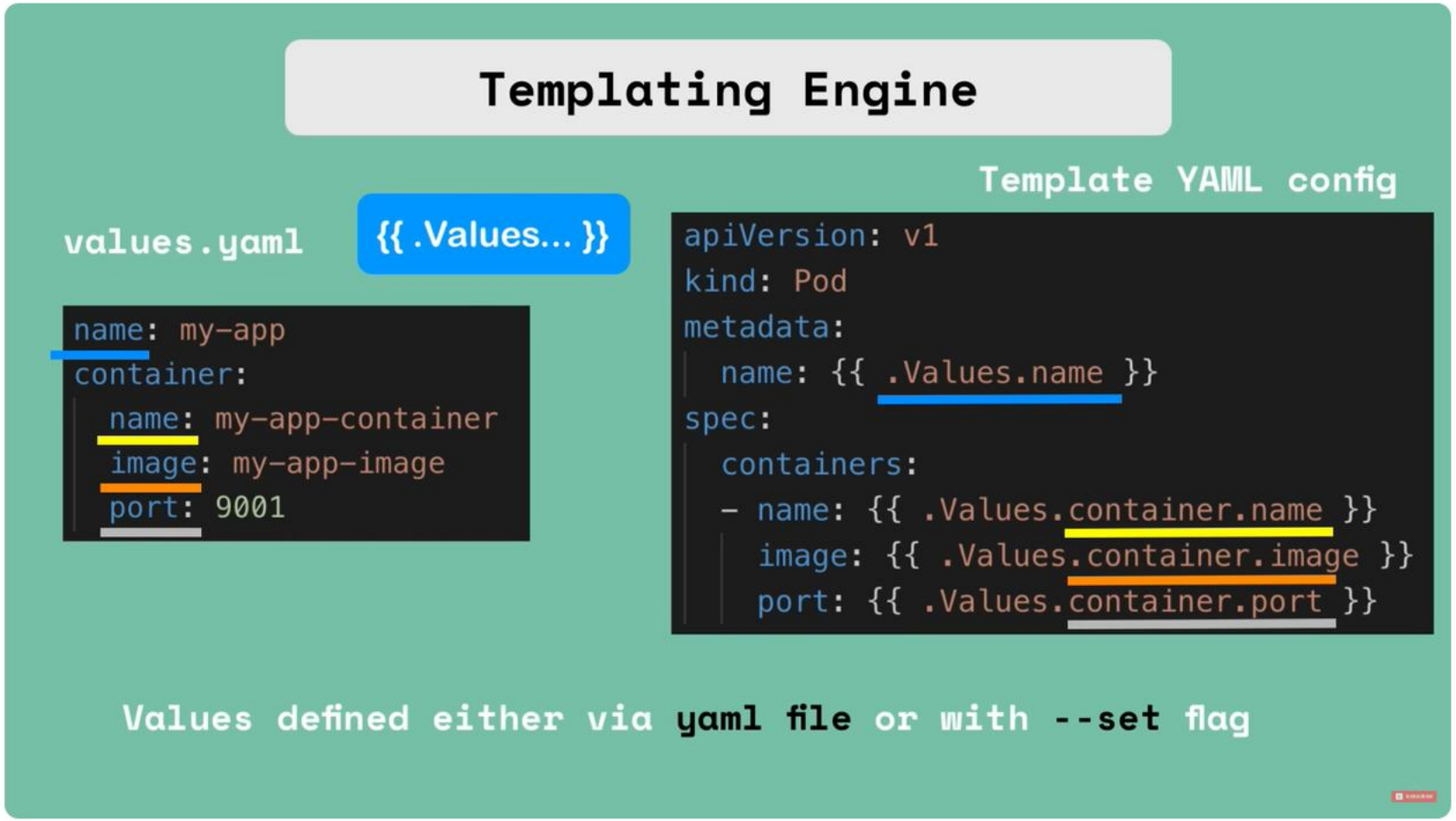


Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

Helm is also a templating Engine

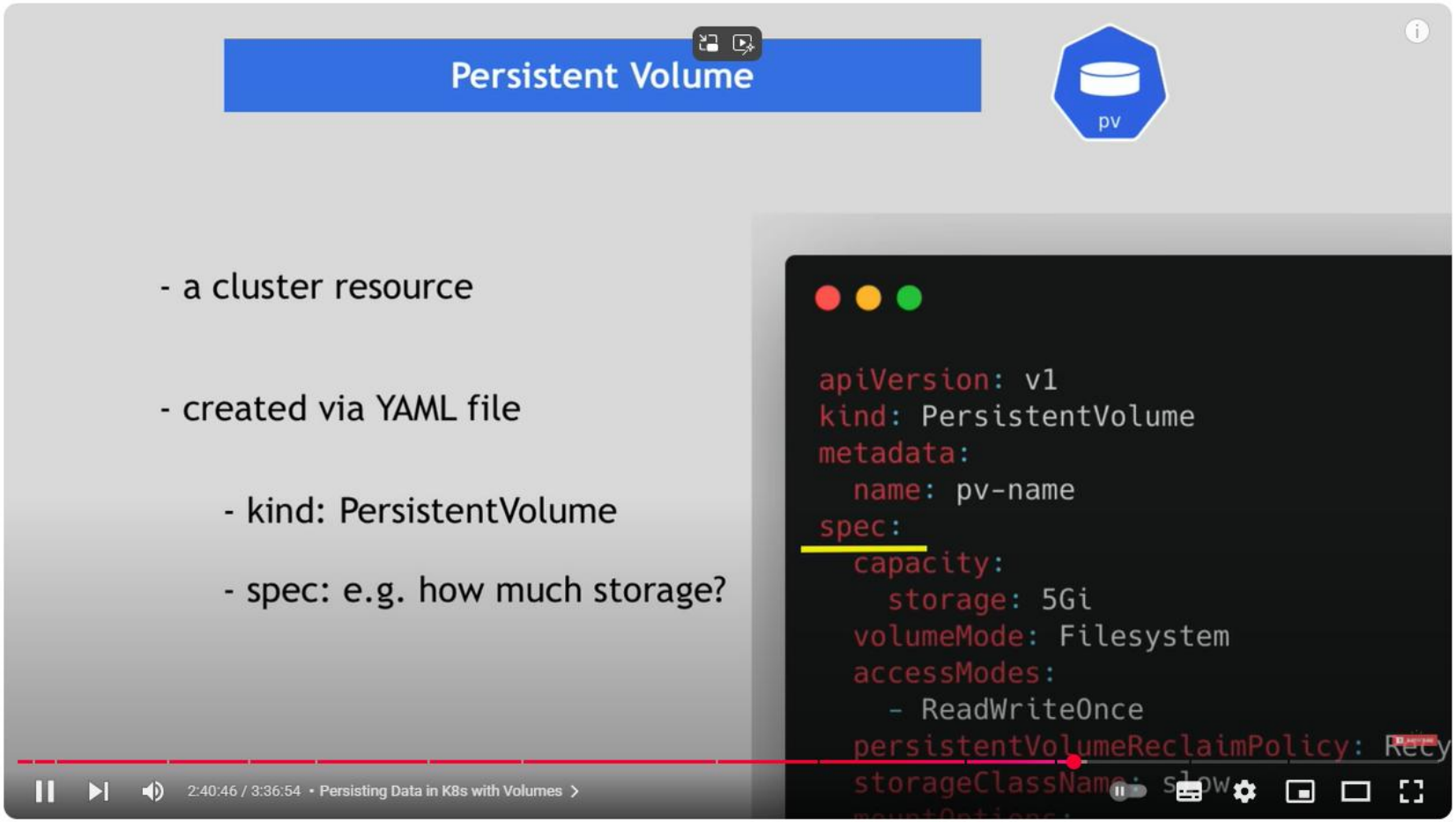


Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

Helm: release management



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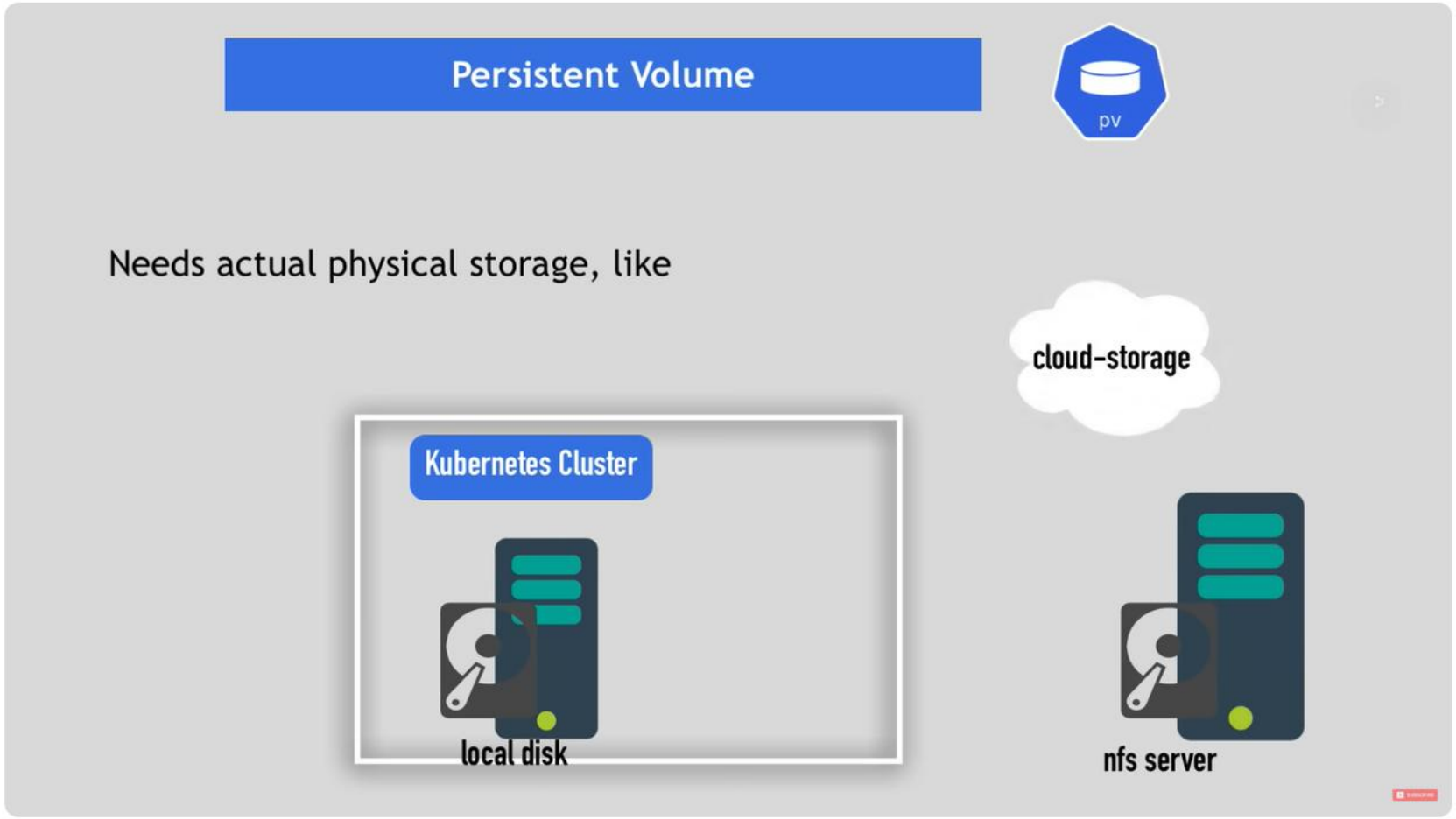

Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

StatefulSet for stateful applications

stateful applications

- examples of stateful applications:

stateless applications

- don't keep record of state

- each request is completely new

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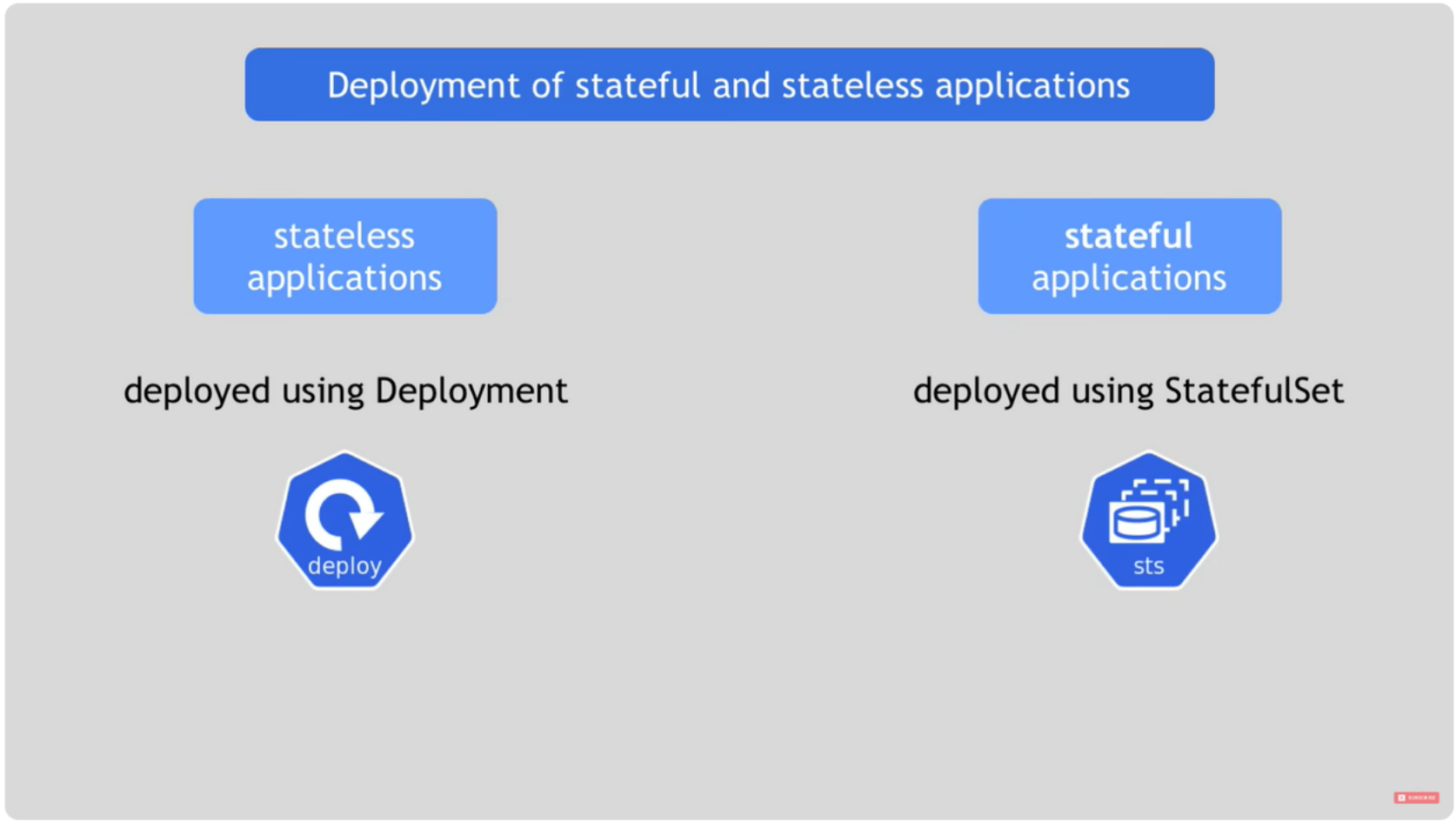
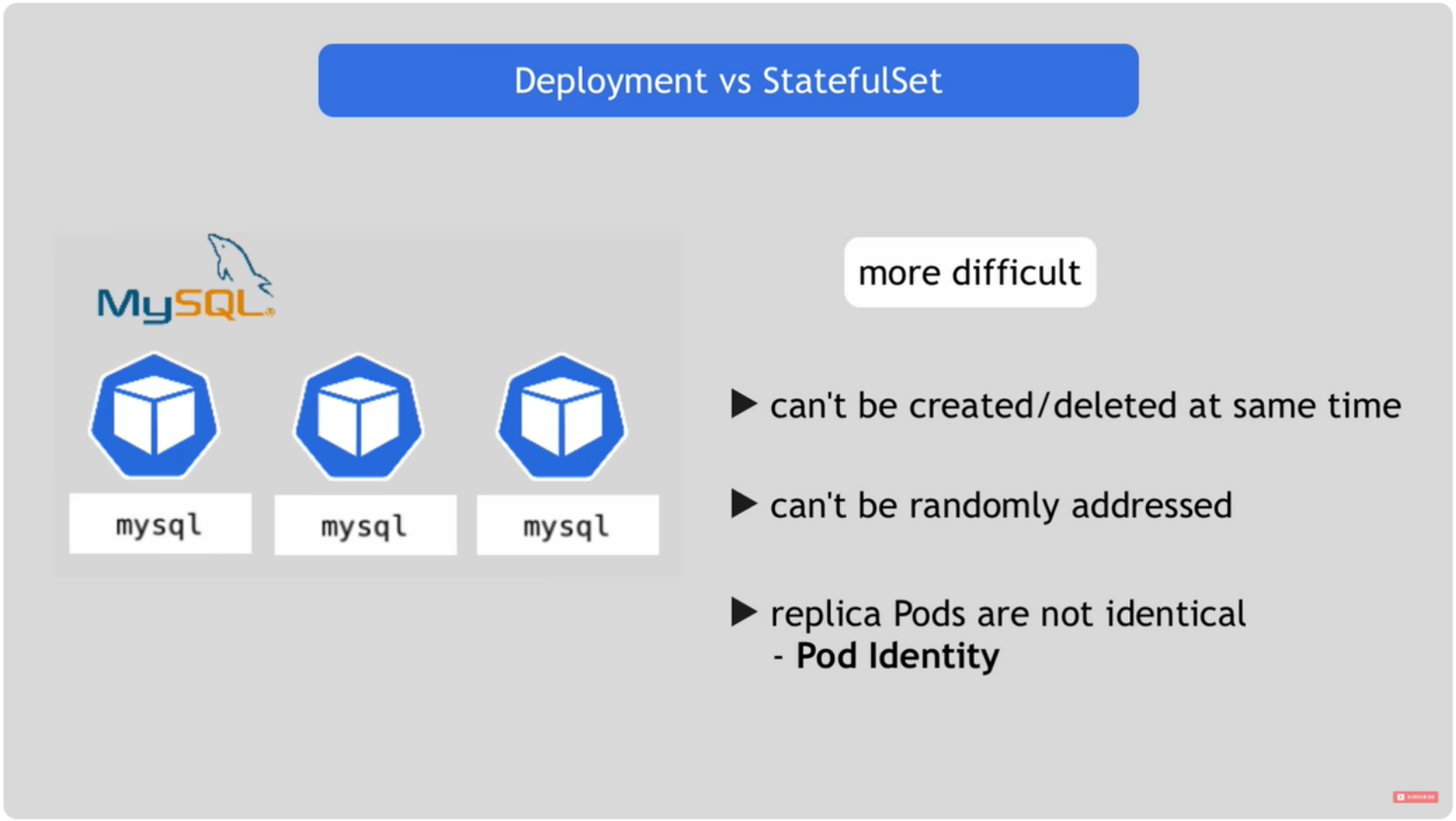


Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



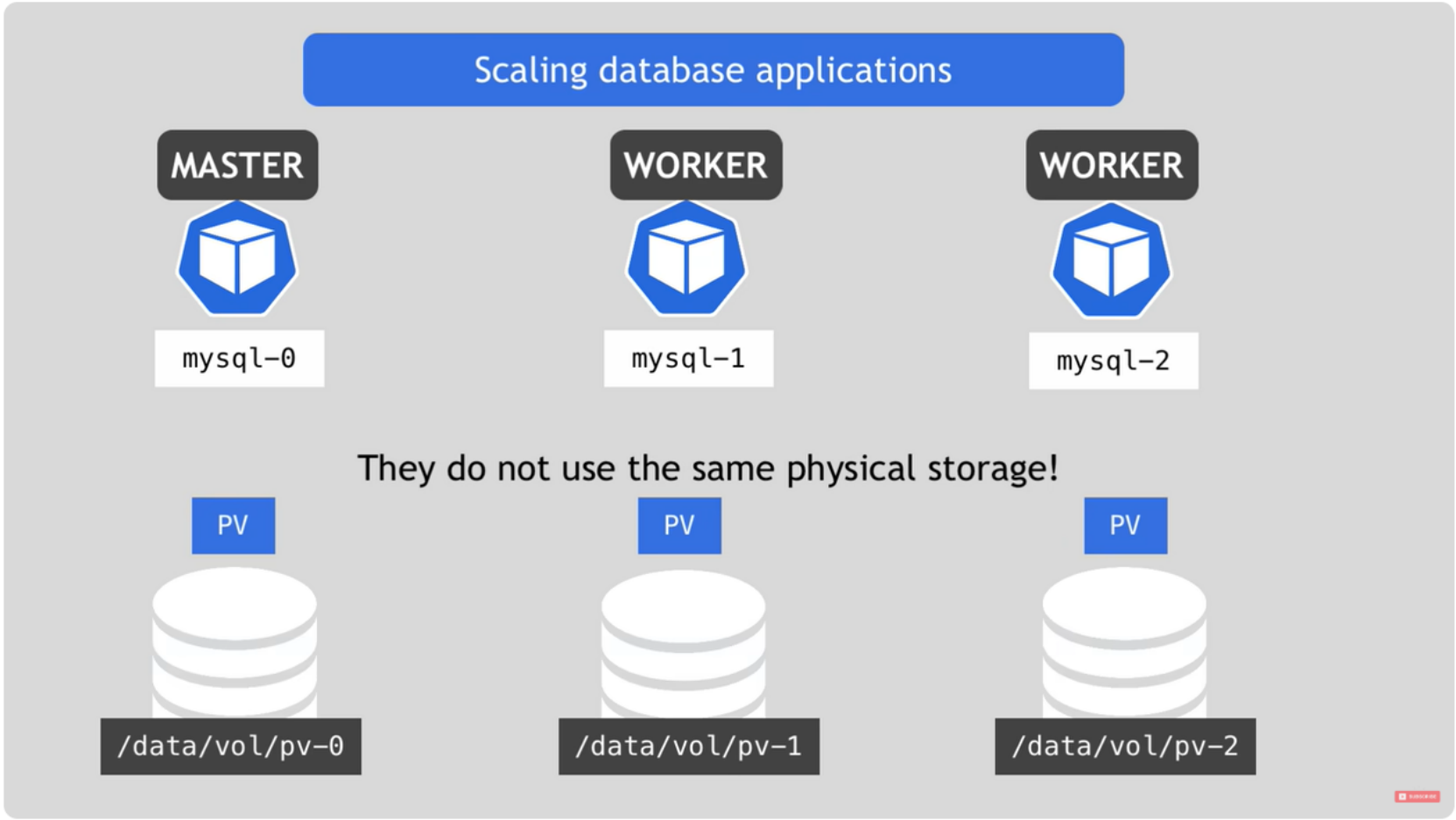
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

Pod Identity

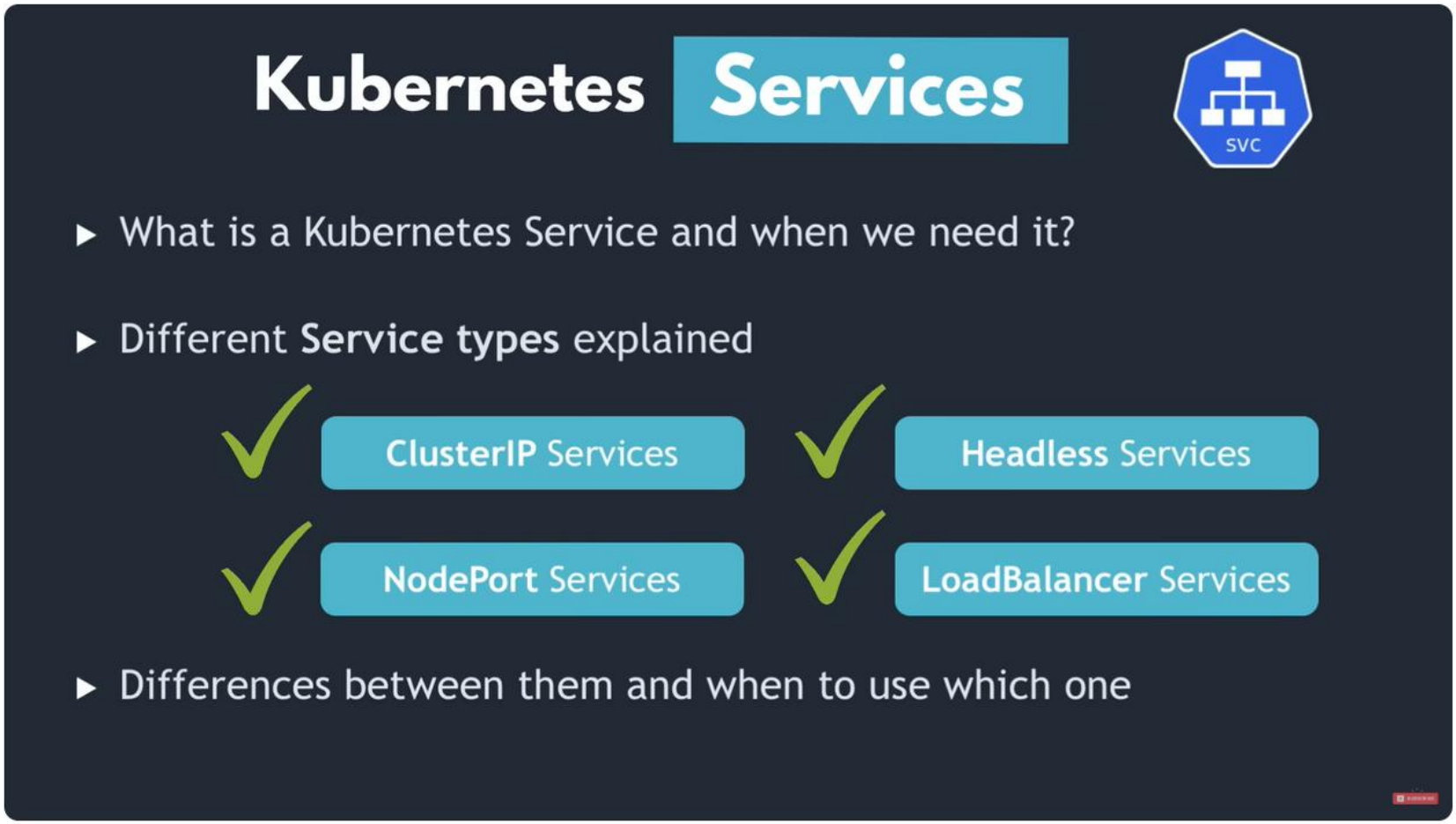
- sticky identity for each pod

- created from **same specification**, but **not interchangeable**!

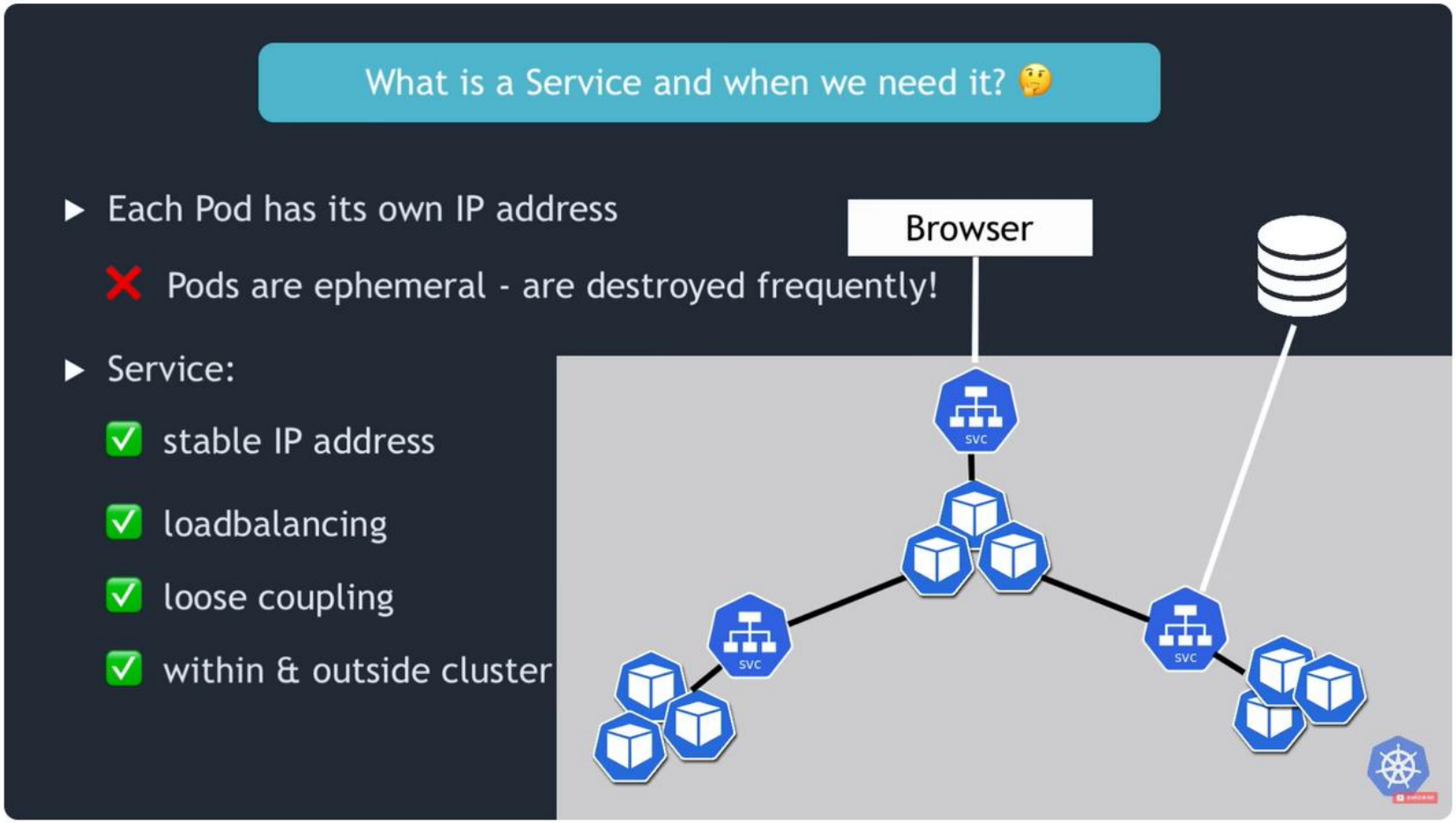
- persistent identifier across any re-scheduling



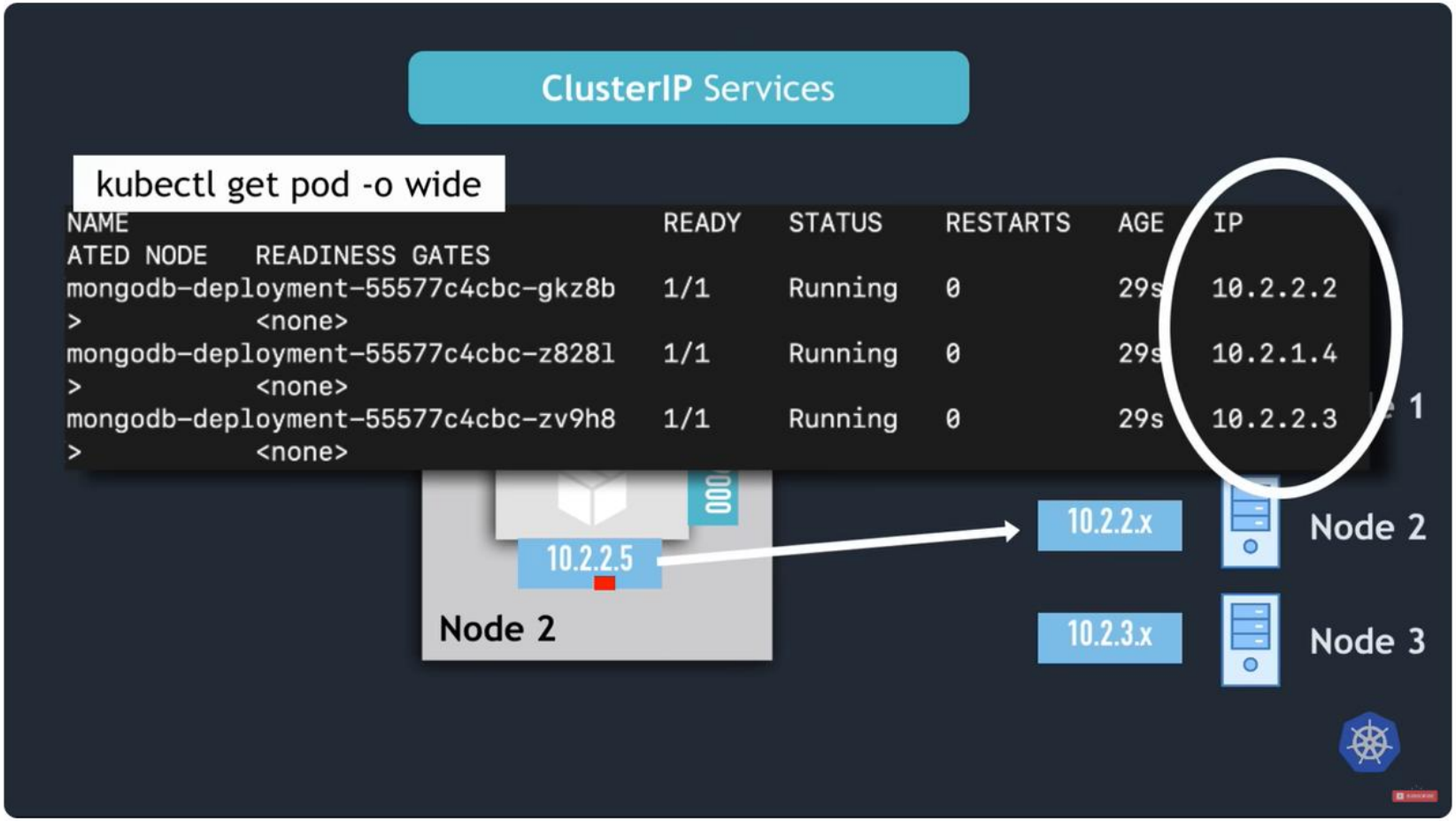
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



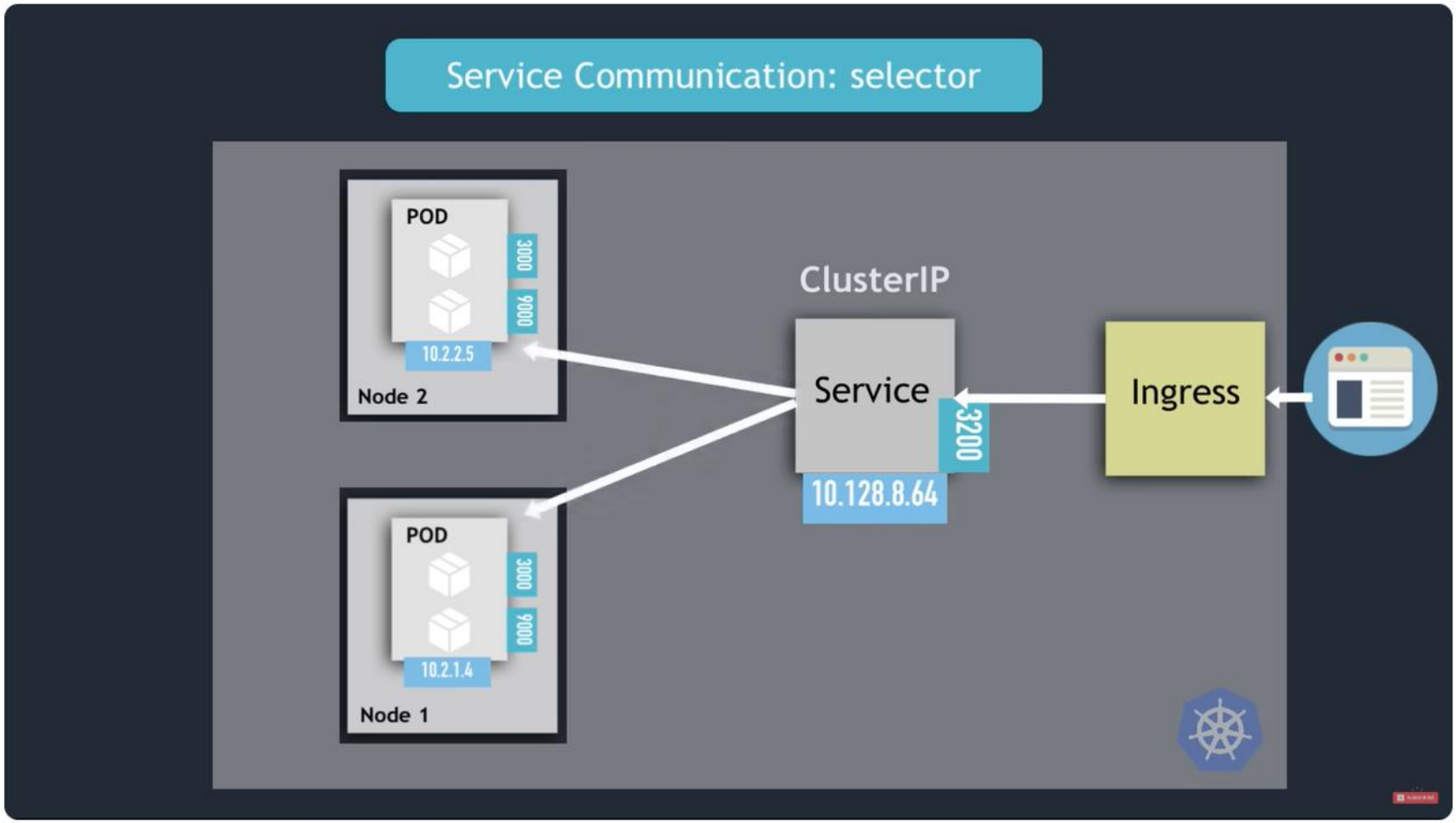
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



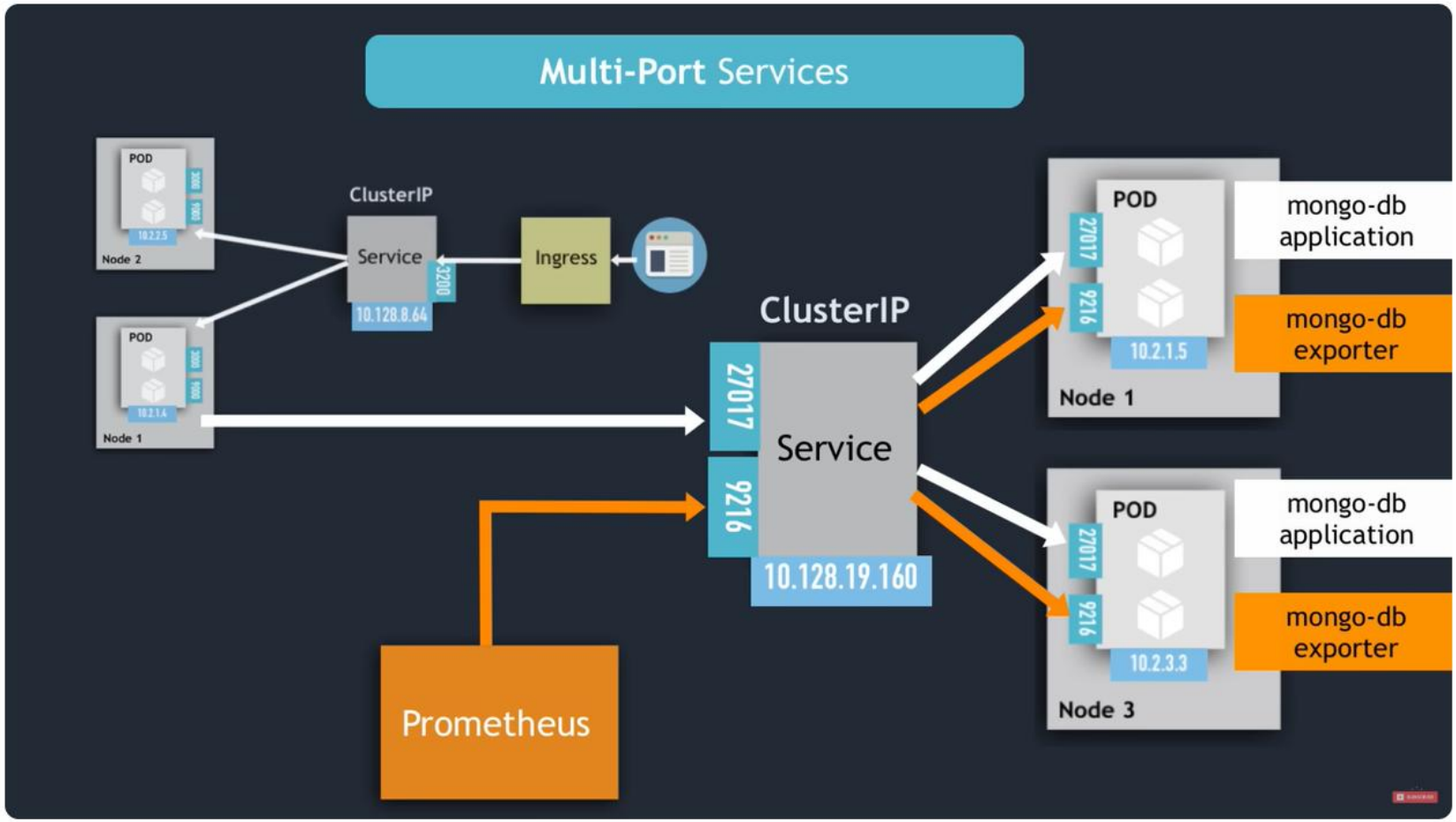
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



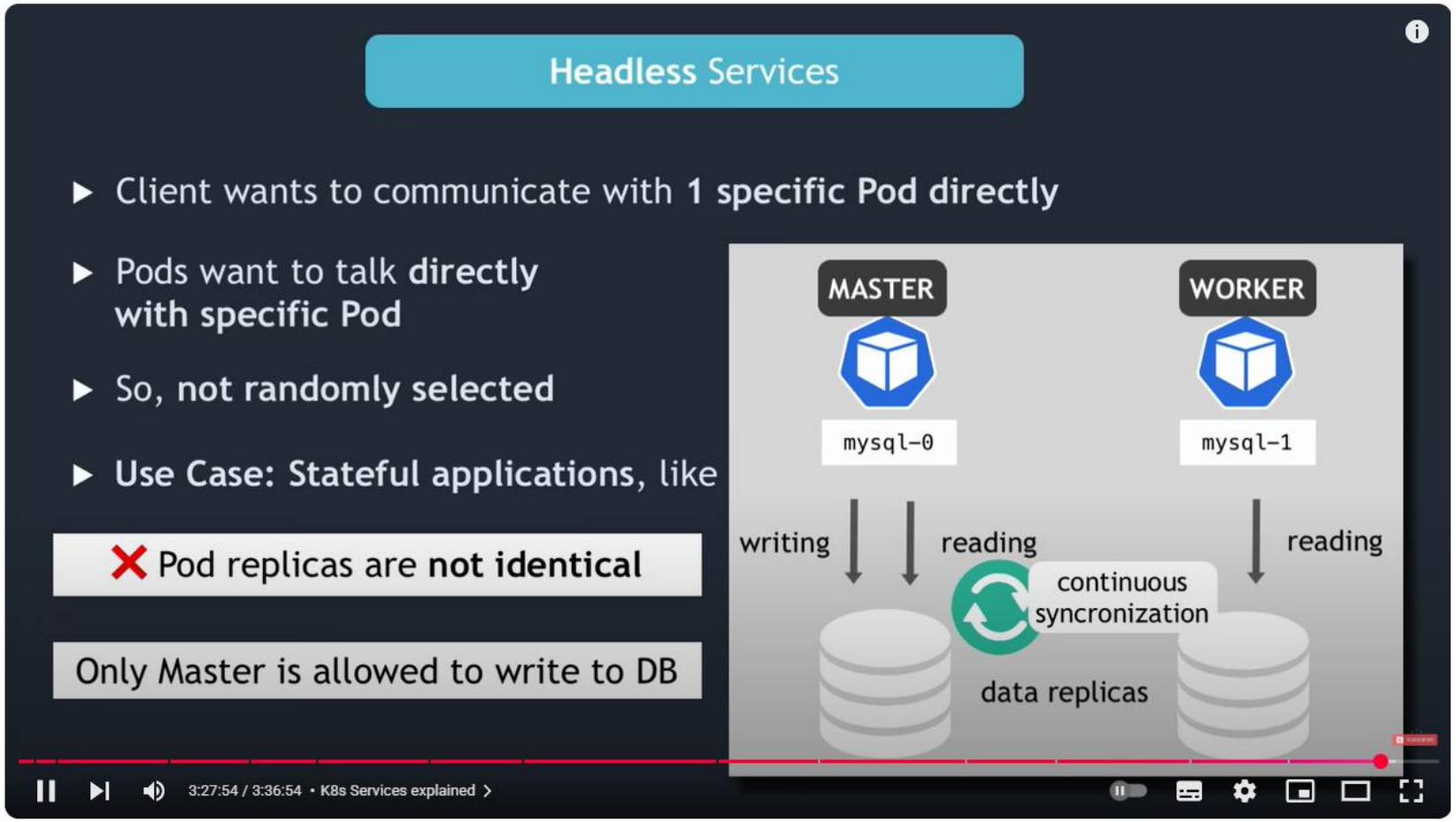
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



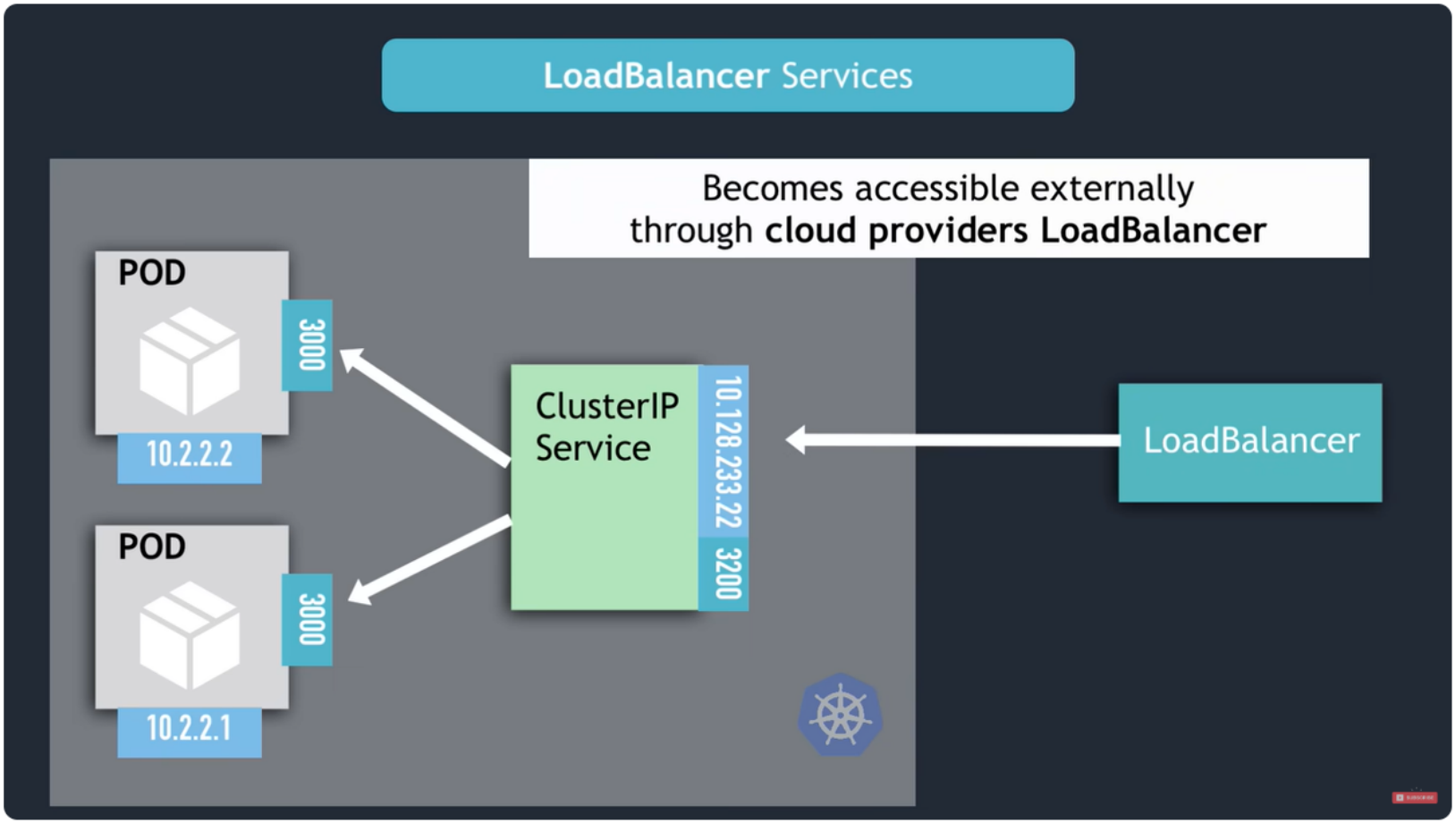
Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]



Kubernetes Tutorial for Beginners [FULL COURSE in 4 Hours]

- https://gitlab.com/nanuchi/youtube-tutorial-series/-/tree/master
- Availability / Scalability / Restore & recover
- DB URL usually in the built application
- Data must be managed (backup, recover, etc.), k8s do Not manage them

- StatefulSet used for stateful application like database
- Worker Nodes are mandatory
- Must be installed in every node
    - Container runtime
    - Kubelet
    - Kube proxy
- Master Nodes
    - Api server (load balanced between all master nodes)
        - Through UI, API and/or CLI (kubectl)
    - Scheduler
    - Controller manager
    - etcd (distributed to all master nodes)
- on local machine, how to?
    - Use minikube
        - 1 node k8s cluster
        - Virtual box
        - Docker pre-installed
        - For testing purposes
- kubectl
- marter node ~ control-plane rôle ?

```
v_tu@PN-N-0237:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

v_tu@PN-N-0237:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   13m   v1.33.1
v_tu@PN-N-0237:~$ kubectl get pod
No resources found in default namespace.
v_tu@PN-N-0237:~$ kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   17m
v_tu@PN-N-0237:~$
```

- In practice we do not handle pods directly
    - But its abstraction: deployment
- Need image, exemple nginx

```
v_tu@PN-N-0237:~$ kubectl create deployment nginx-depl --image=nginx
deployment.apps/nginx-depl created
v_tu@PN-N-0237:~$ kubectl get deployment
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-depl   0/1     1            0           10s
v_tu@PN-N-0237:~$ kubectl get pod
NAME                          READY   STATUS    RESTARTS   AGE
nginx-depl-68c944fcbc-crxhv   1/1     Running   0          20s
v_tu@PN-N-0237:~$ kubectl get pod
NAME                          READY   STATUS    RESTARTS   AGE
nginx-depl-68c944fcbc-crxhv   1/1     Running   0          51s
v_tu@PN-N-0237:~$ kubectl get deployment
NAME         READY   UP-TO-DATE   AVAILABLE   AGE
nginx-depl   1/1     1            1           59s
v_tu@PN-N-0237:~$ kubectl get replicaset
NAME                    DESIRED   CURRENT   READY   AGE
nginx-depl-68c944fcbc   1         1         1       2m34s
v_tu@PN-N-0237:~$
```

- Replicas are handled automatically; we don’t handle them at all
-
```
$ kubectl edit deployment nginx-depl
```

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
  creationTimestamp: "2025-05-28T09:21:41Z"
  generation: 1
  labels:
    app: nginx-depl
  name: nginx-depl
  namespace: default
  resourceVersion: "2051"
  uid: 0f1b45b3-525a-4d82-9798-53ec64a921b7
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      app: nginx-depl
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: nginx-depl
    spec:
      containers:
      - image: nginx
        imagePullPolicy: Always
        name: nginx
        resources: {}
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
      dnsPolicy: ClusterFirst
      restartPolicy: Always
      schedulerName: default-scheduler
      securityContext: {}
      terminationGracePeriodSeconds: 30
status:
  availableReplicas: 1
  conditions:
"/tmp/kubectl-edit-2425763798.yaml" 66L, 1797B                                2,1
```

- Autogenerated file (default since we didn’t specify anything
    - kubctl create deployment <name of the deployment> --image=<name of the image>
    - kubctl get pod
    - kubectl logs
    - kubectl describe pod <name of the pod>
- delete deployment (delete pod and replicas)

```
v_tu@PN-N-0237:~$ kubectl delete deployment mongo-depl
deployment.apps "mongo-depl" deleted
v_tu@PN-N-0237:~$ kubectl get pod
NAME                           READY   STATUS    RESTARTS   AGE
nginx-depl-68c944fcbc-crxhv    1/1     Running   0          17m
v_tu@PN-N-0237:~$
```

-
- Configuration
    -

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16
        ports:
        - containerPort: 80
~
~
```

```
v_tu@PN-N-0237:~$ touch nginx-deployment.yaml
v_tu@PN-N-0237:~$ vim nginx-deployment.yaml
v_tu@PN-N-0237:~$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
v_tu@PN-N-0237:~$ ^C
v_tu@PN-N-0237:~$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
nginx-depl-68c944fcbc-crxhv         1/1     Running   0          27m
nginx-deployment-7f65fcf556-7s62t   1/1     Running   0          30s
v_tu@PN-N-0237:~$
```

    - 3 parts
        - Metadata
        - Specification
        - Status (will be completed by k8s: Desired vs Actual)

- Service

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
~
```

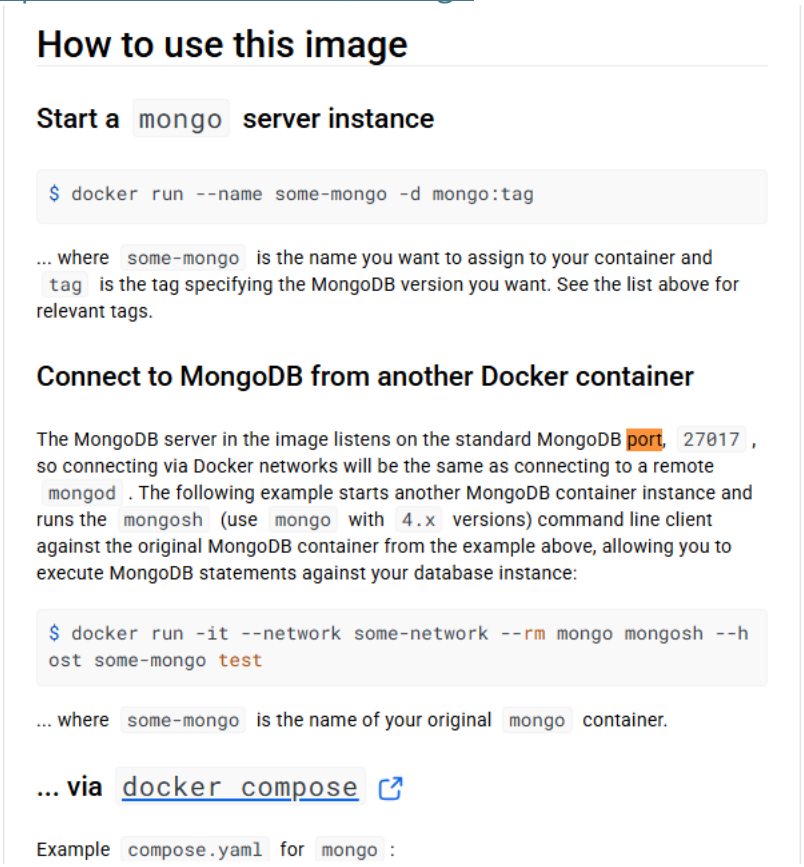
```
v_tu@PN-N-0237:~$ vim nginx-deployment.yaml
v_tu@PN-N-0237:~$ touch nginx-service.yaml
v_tu@PN-N-0237:~$ vim nginx-service.yaml
v_tu@PN-N-0237:~$ vim nginx-service.yaml
v_tu@PN-N-0237:~$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment unchanged
v_tu@PN-N-0237:~$ kubectl apply -f nginx-service.yaml
service/nginx-service created
v_tu@PN-N-0237:~$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
nginx-depl-68c944fcbc-crxhv         1/1     Running   0          115m
nginx-deployment-7f65fcf556-7s62t   1/1     Running   0          88m
nginx-deployment-7f65fcf556-dz7dv   1/1     Running   0          86m
v_tu@PN-N-0237:~$ kubectl get service
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP   150m
nginx-service   ClusterIP   10.105.94.207   <none>        80/TCP    17s
v_tu@PN-N-0237:~$
```

```
v_tu@PN-N-0237:~$ kubectl describe service nginx-service
Name:                     nginx-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.105.94.207
IPs:                      10.105.94.207
Port:                     <unset>  80/TCP
TargetPort:               8080/TCP
Endpoints:                10.244.0.5:8080,10.244.0.6:8080
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
v_tu@PN-N-0237:~$ kubectl get pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE    IP           NODE       NOMINATED NODE   READINESS GATES
nginx-depl-68c944fcbc-crxhv         1/1     Running   0          120m   10.244.0.3   minikube   <none>           <none>
nginx-deployment-7f65fcf556-7s62t   1/1     Running   0          93m    10.244.0.5   minikube   <none>           <none>
nginx-deployment-7f65fcf556-dz7dv   1/1     Running   0          91m    10.244.0.6   minikube   <none>           <none>
v_tu@PN-N-0237:~$
```

- kubectl get deployment nginx-deployment -o yaml

-
```
v_tu@PN-N-0237:~$ kubectl get deployment nginx-deployment -o yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"labels":{"app":"nginx"},"name":"nginx-deployment","namespace":"default"},"spec":{"replicas":2,"selector":{"matchLabels":{"
app":"nginx"}},"template":{"metadata":{"labels":{"app":"nginx"}},"spec":{"containers":[{"image":"nginx:1.16","name":"nginx","ports":[{"containerPort":80}]}]}}}}
  creationTimestamp: "2025-05-28T09:48:27Z"
  generation: 2
  labels:
    app: nginx
  name: nginx-deployment
  namespace: default
  resourceVersion: "3495"
  uid: 87a0f97e-8c52-47b1-abee-24f48988cd13
spec:
  progressDeadlineSeconds: 600
  replicas: 2
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      app: nginx
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16
        imagePullPolicy: IfNotPresent
        name: nginx
        ports:
        - containerPort: 80
          protocol: TCP
        resources: {}
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
      dnsPolicy: ClusterFirst
      restartPolicy: Always
      schedulerName: default-scheduler
      securityContext: {}
      terminationGracePeriodSeconds: 30
status:
  availableReplicas: 2
  conditions:
  - lastTransitionTime: "2025-05-28T09:48:27Z"
    lastUpdateTime: "2025-05-28T09:48:43Z"
    message: ReplicaSet "nginx-deployment-7f65fcf556" has successfully progressed.
    reason: NewReplicaSetAvailable
    status: "True"
    type: Progressing
  - lastTransitionTime: "2025-05-28T09:50:53Z"
    lastUpdateTime: "2025-05-28T09:50:53Z"
    message: Deployment has minimum availability.
    reason: MinimumReplicasAvailable
    status: "True"
    type: Available
  observedGeneration: 2
  readyReplicas: 2
  replicas: 2
  updatedReplicas: 2
v_tu@PN-N-0237:~$
```
- Check containers configuration on https://hub.docker.com/

- https://hub.docker.com/_/mongo



How to use this image

Start a `mongo` server instance

```
$ docker run --name some-mongo -d mongo:tag
```

... where `some-mongo` is the name you want to assign to your container and `tag` is the tag specifying the MongoDB version you want. See the list above for relevant tags.

Connect to MongoDB from another Docker container

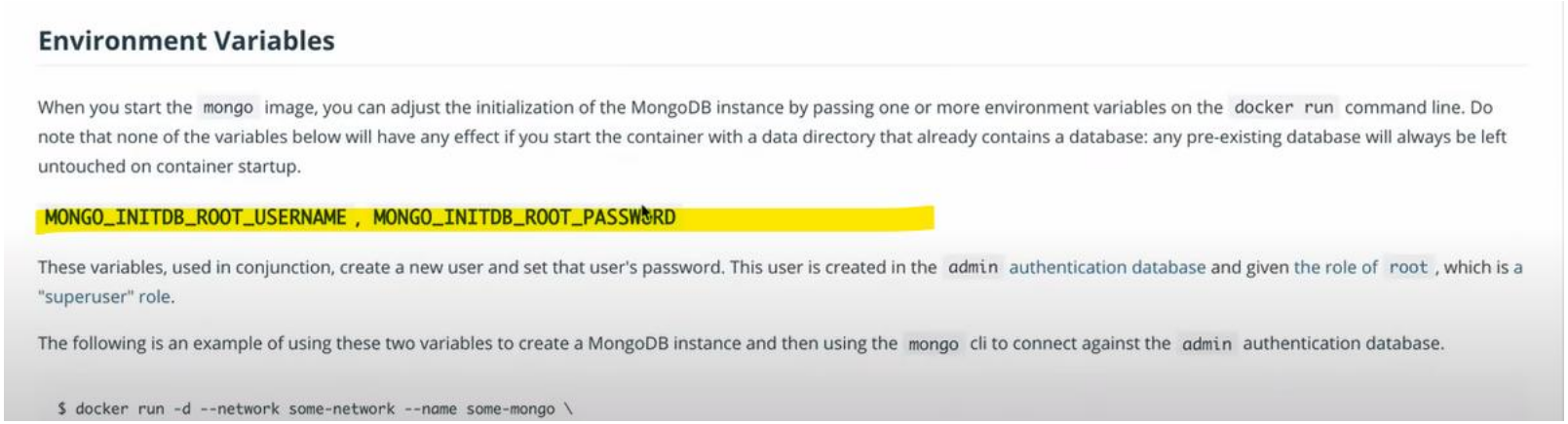
The MongoDB server in the image listens on the standard MongoDB port, `27017`, so connecting via Docker networks will be the same as connecting to a remote `mongod`. The following example starts another MongoDB container instance and runs the `mongosh` (use `mongo` with `4.x` versions) command line client against the original MongoDB container from the example above, allowing you to execute MongoDB statements against your database instance:

```
$ docker run -it --network some-network --rm mongo mongosh --h
ost some-mongo test
```

... where `some-mongo` is the name of your original `mongo` container.

... via `docker compose`

Example `compose.yaml` for `mongo`:

-


Environment Variables

When you start the `mongo` image, you can adjust the initialization of the MongoDB instance by passing one or more environment variables on the `docker run` command line. Do note that none of the variables below will have any effect if you start the container with a data directory that already contains a database: any pre-existing database will always be left untouched on container startup.

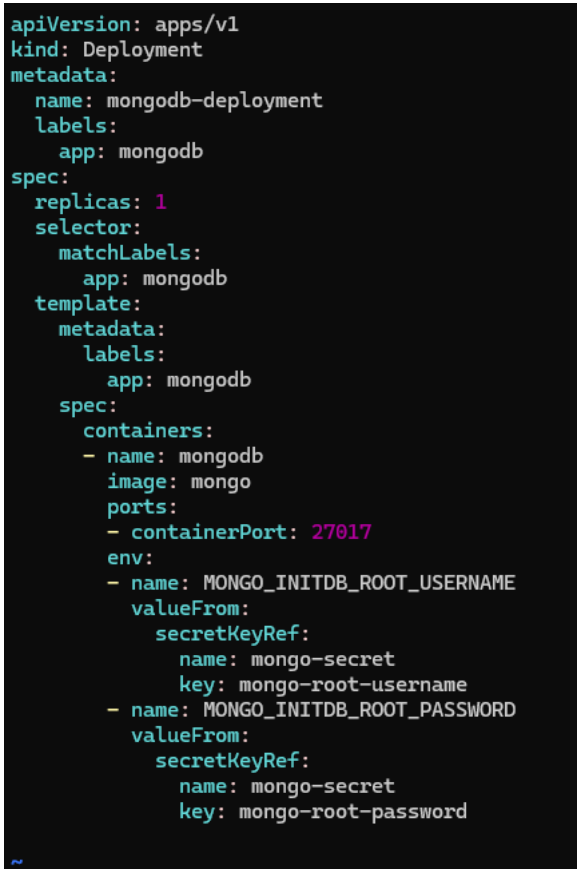
MONGO_INITDB_ROOT_USERNAME , MONGO_INITDB_ROOT_PASSWORD

These variables, used in conjunction, create a new user and set that user's password. This user is created in the `admin` authentication database and given the role of `root` , which is a "superuser" role.

The following is an example of using these two variables to create a MongoDB instance and then using the `mongo` cli to connect against the `admin` authentication database.

```
$ docker run -d --network some-network --name some-mongo \
```

- New appli with secret
  -
```
apiVersion: v1
kind: Secret
metadata:
  name: mongodb-secret
type: Opaque
data:
  mongo-root-username: dXNlcm5hbWU=
  mongo-root-password: cGFzc3dvcmQ=
~
```

- 
```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongodb-deployment
  labels:
    app: mongodb
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongodb
  template:
    metadata:
      labels:
        app: mongodb
    spec:
      containers:
      - name: mongodb
        image: mongo
        ports:
        - containerPort: 27017
        env:
        - name: MONGO_INITDB_ROOT_USERNAME
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-root-username
        - name: MONGO_INITDB_ROOT_PASSWORD
          valueFrom:
            secretKeyRef:
              name: mongo-secret
              key: mongo-root-password
~
```

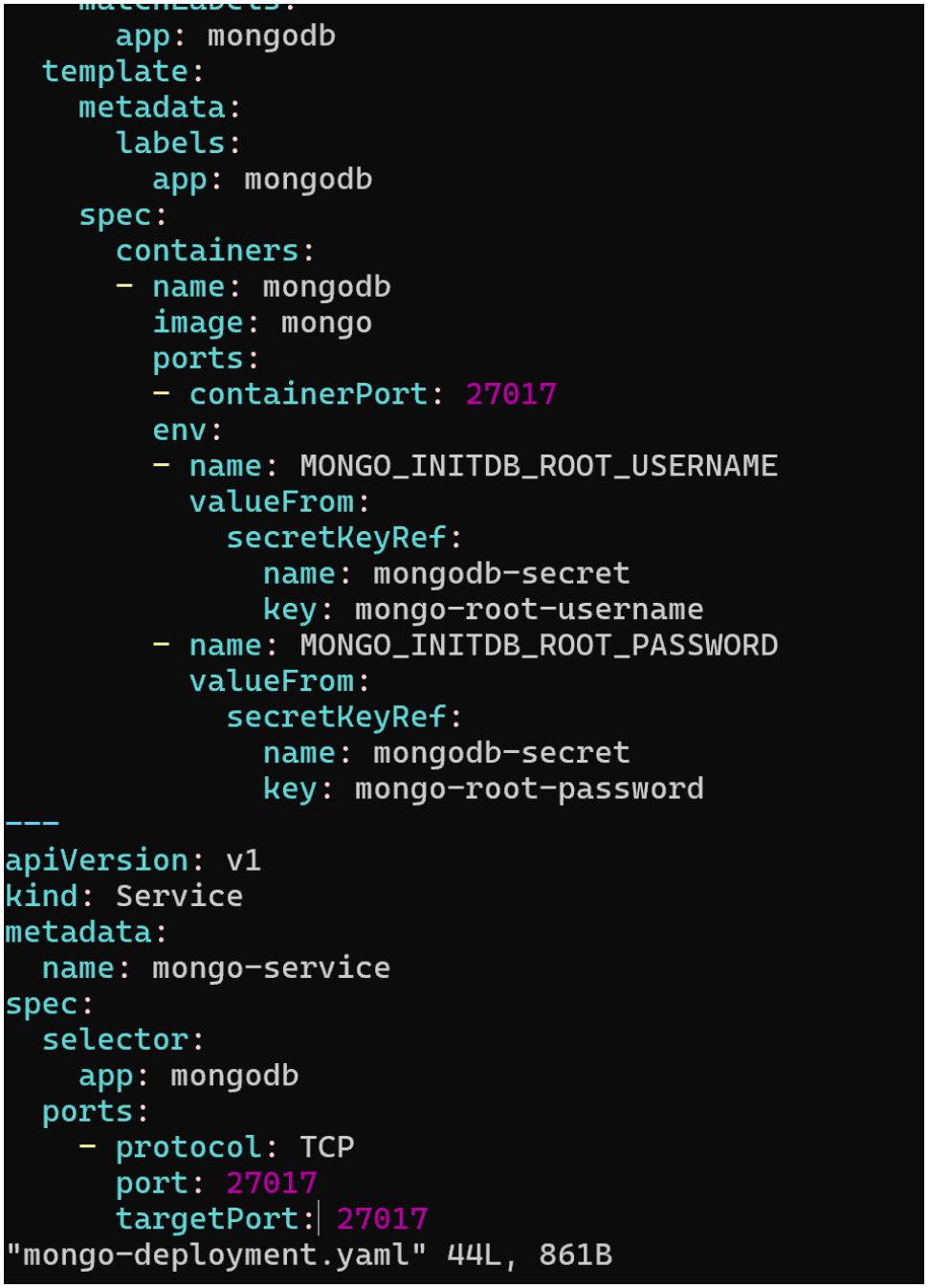
```
v_tu@PN-N-0237:~$ kubectl apply -f mongo-secret.yaml
secret/mongodb-secret created
v_tu@PN-N-0237:~$ kubectl get secret
NAME             TYPE     DATA   AGE
mongodb-secret   Opaque   2      16s
v_tu@PN-N-0237:~$ vim mongo-deployment.yaml
v_tu@PN-N-0237:~$ vim mongo-deployment.yaml
v_tu@PN-N-0237:~$ kubectl apply -f mongo-deployment.yaml
deployment.apps/mongodb-deployment created
v_tu@PN-N-0237:~$ kubectl get all
NAME                                    READY   STATUS                       RESTARTS   AGE
pod/mongodb-deployment-7c8d894848-9zhpp   0/1     CreateContainerConfigError   0          12s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   5h46m

NAME                                 READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/mongodb-deployment   0/1     1            0           12s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/mongodb-deployment-7c8d894848   1         1         0       12s
v_tu@PN-N-0237:~$
```

- Service

- 
```
    matchLabels:
      app: mongodb
  template:
    metadata:
      labels:
        app: mongodb
    spec:
      containers:
      - name: mongodb
        image: mongo
        ports:
        - containerPort: 27017
        env:
        - name: MONGO_INITDB_ROOT_USERNAME
          valueFrom:
            secretKeyRef:
              name: mongodb-secret
              key: mongo-root-username
        - name: MONGO_INITDB_ROOT_PASSWORD
          valueFrom:
            secretKeyRef:
              name: mongodb-secret
              key: mongo-root-password
---
apiVersion: v1
kind: Service
metadata:
  name: mongo-service
spec:
  selector:
    app: mongodb
  ports:
    - protocol: TCP
      port: 27017
      targetPort: 27017
"mongo-deployment.yaml" 44L, 861B
```

```
v_tu@PN-N-0237:~$ vim mongo-deployment.yaml
v_tu@PN-N-0237:~$ kubectl apply -f mongo-deployment.yaml
*deployment.apps/mongodb-deployment unchanged
service/mongo-service created
v_tu@PN-N-0237:~$ vim mongo-deployment.yaml
v_tu@PN-N-0237:~$ kubectl get service
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)     AGE
kubernetes      ClusterIP   10.96.0.1      <none>        443/TCP     5d4h
mongo-service   ClusterIP   10.103.146.8   <none>        27017/TCP   105s
v_tu@PN-N-0237:~$
```

-

```
v_tu@PN-N-0237:~$ kubectl get service
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)     AGE
kubernetes      ClusterIP   10.96.0.1      <none>        443/TCP     5d4h
mongo-service   ClusterIP   10.103.146.8   <none>        27017/TCP   105s
v_tu@PN-N-0237:~$ kubectl describe service mongo-service
Name:                     mongo-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=mongodb
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.103.146.8
IPs:                      10.103.146.8
Port:                     <unset>  27017/TCP
TargetPort:               27017/TCP
Endpoints:                10.244.0.8:27017
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
v_tu@PN-N-0237:~$ kubectl get pod -o wide
NAME                                  READY   STATUS    RESTARTS   AGE     IP           NODE       NOMINATED NODE   READINESS GATES
mongodb-deployment-6d9d7c68f6-grfdv   1/1     Running   0          4d22h   10.244.0.8   minikube   <none>           <none>
v_tu@PN-N-0237:~$
```

```
v_tu@PN-N-0237:~$ kubectl get all | grep mongodb
pod/mongodb-deployment-6d9d7c68f6-grfdv   1/1     Running   0          4d22h
deployment.apps/mongodb-deployment   1/1     1            1           4d23h
replicaset.apps/mongodb-deployment-6d9d7c68f6   1         1         1       4d22h
replicaset.apps/mongodb-deployment-7c8d894848   0         0         0       4d23h
v_tu@PN-N-0237:~$
```

- Mongodb-express / configMap
  - database_url refers to the service:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: mongodb-configmap
data:
  database_url: mongodb-service
~
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-express
  labels:
    app: mongo-express
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo-express
  template:
    metadata:
      labels:
        app: mongo-express
    spec:
      containers:
      - name: mongo-express
        image: mongo-express
        ports:
        - containerPort: 8081
        env:
        - name: ME_CONFIG_MONGODB_AUTH_USERNAME
          valueFrom:
            secretKeyRef:
              name: mongodb-secret
              key: mongo-root-username
        - name: ME_CONFIG_MONGODB_AUTH_PASSWORD
          valueFrom:
            secretKeyRef:
              name: mongodb-secret
              key: mongo-root-username
        - name: ME_CONFIG_MONGODB_AUTH_DATABASE
          valueFrom:
            configMapKeyRef:
              name: mongodb-configmap
              key: database_url
~
```

    - kubectl apply -f mongodb-configmap.yaml
    - kubectl apply -f mongo-express.yaml
- external service

-
```
            configMapKeyRef:
              name: mongodb-configmap
              key: database_url
---
apiVersion: v1
kind: Service
metadata:
  name: mongo-express-service
spec:
  selector:
    app: mongo-express
  type: LoadBalancer
  ports:
    - protocol: TCP
      port: 8081
      targetPort: 8081
      nodePort: 30000
"mongo-express.yaml" 51L, 1118B
```
-
```
v_tu@PN-N-0237:~$ vim mongo-express.yaml
v_tu@PN-N-0237:~$ kubectl apply -f mongo-express.yaml
deployment.apps/mongo-express unchanged
service/mongo-express-service created
v_tu@PN-N-0237:~$ kubectl get pod
NAME                                READY   STATUS    RESTARTS      AGE
mongo-express-78fc69476c-76sgw      1/1     Running   4 (85s ago)   8m24s
mongodb-deployment-6d9d7c68f6-grfdv   1/1     Running   0             6d16h
v_tu@PN-N-0237:~$ kubectl get service
NAME                    TYPE           CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes              ClusterIP      10.96.0.1       <none>        443/TCP          6d22h
mongo-express-service   LoadBalancer   10.99.163.203   <pending>     8081:30000/TCP   15s
mongo-service           ClusterIP      10.103.146.8    <none>        27017/TCP        42h
v_tu@PN-N-0237:~$
```
-
```
v_tu@PN-N-0237:~$ minikube service mongo-express-service
|-----------|----------------------|-------------|---------------------------|
| NAMESPACE |         NAME         | TARGET PORT |            URL            |
|-----------|----------------------|-------------|---------------------------|
| default   | mongo-express-service |        8081 | http://192.168.39.44:30000 |
|-----------|----------------------|-------------|---------------------------|
🎉  Opening service default/mongo-express-service in default browser...
👉  http://192.168.39.44:30000
v_tu@PN-N-0237:~$
```

- Namespace
  - Virtual cluster inside the cluster

    - Example of namespace
        - Resources grouped: a namespace for DB, a namespace for mentoring, etc.
        - Teams grouped: for example, several teams use the same naming, creating conflict
        - Resource sharing:
            - Environment: staging, development + common resources (like elastic stack for example)
            - Blue/Green deployment: blue deploy, green deploy + common resources
        - Limit resources and/or access (example rights control, resource limitation, etc)
    - Cannot be shared between namespace: configMap or secret, even for a same resource, need to be duplicated for each resource needed
    - Can be shared between namespace: service
- External service VS Ingress: to access to the application
    - External service
        - Simple to use
        - useful for quick tests,
        - Expose port, example: 192.60.11.5:4123
        - Not secure (http)
    - Ingress
        - Expected result
        - Looks like a proper address: my-app/com
        - Can be secured (https), default http
- Ingress
    - https://dev-k8sref-io.web.app/docs/services/ingress-v1/#ingressspec

```
v_tu@PN-N-0237:~$ minikube addons enable ingress
💡  ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
    ▪ Using image registry.k8s.io/ingress-nginx/controller:v1.12.2
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.5.3
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.5.3
🔎  Verifying ingress addon...
🌟  The 'ingress' addon is enabled
v_tu@PN-N-0237:~$ kubectl get ns
NAME              STATUS   AGE
default           Active   6d23h
ingress-nginx     Active   59s
kube-node-lease   Active   6d23h
kube-public       Active   6d23h
kube-system       Active   6d23h
v_tu@PN-N-0237:~$ kubectl create namespace kubernetes-dashboard
namespace/kubernetes-dashboard created
v_tu@PN-N-0237:~$ kubectl get ns
NAME                   STATUS   AGE
default                Active   6d23h
ingress-nginx          Active   99s
kube-node-lease        Active   6d23h
kube-public            Active   6d23h
kube-system            Active   6d23h
kubernetes-dashboard   Active   2s
v_tu@PN-N-0237:~$
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: dashboard-ingress
  namespace: kubernetes-dashboard
spec:
  rules:
  - host: dashboard.com
    http:
      paths:
        - pathType: Prefix
          path: "/"
          backend:
            service:
              name: kubernetes-dashboard
              port:
                number: 80
~
~
~
```

-
```
v_tu@PN-N-0237:~$ kubectl apply -f dashboard-ingress.yaml
ingress.networking.k8s.io/dashboard-ingress created
```
-
```
v_tu@PN-N-0237:~$ kubectl get Ingress --namespace=kubernetes-dashboard
NAME                CLASS   HOSTS           ADDRESS         PORTS   AGE
dashboard-ingress   nginx   dashboard.com   192.168.39.44   80      116s
v_tu@PN-N-0237:~$ ^C
v_tu@PN-N-0237:~$ kubectl get Ingress -n kubernetes-dashboard
NAME                CLASS   HOSTS           ADDRESS         PORTS   AGE
dashboard-ingress   nginx   dashboard.com   192.168.39.44   80      2m51s
v_tu@PN-N-0237:~$
```
-
- sudo vim /etc/hosts
-
```
# This file was automatically generated by WSL. To stop automatic generation of this file, add the following entry to /etc/wsl.conf:
# [network]
# generateHosts = false
127.0.0.1       localhost
127.0.1.1       PN-N-0237.sesame.infotel.com    PN-N-0237

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
192.168.39.44   dashboard.com
~
```

- Volume
  - Storage that doesn’t depend on the pod lifecycle (like a pod for mysql for e.g.)
- StatefulSet
  - for stateful application
  - can replicate pod
  - can run multiple replicates
- Service type
  - ClusterIP
    - Default
  - Headless
    - For statefull application (like db)

    - NodePort
        - Call the port directly
        - Not secure
    - LoadBalancer
        - Accessible and more secure than nodeport

Installer et configurer kubectl | Kubernetes

- Vous devez utiliser une version de kubectl qui diffère seulement d'une version mineure de la version de votre cluster. Par exemple, un client v1.2 doit fonctionner avec un master v1.1, v1.2 et v1.3. L'utilisation de la dernière version de kubectl permet d'éviter des problèmes imprévus.
- Installer **KVM** (hyperviseur) sur Ubuntu
    - https://computingforgeeks.com/install-kvm-virtualization-on-ubuntu-noble-numbat/
    - Configuration / Ajouter les permissions : https://help.ubuntu.com/community/KVM/Installation

```
v_tu@PN-N-0237:~$ groups
v_tu adm dialout cdrom floppy sudo audio dip video plugdev users netdev libvirt docker kvm
v_tu@PN-N-0237:~$ minikube start --driver=kvm
😄  minikube v1.36.0 on Ubuntu 24.04 (amd64)
✨  Using the kvm2 driver based on user configuration
💾  Downloading driver docker-machine-driver-kvm2:
    > docker-machine-driver-kvm2-...:  65 B / 65 B [---------] 100.00% ? p/s 0s
    > docker-machine-driver-kvm2-...:  15.20 MiB / 15.20 MiB  100.00% 23.01 MiB
💿  Downloading VM boot image ...
    > minikube-v1.36.0-amd64.iso....:  65 B / 65 B [---------] 100.00% ? p/s 0s
    > minikube-v1.36.0-amd64.iso:  360.83 MiB / 360.83 MiB  100.00% 22.70 MiB p
👍  Starting "minikube" primary control-plane node in "minikube" cluster
💾  Downloading Kubernetes v1.33.1 preload ...
    > preloaded-images-k8s-v18-v1...:  347.04 MiB / 347.04 MiB  100.00% 24.12 M
🔥  Creating kvm2 VM (CPUs=2, Memory=2200MB, Disk=20000MB) ...
❗  Image was not built for the current minikube version. To resolve this you can delete and recreate
your minikube cluster using the latest images. Expected minikube version: v1.35.0 -> Actual minikube v
ersion: v1.36.0
🐳  Preparing Kubernetes v1.33.1 on Docker 28.0.4 ...
    ▪ Generating certificates and keys ...
    ▪ Booting up control plane ...
    ▪ Configuring RBAC rules ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
v_tu@PN-N-0237:~$ virsh list --all
 Id   Name       State
-------------------------
 1    minikube   running

v_tu@PN-N-0237:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

v_tu@PN-N-0237:~$
```

-